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Part I

Time Series

Chapter 1

Introduction

1.1 Informal introduction

Definition 1.1.1 (Informal). A time series is a collection of observations $\{x_1, x_2, \dots, x_n\}$ on a given phenomenon, ordered in time ($t = 1, \dots, n$).

Example 1.1.1. Hourly temperature of a patient subject to a treatment, daily price of an asset return, monthly index of industrial production, quarterly U.S. GDP, annual immigrations in Italy.

Remark 1. There are two ways of tackling time series analysis:

1. **decomposition** (classical time series analysis, 1930's): what we observe is the combination of some unobserved components we can estimate making some assumption.
Models can be unobserved component models, state space methods or non-parametric (filtering and smoothing methods, local polynomial regression, kernel estimation);
2. **modelling** (modern time series analysis, 1970's, Box Jenkins): we specify a hypothetical probability model to represent the data (which describes the dependency of observation in time). After an appropriate family of models has been chosen, it is then possible to estimate parameters, check goodness of fit etc.
Model can be linear (ARMA and generalization) or non-linear (GARCH, TAR, ARFIMA, GAS)¹

Remark 2. Some remarks:

- these observations come from random variables that are likely to be *dependent*;
- *frequency of observation* and *length of the series*/number of observation depends on nature of the phenomenon (eg it doesn't make sense to measure temperature of patient yearly); this typically leads to using different methods for different kind of series

¹In un corso base si fanno le decomposizioni classiche e la modellazione lineare, la modellazione non lineare è advanced

- aims of ts analysis is to analyse the dependence within time series (univariate) or between series (multivariate), and/or to produce nowcasts (real time predictions) and forecasts

Remark 3. First steps

1. Plot the series and describe the main features of the graph, checking in particular whether is there:
 - an increasing/decreasing trend,
 - a seasonal component/a cyclical behaviour (if so the periods seems to be constant or varying?)
 - any apparent sharp changes in behavior/discontinuities in the series: such as a sudden change of level, it may be advisable to analyze the series by first breaking it into homogeneous segments;
 - any outlying observations: they should be studied carefully to check whether there is any justification for discarding them (as for example if an observation has been incorrectly recorded)

Inspection of a graph may also suggest the possibility of representing the data as a realization of the process (the classical decomposition model)

$$X_t = m_t + s_t + \varepsilon_t \quad (1.1)$$

with

- m_t is a slowly changing function known as a **trend component**,
- s_t is a function with known period d referred to as a **seasonal component**
- ε_t a random **noise component** that is stationary

The model could be even specified by pre-logging the observed variable in order to obtain a relation which is multiplicative in the three component

$$\log X_t = m_t + s_t + \varepsilon_t$$

2. one typically does some **transformation** in order to prepare the sequence to be analyzed. The idea is to remove the trend and seasonal components (ITSF 1.5) to get stationary residuals (sec 1.4 ITSF).
If seasonal and noise fluctuations appear to *increase* with the level of the process, then a preliminary transformation of the data (eg log) is often used to make the transformed data more compatible with model 1.1.
3. Choose a model to fit the residuals, making use of various sample statistics including the sample autocorrelation
4. Forecasting will be achieved by forecasting the residuals and then inverting the transformations described above to arrive at forecasts of the original series

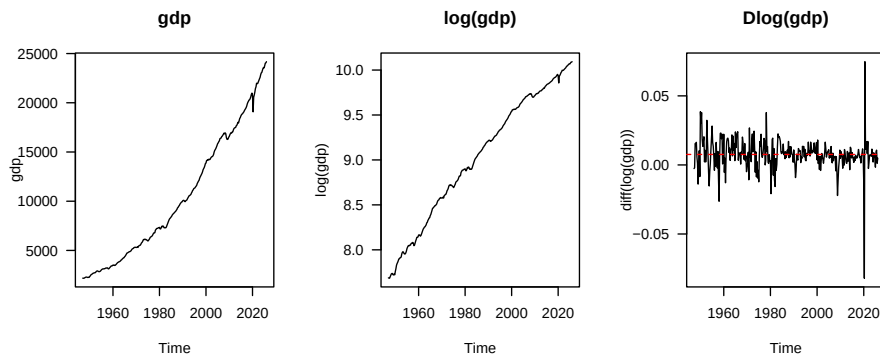


Figure 1.1: US GDP (1947-2026)

1.2 Data examples

Example 1.2.1. eg the gdp follow an exponential trend, but if we take the logarithm the increase become linear and if we further take the differences of the logarithm of the series we have an estimate of the *growth rate*² which become approximately constant in time. So the growth rate is stationary in mean; there seems to two main phases (the first up to 1980 with greater variability, then the following more stable but a spike due to covid) so we could think of two different generating process

```
library(lbts)
head(gdp)

##      Qtr1 Qtr2 Qtr3 Qtr4
## 1947 2183 2177 2172 2206
## 1948 2240 2277

par(mfrow = c(1,3))
plot(gdp, main = "gdp", las=1)
plot(log(gdp), main = "log(gdp)", las=1)
plot(diff(log(gdp)), main = "Dlog(gdp)", las=1)
abline(h=mean(diff(log(gdp))), lty="dashed", col="red")
```

For plotting

- a raw R vector as time series or multiple time series in the same graph use `ts.plot`
- for a single time series object standard `plot.ts` is better programmable

Example 1.2.2. `library(TSA)`
`options(width=90)`

²This is discrete equivalent of derivative (and we know that if we have a linear increasing function and take the derivative it becomes a constant)

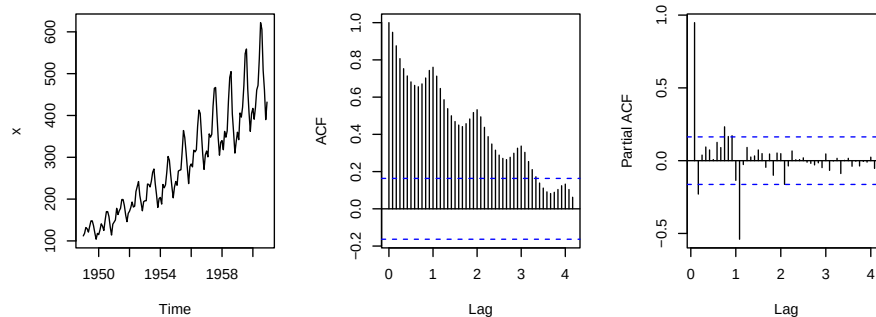
```
h <- function(x) round(head(x, n=20), 3)

## a) Raw serie (monthly data)
x <- AirPassengers
h(x)

##      Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
## 1949 112 118 132 129 121 135 148 148 136 119 104 118
## 1950 115 126 141 135 125 149 170 170

ts_plot(x, main="a) raw series")
```

a) raw series

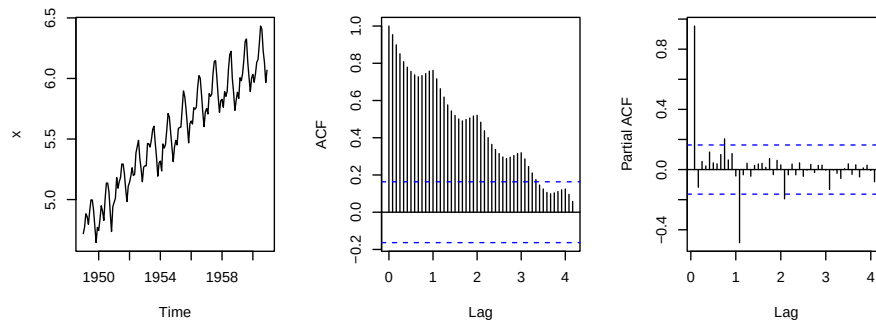


```
## b) Log series
h(log(x))

##      Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
## 1949 4.718 4.771 4.883 4.860 4.796 4.905 4.997 4.997 4.913 4.779 4.644 4.771
## 1950 4.745 4.836 4.949 4.905 4.828 5.004 5.136 5.136

ts_plot(log(x), main="b) log series")
```

b) log series

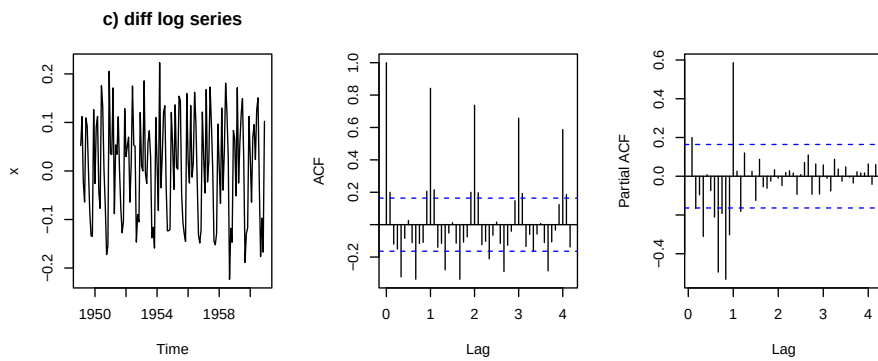


```
## c) diff log series
h(diff(log(x)))
```



```
##           Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct      Nov      Dec
## 1949           0.052  0.112 -0.023 -0.064  0.109  0.092  0.000 -0.085 -0.134 -0.135  0.126
## 1950 -0.026  0.091  0.112 -0.043 -0.077  0.176  0.132  0.000 -0.073
```

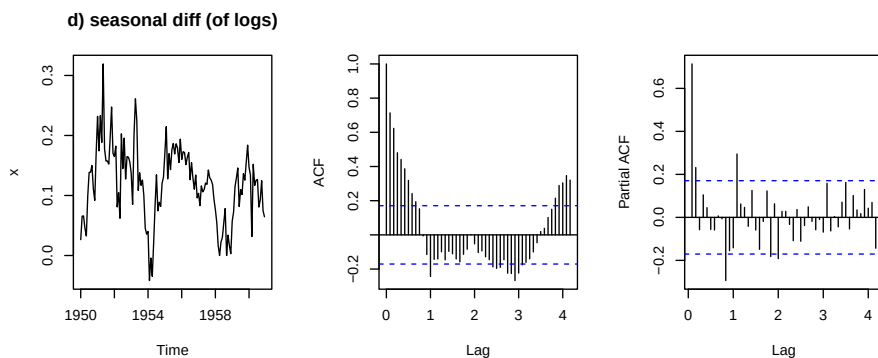
```
ts_plot(diff(log(x)), main = "c) diff log series")
```



```
## d) seasonal difference (of logs)
h(diff(log(x), 12))
```

```
##           Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct      Nov      Dec
## 1950  0.026  0.066  0.066  0.045  0.033  0.099  0.139  0.139  0.150  0.111  0.092  0.171
## 1951  0.232  0.174  0.233  0.188  0.319  0.178  0.158  0.158
```

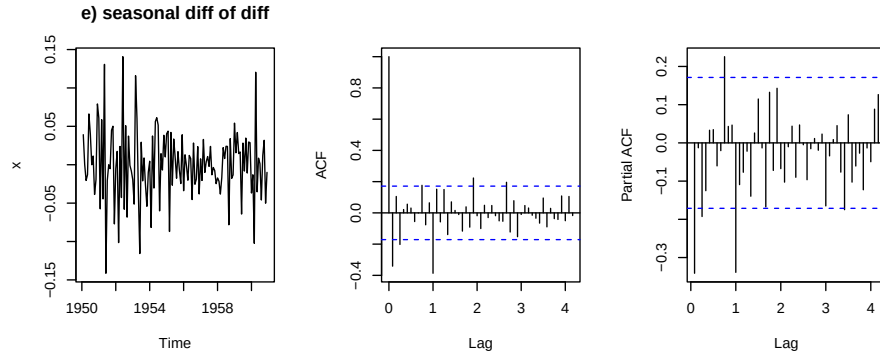
```
ts_plot(diff(log(x), 12), main = "d) seasonal diff (of logs)")
```



```
## e) seasonal differences of the differences
y <- diff( diff(log(x)), 12)
h(y)
```

```
##           Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct      Nov      Dec
## 1950           0.039  0.000 -0.020 -0.013  0.066  0.040  0.000  0.011 -0.039 -0.019  0.079
## 1951  0.061 -0.057  0.059 -0.045  0.131 -0.141 -0.020  0.000 -0.005
```

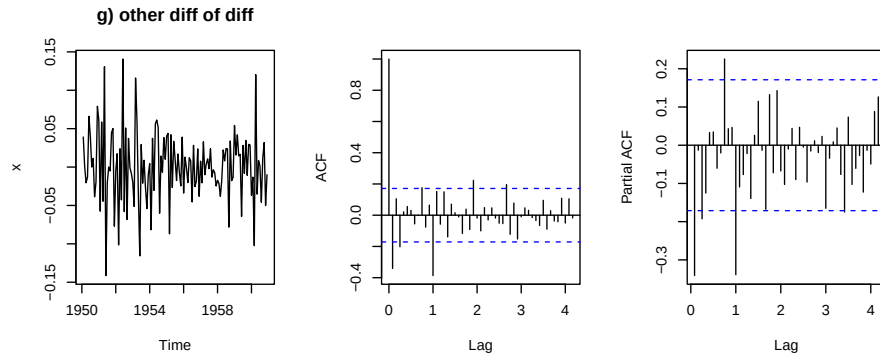
```
ts_plot(y, main="e) seasonal diff of diff")
```



```
# g) the other diff in diff
y <- diff( diff(log(x), 12) )
h(y)
```

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
## 1950	0.039	0.000	-0.020	-0.013	0.066	0.040	0.000	0.011	-0.039	-0.019	
## 1951	0.061	-0.057	0.059	-0.045	0.131	-0.141	-0.020	0.000	-0.005		

```
ts_plot(y, main="g) other diff of diff")
```



1.3 First definitions

Remark 4. Formally a time series is a single realisation of a stochastic process.

Definition 1.3.1 (Time series). A time series $(x_1, \dots, x_n)$, with $n < \infty$ ³ is *one, finite*, realisation from a stochastic process $(X_t)_{t \in T}$, where the single random variable are $X_t : (\Omega, \mathcal{A}, \mathbb{P}) \rightarrow (\mathbb{R}, \beta(\mathbb{R}), \mathbb{P}_X)$.

Example 1.3.1. Prof uses $T = \mathbb{Z}$ so we have a huge, infinity, number of random variables $\{\dots, X_{t-1}, X_t, X_{t+1}, \dots\}$, of which we observe only a *single* realization for

³In general T is a numeric set of times: if finite or countable we have *discrete-time* time series, if continuous we have *continuous-time* time series. Here as in ITSf, focus seems more on discrete-time time series.

a *small subsequent subset* of the random variables coming from the random process eg $(x_t, x_{t+1}, \dots, x_{t+n})$.

Definition 1.3.2 (Law of stochastic process). It's the family of multivariate distribution generated by the process $\{F_{\mathbf{t}}(\mathbf{x})\}_{\mathbf{t} \in \mathcal{T}}$ such that for $n = 1, 2, \dots$

$$F_{\mathbf{t}}(\mathbf{x}) = \mathbb{P}(X_{t_1} \leq x_1, \dots, X_{t_n} \leq x_n,)$$

where $\mathcal{T} = \{\mathbf{t} = (t_1, \dots, t_n) \in T^n, t_1 < t_2 < \dots < t_n\}$ and $\mathbf{x} = (x_1, \dots, x_n)$.

Remark 5. So it's the collection of all the univariate cumulative distributions at each time, the bivariate distribution etc.

Important remark 1 (Problem with ts). Some remarks:

- our aim is describing the dependence among the random variables in the process: to get information on the stochastic process that generated the time series we see.
- we would reach this perfectly if we could access the law above, but often this is not known (even if we can prove its existence, via Kolmogorov or Tulcea/Ionescu) or too difficult to be obtained (too many parameters to be estimated from the limited available data). That is, based on one single realization (and observed for limited amount of time) it's impossible to describe the data generating process (unless we make further assumptions).
- Thus we often *restrict* the class of stochastic process investigated to those whose stochastic dependence can be described to something we can observe, that is moments (*unconditional* and *conditional*); first and second order moments are of particular interest and we focus on properties of sequence of random variables depending only on these (*second-order properties*).
- to guarantee the validity of inference based on moments (and limited observation) however, process has to possess some homogeneity conditions, such as *stationarity*⁴, *invertibility* and *ergodicity*
- we may lose a certain amount of information by looking at second-order properties of random process but as in chapter 2 of ITSF theory of minimum MSE linear predictor depends only on the second-order properties, thus providing further justification for the use of the second-order characterization of time series models.⁵

Definition 1.3.3 (Moments of a stochastic process). First and second moments of $\{X_t\}_{t \in T}$ are

$$\begin{aligned}\mu_X(t) &= \mu_t = \mathbb{E}[X_t] = \int_{\mathbb{R}} x_t \, dF_{(X_t)}(x) \\ \gamma_X(0, t) &= \gamma_{0,t} = \text{Var}[X_t] = \mathbb{E}[(X_t - \mu_t)^2] \\ \gamma_X(r, s) &= \gamma_{r,s} = \text{Cov}(X_r, X_s) = \mathbb{E}[(X_r - \mu_r)(X_s - \mu_s)] \\ &= \mathbb{E}[X_r X_s] - \mathbb{E}[X_r] \mathbb{E}[X_s]\end{aligned}$$

⁴Intended weak stationarity, which we will assume for all the course except here and there where stronger assumption is needed

⁵Btw/Furthermore when all the joint distributions are multivariate normal, the second order properties of the process completely determine the joint distribution and thus thive a complete probabilistic characterization of the sequence of rvs.

NB: Should be the finite dimensional distributions in Rigo's terms.

with the last called *autocovariance function (ACF)*.

Definition 1.3.4 (Strong/strict stationarity). A process if joint distributions does not change for shift of times, that is

$$(X_{t_1}, \dots, X_{t_k}) \sim (X_{t_1+h}, \dots, X_{t_k+h})$$

for all $k \geq 1$, and for all $t_1, \dots, t_h, h \in \mathbb{Z}$. Or in other terms

$$F_t(\mathbf{x}) = F_{t+\delta}(\mathbf{x}), \quad \forall \delta$$

NB: If in general we speak about stationarity, weakly is intended (ITSF p. 13)

Definition 1.3.5 (Weakly stationarity). A process is weakly-stationary if the *first and second order moments exist* (finite) and are constant/do not depend on the time. In this case the moments:

$$\begin{aligned} \mu_X &= \mu = \mathbb{E}[X_t] < \infty, \quad \forall t \in T \\ \gamma_X(0) &= \gamma(0) = \gamma_0 = \text{Var}[X_t] < \infty, \quad \forall t \in T \\ \gamma_X(k) &= \gamma(k) = \gamma_k = \text{Cov}(X_t, X_{t+k}) < \infty, \quad \forall t \in T, k = 0, \pm 1, \pm 2, \dots \end{aligned}$$

The first two above are constant while the third, called autocovariance function, does not depend on the time chosen (for one of the two variables, say the first) but only on the distance $k = |t - s|$ between the variables. Furthermore by properties of covariance $\gamma_{-k} = \text{Cov}(X_{t+k}, X_t) = \text{Cov}(X_t, X_{t+k}) = \gamma_k$.

Remark 6 (Autocovariance function importance). So in stationary process we may have different realizations and spaghetti like movement but the vertical means is the same, as well as vertical spread in time; since mean and variance for these process is constant we look at *autocovariance function as key tool* to say something on process.

Auto means covariance of the process with self; we call it *function* because at each $k = 0, \pm 1, \pm 2, \dots$ we have an estimate which we can plot (stick function). Since the function $\gamma_k = \gamma_{-k}$ we typically plot only the positive side (first and fourth quadrant) and looking at it we can try to tell which model generated the observed values

Proposition 1.3.1 (Covariance useful property). If $\mathbb{E}[X^2] < \infty, \mathbb{E}[Y^2] < \infty, \mathbb{E}[Z^2] < \infty$ and $a, b, c \in \mathbb{R}$

$$\text{Cov}(aX + bY + c, Z) = a \text{Cov}(X, Z) + b \text{Cov}(Y, Z) \quad (1.2)$$

Definition 1.3.6 (Autocorrelation function). For a stationary stochastic process, it's an adimensional version of the ACV, defined as

$$\rho_X(k) = \rho_k = \frac{\gamma_k}{\gamma_0}, \quad k = 0, \pm 1, \pm 2$$

with $|\rho_k| \leq 1$ ⁶.

Important remark 2. We have that:

- weak stationarity $\nRightarrow$ strong stationarity (obviously, as moment conditions do not imply, except for densities completely characterised by the first two moments, like the Gaussian, distributional condition)

⁶Due to Cauchy-Schwartz inequality where we define the inner product $\langle X, Y \rangle$ like the expectation $\mathbb{E}[XY]$

- if $\{X_t\}$ is strictly/strong stationary AND first and second moments are finite ($\mathbb{E}[X_t^2] < \infty, \forall t$), then $\{X_t\}$ is also weakly stationary

Important remark 3 (Stationarity and ACV/ACF). It is not required that the process is stationary in order for autocovariance and autocorrelation to be defined but typically they have sense for stationary processes. Indeed if the process is non stationary the autocovariance function does not give us that much information on the underlined data generating process.

1.4 Examples

Remark 7. In this section we see examples of stochastic processes and their autocorrelation function.

1.4.1 White noise

Example 1.4.1 (White noise). $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ (read ε_t is a white noise process $0 \sigma^2$) if composed by random random variables with uncorrelated zero mean σ^2 variance random variables: $\forall t \in \mathbb{Z}$

$$\begin{aligned}\mathbb{E}[\varepsilon_t] &= 0 \\ \text{Var}[\varepsilon_t] &= \mathbb{E}[\varepsilon_t^2] = \sigma^2 \\ \text{Cov}(\varepsilon_s, \varepsilon_t) &= \mathbb{E}[\varepsilon_s \varepsilon_t] = 0, \quad \forall t \neq s\end{aligned}$$

WN is a weakly stationary process, since fullfill requirements of definition (being the three above finite $\forall t \in \mathbb{Z}$).

1.4.2 IID noise

Example 1.4.2 (IID noise). Differently from the previous not only variables are *uncorrelated* but even independent and identically distributed. In IID noise random variables of the process are *independent and identically distributed with zero mean*:

$$(X_1, \dots, X_n), \quad X_t \sim \text{IID}(0, \sigma^2)$$

Thus

- IID $(0, \sigma^2)$ is a WN $(0, \sigma^2)$ but the converse does not hold (independence implies zero correlation but not the converse);
- any finite dimensional distribution is just the product of the marginals of each time (with F being the common cumulative distribution function):

$$\begin{aligned}\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) &= \mathbb{P}(X_1 \leq x_1) \cdot \dots \cdot \mathbb{P}(X_n \leq x_n) \\ &= F(x_1) \cdot \dots \cdot F(x_n)\end{aligned}$$

- there's no dependence between observations and knowledge of $X_1, \dots, X_n$ is of no value for predicting the behaviour of X_{n+h}

$$\mathbb{P}(X_{n+h} \leq x | X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_{n+h} \leq x), \quad \forall h \geq 1$$

It's

- a simple model for a time series (with no trend or seasonal component);
- an important as building block for more complicated models;
- weakly stationary (if $\sigma^2 < \infty$) since $\forall t \in \mathbb{Z}$:

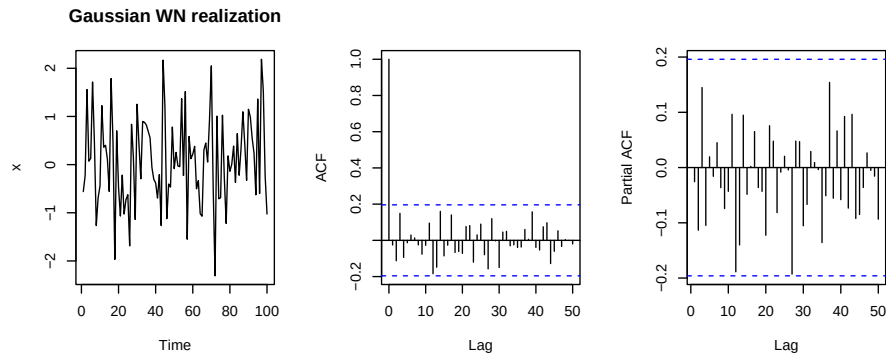
$$\begin{aligned}\mathbb{E}[X_t] &= 0 \\ \text{Var}[X_t] &= \mathbb{E}[X_t^2] = \sigma^2 \\ \text{Cov}(X_t, X_{t+h}) &= \begin{cases} \sigma^2 & \text{if } h = 0 \\ 0 & \text{if } h \neq 0 \end{cases}\end{aligned}$$

Not necessarily an IID process is weakly stationary: eg $\varepsilon_t \sim T(\nu = 1)$ is IID process but it's not stationary because has second moment infinite.

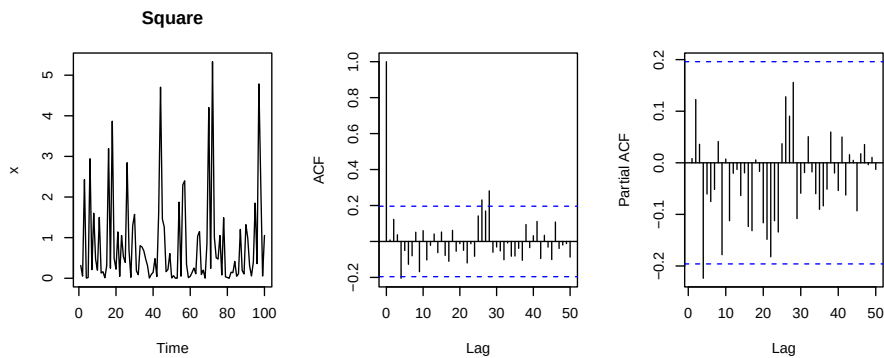
Example 1.4.3 (Gaussian white noise ($NID(0, 1)$)). As example of IID RP gaussian white noise (means that the process is actually IID Gaussian), where $\varepsilon_t \sim N(0, 1)$ are iid normal;

- process is weakly stationary because $\mathbb{E}[\varepsilon_t] = 0$, $\text{Var}[\varepsilon_t] = 1$ and $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ (due to independence)
- this normal IID process is also *strongly stationary*.

```
set.seed(123)
y <- rnorm(100) # Gaussian White Noise process simulation
ts_plot(y, main="Gaussian WN realization")
```



```
# non linear transformations of the data (eg squares) do not change the
# autocorrelation of the series (as it is an IID series)
ts_plot(y^2, main="Square")
```

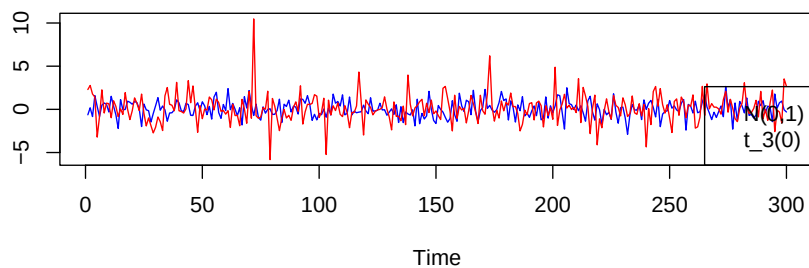


Example 1.4.4. Other simulations: T vs normal white noise

```
n <- 300

## Gaussian and student t white noise processes
set.seed(1)
wnn <- rnorm(n=n)
wnt <- rt(n=n, df=3) # with lower df the process have larger
# noise/variability and extreme observation

mts_plot(cbind(wnn, wnt), col = c("blue", "red"))
legend("bottomright", legend=c("N(0,1)", "t_3(0)"))
```



1.4.3 Random walk

Example 1.4.5 (Random walk). Defined as

$$(S_0, S_1, S_2, \dots)$$

with

$$S_n = X_0 + \sum_{t=1}^n X_t$$

where $X_0 \perp\!\!\!\perp X_t$, X_t are iid, $\forall t \geq 1$. For example if $X_0 = 0$ and $X_t \sim \text{IID}(0, \sigma^2)$ we obtain a random walk with zero mean (that is $\mathbb{E}[S_n] = 0, \forall n$). It's *not stationary*: eg if we consider the zero mean random walk

$$\mathbb{E}[S_t] = 0$$

$$\text{Var}[S_t] = \mathbb{E}[S_t^2] - [\mathbb{E}[S_t]]^2 = \mathbb{E}\left[\left(\sum_{i=1}^t X_i\right)^2\right] \stackrel{(\perp\!\!\!\perp)}{=} \sum_{i=1}^t \mathbb{E}[X_i^2] = t\sigma^2$$

$$\text{Cov}(S_{t+h}, S_t) = \text{Cov}(S_t + X_{t+1} + \dots + X_{t+h}, S_t) \stackrel{(1)}{=} \text{Cov}(S_t, S_t) = t\sigma^2$$

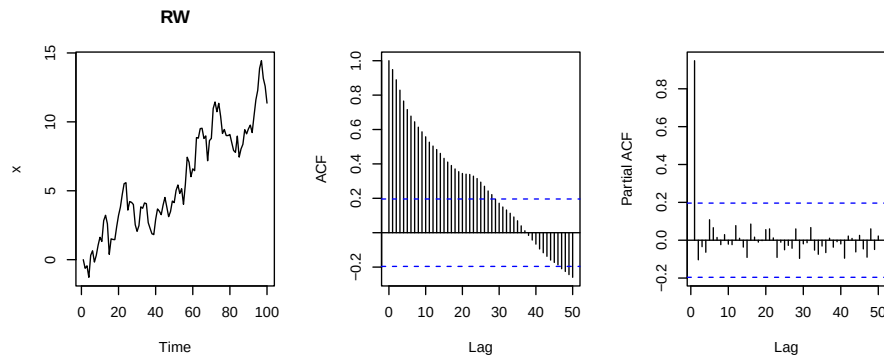
where (1) by applying 1.2.

Thus both second moment and covariance, depending on t , may not be finite.

```
sim_rw

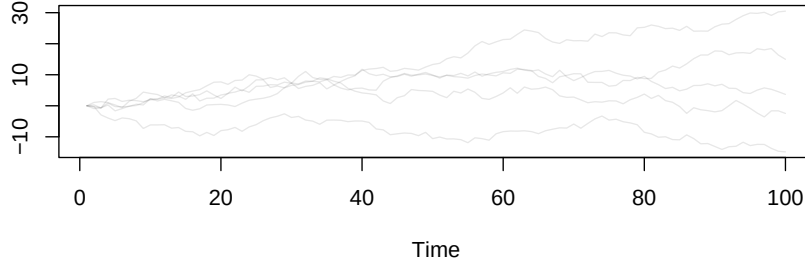
## function (n = 100, x0 = 0, xt = rnorm)
## {
##   x <- x0
##   xts <- xt(n = n)
##   rw <- cumsum(c(x0, xts))
##   rw[-length(rw)]
## }
## <bytecode: 0x577441c8e2a8>
## <environment: namespace:lbts>

set.seed(1)
x <- sim_rw()
ts_plot(x, main="RW")
```



```
# multiple ts
set.seed(2)
mts_plot(replicate(5, sim_rw()), main="5 RW")
```


5 RW



1.4.4 MA

Example 1.4.6 (First order moving average, MA(1)). Series with term defined⁷ as

$$X_t = \theta \varepsilon_{t-1} + \varepsilon_t, \quad t = 0, \pm 1, \dots$$

with $\{\varepsilon_t\} \sim \text{WN}(0, \sigma^2)$ and $\theta \in \mathbb{R}$.

The process is *stationary* since

$$\begin{aligned} \mathbb{E}[X_t] &= \mathbb{E}[\varepsilon_t + \theta \varepsilon_{t-1}] = \mathbb{E}[\varepsilon_t] + \theta \mathbb{E}[\varepsilon_{t-1}] = 0 \\ \text{Var}[X_t] &= \mathbb{E}[X_t^2] = \mathbb{E}[(\varepsilon_t + \theta \varepsilon_{t-1})^2] = \mathbb{E}[\varepsilon_t^2 + \theta^2 \varepsilon_{t-1}^2 + 2\varepsilon_t \varepsilon_{t-1}] \\ &= \sigma^2 + \theta^2 \sigma^2 + 0 = \sigma^2(1 + \theta^2) < \infty \\ \text{Cov}(X_t, X_{t+h}) &= \begin{cases} \sigma^2(1 + \theta^2) & \text{if } h = 0 \\ \theta \sigma^2 & \text{if } h = \pm 1 \\ 0 & \text{if } |h| > 1 \end{cases} \end{aligned}$$

For covariance, the first case is the already developed variance, the second and the third can be verified, eg for $h = 1$ and $h = 2$

$$\begin{aligned} \text{Cov}(X_t, X_{t+1}) &= \mathbb{E}[X_t X_{t+1}] - \mathbb{E}[X_t] \mathbb{E}[X_{t+1}] = \mathbb{E}[(\varepsilon_t + \theta \varepsilon_{t-1})(\varepsilon_{t+1} + \theta \varepsilon_t)] \\ &= \mathbb{E}[\varepsilon_t \varepsilon_{t+1} + \theta \varepsilon_t^2 + \theta \varepsilon_{t-1} \varepsilon_{t+1} + \theta^2 \varepsilon_{t-1} \varepsilon_t] = 0 + \theta \sigma^2 + 0 + 0 = \theta \sigma^2 \end{aligned}$$

$$\begin{aligned} \text{Cov}(X_t, X_{t+2}) &= \dots = \mathbb{E}[(\varepsilon_t + \theta \varepsilon_{t-1})(\varepsilon_{t+2} + \theta \varepsilon_{t+1})] \\ &= \mathbb{E}[\varepsilon_t \varepsilon_{t+2} + \theta \varepsilon_t \varepsilon_{t+1} + \theta \varepsilon_{t-1} \varepsilon_{t+2} + \theta^2 \varepsilon_{t-1} \varepsilon_{t+1}] = 0 + 0 + 0 + 0 \end{aligned}$$

The autocorrelation function of $\{X_t\}$, considering $h = 0, \pm 1, \pm 2, \dots$

$$\rho(h) = \text{Corr}(X_t, X_{t+h}) = \begin{cases} 1 & \text{if } h = 0 \\ \frac{\theta}{(1 + \theta^2)} & \text{if } h = \pm 1 \\ 0 & \text{if } |h| > 1 \end{cases}$$

Thus it's a vertical stick like function, where in -1 and $+1$ sign is the same as θ , is 1 in 0 and 0 elsewhere. So MA(1) *autocorrelation become 0 after first lag/a*

⁷La prof la definisce forzando un $-\theta$; qua per ora tengo la stessa notazione di ITSF che mi pare più semplice

single time period (the random variable are not anymore correlated). If $\theta > 0$ then X_t and X_{t-1} are positively autocorrelated, otherwise negatively correlated. For calculating theoretical ACF R becomes handy, eg for $\theta = 0.9$ $\rho_1 = 0.9/(1 + 0.9^2) = 0.49$ as follows

```
# calculate theoretical ACF for MA(1) model with theta=0.9
ARMAacf(ma=0.9, lag.max=10)

##      0      1      2      3      4      5      6      7      8      9     10
## 1.0000 0.4972 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000
```

Example 1.4.7 (General MA(1)). We can add a generic $\omega \in \mathbb{R}$ (above fixed to 0) at beginning

$$X_t = \omega + \theta\varepsilon_{t-1} + \varepsilon_t$$

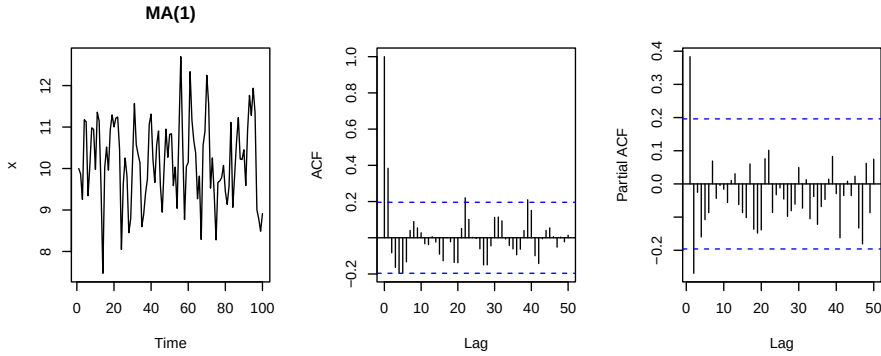
One have $\mathbb{E}[X_t] = \omega$ and then for variance/covariance

Generally speaking an MA model expresses the current value of a time series as a linear function of current and past random shocks (error terms) with finite lag length. The moving-average model relies solely on the dependency structure of the error terms not on values of the process (differently from AR)

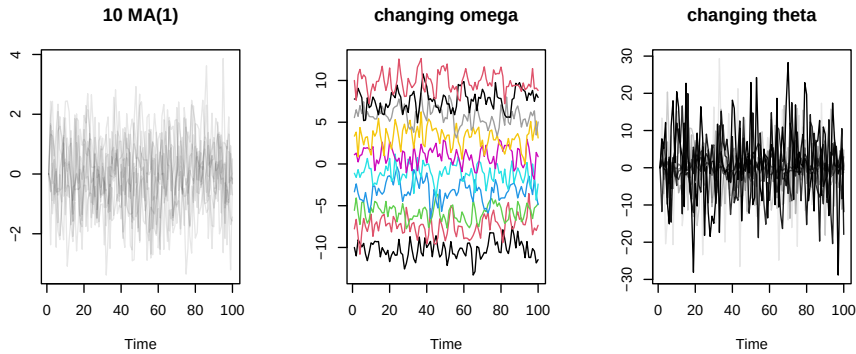
```
# function to generate data
sim_ma1

## function (n = 100, omega = 0, theta = 0.5, sigma2 = 1)
## {
##   e <- sqrt(sigma2) * rnorm(n)
##   x <- omega
##   for (t in seq(1, n - 1)) {
##     x[t + 1] <- omega + theta * e[t] + e[t + 1]
##   }
##   x
## }
## <bytecode: 0x57743eab85c0>
## <environment: namespace:lbts>

set.seed(1)
x <- sim_ma1(omega=10)
ts_plot(x, main="MA(1)")
```



```
par(mfrow=c(1,3))
mts_plot(replicate(10, sim_ma1()), main="10 MA(1)") # 10 MA(1) starting from 0
mts_plot(lapply(seq(-10, 10, length=10), function(o) sim_ma1(omega=o)),
          main = "changing omega",
          col = 1:10)
mts_plot(lapply(seq(-10, 10, length=10), function(t) sim_ma1(theta=t)),
          main = "changing theta",
          # chiara theta negativo, scuri positivo
          col = lbmisc::col2hex("black", seq(0.1, 1, length=10)))
```



Example 1.4.8 (Generic MA(q)). Defined as linear combination of q past random shock plus lift (ω) and a current error as follows:

$$X_t = \omega + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q}$$

with $\{\varepsilon_t\} \sim \text{WN}(0, \sigma^2)$ has **ACF that becomes zero** after the q -th lag

Proposition 1.4.1. *If one has a stationary process and*

- *the autocorrelation function becomes 0 after the first lag then one can conclude that the process is an MA(1).*
- *the autocorrelation function becomes 0 after q lag then one can conclude that the process is an MA(q).*

1.4.5 AR

Example 1.4.9 (First order autoregression, AR(1)). Assume X_t is a stationary series satisfying

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad t = 0, \pm 1, \dots$$

with

- $\varepsilon_t \sim \text{WN}(0, \sigma)$,
- $|\phi| < 1$ (we will see that *only if* $|\phi| < 1$ then X_t is *stationary*)
- ε_t uncorrelated with X_s , for $t > s$

Taking expectation to both

$$\begin{aligned} \mathbb{E}[X_t] &= \phi \mathbb{E}[X_{t-1}] + \mathbb{E}[\varepsilon_t] \\ \mathbb{E}[X_t] &= \phi \mathbb{E}[X_{t-1}] \end{aligned}$$

Thus by stationarity $\mathbb{E}[X_t] = \mathbb{E}[X_{t-1}]$ and we conclude that $\mathbb{E}[X_t] = 0$. To obtain autocorrelation we multiply each side by X_{t-h} and take expectation obtaining a relationship between autocovariances, and then develop using covariance properties:

$$\begin{aligned} X_t X_{t-h} &= \phi X_{t-1} X_{t-h} + \varepsilon_t X_{t-h} \\ \mathbb{E}[X_t X_{t-h}] &= \mathbb{E}[\phi X_{t-1} X_{t-h} + \varepsilon_t X_{t-h}] \\ \text{Cov}(X_t, X_{t-h}) &= \phi \cdot \mathbb{E}[\phi X_{t-1} X_{t-h}] + \underbrace{\mathbb{E}[\varepsilon_t X_{t-h}]}_{\text{Cov}(\varepsilon_t, X_{t-h})=0} \\ &= \phi \cdot \text{Cov}(X_{t-1}, X_{t-h}) \\ \gamma(h) &= \phi \cdot \gamma(h-1) \\ &= \phi \cdot \phi \cdot \gamma(h-2) = \dots \\ &= \phi^h \cdot \underbrace{\gamma(0)}_{\text{Var}[X_t]} \end{aligned}$$

thus the autocorrelation

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \phi^h$$

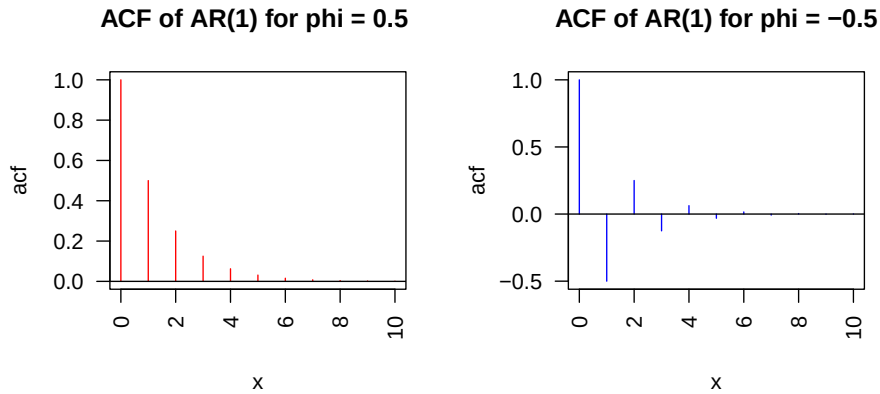
and being in general that covariance is symmetric $\gamma(h) = \gamma(-h)$ we can write/it must be that the autocorrelation

$$\rho(h) = \phi^{|h|}, \quad h = 0, \pm 1, \pm 2, \dots$$

So for $|\phi| < 1$ shape of autocorrelation with lag h is a stick function which is decreasing if $\phi > 0$ and is oscillating around 0 if $\phi < 0$. In both cases never goes to 0, differently from MA(1). For AR the most common case is having $0 < \phi < 1$ (positive decreasing autocorrelation).

```
ar1_rho <- function(phi=0.5, h=0:10) phi^(abs(h))

par(mfrow=c(1,2))
stick_plot(ar1_rho(), main= "ACF of AR(1) for phi = 0.5 ", col = "red",
           xlab="Time", ylab="acf", las=2)
stick_plot(ar1_rho(phi = -0.5), main= "ACF of AR(1) for phi = -0.5", col = "blue",
           xlab="Time", ylab="acf", las=2)
```



```
## for theoretical acf one can use ARMAacf:
# ACF of AR(1) with parameter phi = 0.5
ARMAacf(ar=0.5, lag.max=10)

##          0          1          2          3          4          5          6          7          8
## 1.0000000 0.5000000 0.2500000 0.1250000 0.0625000 0.0312500 0.0156250 0.0078125 0.0039062
##          9         10
## 0.0019531 0.0009766

# ACF of AR(1) with parameter phi = -0.5
ARMAacf(ar=-0.5, lag.max=10)

##          0          1          2          3          4          5          6          7
## 1.0000000 -0.5000000 0.2500000 -0.1250000 0.0625000 -0.0312500 0.0156250 -0.0078125
##          8          9         10
## 0.0039062 -0.0019531 0.0009766
```

Regarding variance:

$$\begin{aligned}
 \text{Var}[X_t] &= \text{Cov}(X_t, X_t) = \text{Cov}(\phi X_{t-1} + \varepsilon_t, \phi X_{t-1} + \varepsilon_t) \\
 &= \phi \text{Cov}(X_{t-1}, \phi X_{t-1} + \varepsilon_t) + \text{Cov}(\varepsilon_t, \phi X_{t-1} + \varepsilon_t) \\
 &= \phi \left[\underbrace{\text{Cov}(X_{t-1}, X_{t-1})}_{\text{Var}[X_{t-1}]} + \underbrace{\text{Cov}(X_{t-1}, \varepsilon_t)}_{=0} \right] + \phi \underbrace{\text{Cov}(\varepsilon_t, X_{t-1})}_{=0} + \underbrace{\text{Cov}(\varepsilon_t, \varepsilon_t)}_{=\text{Var}[\varepsilon_t] = \sigma^2} \\
 &= \phi^2 \text{Var}[X_{t-1}] + \sigma^2
 \end{aligned}$$

Being stationary by assumption $\text{Var}[X_{t-1}] = \text{Var}[X_t]$ thus

$$\begin{aligned}\text{Var}[X_t] &= \phi^2 \text{Var}[X_t] + \sigma^2 \\ \text{Var}[X_t] (1 - \phi^2) &= \sigma^2\end{aligned}$$

and thus

$$\gamma_0 = \text{Var}[X_t] = \frac{\sigma^2}{1 - \phi^2}$$

so we need $\phi \neq 1$ for γ_0 to exist

Example 1.4.10 (General AR(1)). Adding a generic $\omega \in \mathbb{R}$

$$X_t = \omega + \phi X_{t-1} + \varepsilon_t$$

Thus here model specifies current value of the process as dependent linearly on past values plus error terms.

We have that and for (hypothysed) stationarity $\mathbb{E}[X_t] = \mathbb{E}[X_{t-1}] = \mu_t = \mu$ so

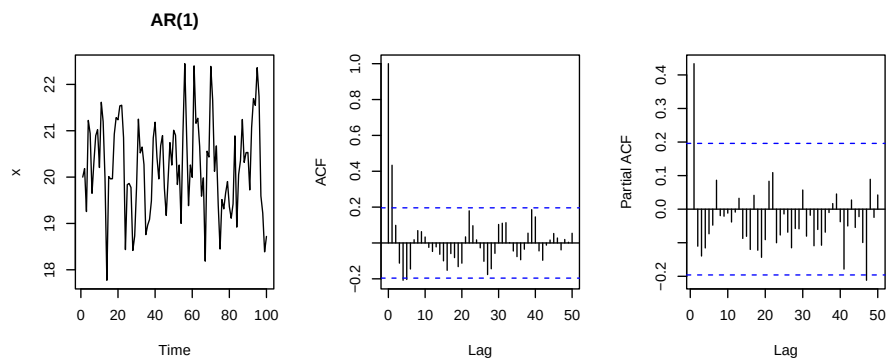
$$\begin{aligned}\mathbb{E}[X_t] &= \omega + \phi \mathbb{E}[X_{t-1}] \\ \mu &= \omega + \phi \mu \\ \mu &= \frac{\omega}{1 - \phi}\end{aligned}$$

with $\phi \neq 1$. In the following we assume $\omega = 0$ thus $\mathbb{E}[X_t] = 0$.

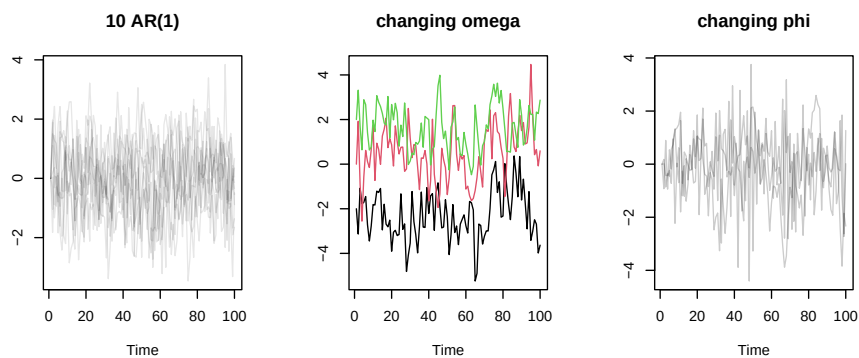
```
sim_ar1

## function (n = 100, omega = 0, phi = 0.5, sigma2 = 1)
## {
##   e <- sqrt(sigma2) * rnorm(n)
##   x <- omega/(1 - phi)
##   for (t in seq(1, n - 1)) {
##     x[t + 1] <- omega + phi * x[t] + e[t + 1]
##   }
##   x
## }
## <bytecode: 0x577441e44460>
## <environment: namespace:lbts>

set.seed(1)
x <- sim_ar1(omega=10)
ts_plot(x, main="AR(1)")
```

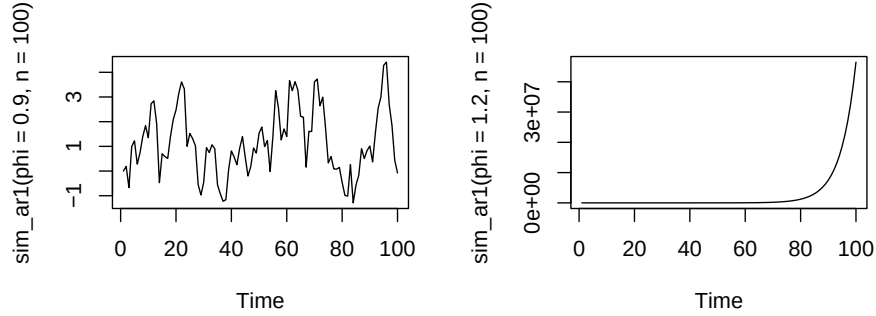


```
par(mfrow=c(1,3))
mts_plot(replicate(10, sim_ar1()), main="10 AR(1)") # 10 MA(1) starting from 0
mts_plot(lapply(c(-1,0,1), function(o) sim_ar1(omega=o)),
          main = "changing omega",
          col = 1:3)
mts_plot(lapply(c(-0.8,0,0.8), function(p) sim_ar1(phi=p)),
          main = "changing phi",
          col = lbmisc::col2hex("black", seq(0.2, 0.8, 3)))
```



Finally let's see for $\omega = 0$ the impact of changing $\theta = 0.99$ to 1 (the series is no more stationary)

```
set.seed(1)
par(mfrow=c(1,2))
ts.plot(sim_ar1(phi=0.90, n=100))
ts.plot(sim_ar1(phi=1.2, n=100))
```



Example 1.4.11 (Generic AR(p) process). Defined as linear combination of p past values plus lift (ω) and an error as follows:

$$X_t = \omega + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \varepsilon_t$$

with $\{\varepsilon_t\} \sim \text{WN}(0, \sigma^2)$ has an ACF that never goes to zero. So differently from MA, we cannot identify the order of the autoregressive process looking at its ACF

Example 1.4.12 (ARMA(1,1) process). Defined as a fusion of AR(1) and MA(1) a process such that

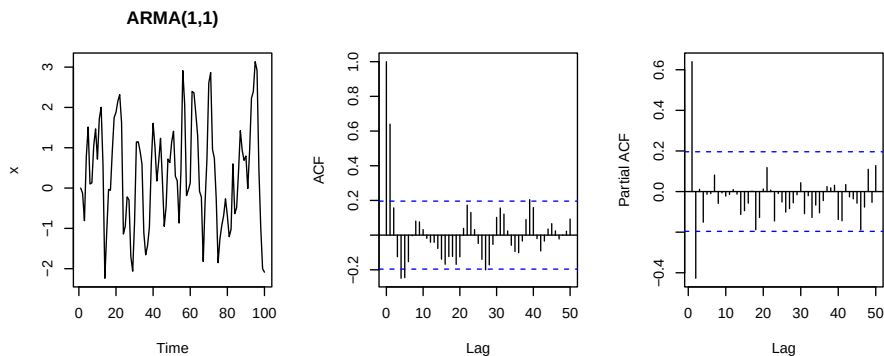
$$X_t = \omega + \phi X_{t-1} + \theta \varepsilon_{t-1} \varepsilon_t$$

with $\{\varepsilon_t\} \sim \text{WN}(0, \sigma^2)$ $\theta \in R$ and the process is stationary for $|\phi| < 1$

```
sim_arma11

## function (n = 100, omega = 0, theta = 0.5, phi = 0.5, sigma2 = 1)
## {
##   e <- sqrt(sigma2) * rnorm(n)
##   x <- omega/(1 - phi)
##   for (t in seq(1, n - 1)) {
##     x[t + 1] = omega + phi * x[t] + theta * e[t] + e[t +
##       1]
##   }
##   x
## }
## <bytecode: 0x57743f453bd0>
## <environment: namespace:lbts>

set.seed(1)
x <- sim_arma11()
ts_plot(x, main="ARMA(1,1)")
```

1.4.6 Other R utilities

```
# -----
# SIMULATE DATA
# -----
## all the process below with mean 1
mu <- 1
set.seed(123)

# MA(1) processes with theta 0.9 and  $e(t) \sim N(0,1)$ 
ma1.sim <- mu + arima.sim(model=list(ma=0.9), n=250)
# MA(1) process with theta 0.9 and  $e(t) \sim N(0,(0.1)^2)$ 
ma1.sim2 <- mu + arima.sim(model=list(ma=0.9), n=250, innov=rnorm(n=250, mean=0, sd=0.1))

# AR(1) with parameter phi=0.9
ar1.sim <- mu + arima.sim(model=list(ar=0.9), n=250)

## -----
## theoretical ACF (examples)
## -----

# ACF for AR(1) model: what happens if phi increase in absolute value
ARMAacf(ar=0.75, lag.max=5)

##      0      1      2      3      4      5
## 1.0000 0.7500 0.5625 0.4219 0.3164 0.2373

ARMAacf(ar=0.99, ma=0, lag.max=5) # corr increase

##      0      1      2      3      4      5
## 1.0000 0.9900 0.9801 0.9703 0.9606 0.9510

ARMAacf(ar=-0.75, lag.max=5)

##      0      1      2      3      4      5
## 1.0000 -0.7500 0.5625 -0.4219 0.3164 -0.2373
```

```

ARMAacf(ar=-0.99, ma=0, lag.max=5) # corr increase in absolute value
##      0      1      2      3      4      5
## 1.0000 -0.9900  0.9801 -0.9703  0.9606 -0.9510

# ACF for AR(2) model with phi_1 = 0.75 phi_2 = 0.2
ARMAacf(ar=c(0.75,.2), lag.max=10)
##      0      1      2      3      4      5      6      7      8      9     10
## 1.0000 0.9375 0.9031 0.8648 0.8293 0.7949 0.7620 0.7305 0.7003 0.6713 0.6435

```

1.5 Sample autocorrelation function

Remark 8. Previously we've seen autocorrelation function for theoretical time series models.

In practical problems we do not start with a model but with observed data $x_1, x_2, \dots, x_n$. To assess the degree of dependence in the data and to select a model for the data that reflects this, one of the important tools we use is the **sample autocorrelation function** (sample ACF) of the data.

If we believe that the data are realized values of a stationary time series $\{X_t\}$ then the sample ACF will provide us with an estimate of the ACF of $\{X_t\}$. This estimate may suggest which of the many possible stationary time series models is a suitable candidate for representing the dependence in the data.

For example, a sample ACF that is close to zero for all nonzero lags suggests that an appropriate model for the data might be iid noise.

Let's see sample analogues for autocovariance and autocorrelation

Definition 1.5.1. If $x_1, \dots, x_n$ are observation of a time series, the sample mean is defined as

$$\bar{x} = \frac{\sum_{t=1}^n x_t}{n}$$

Definition 1.5.2 (Sample autocovariance function). Defined as for h such that $-n < h < n$

$$\hat{\gamma}(h) = \frac{\sum_{t=1}^{n-|h|} (x_{t+|h|} - \bar{x})(x_t - \bar{x})}{n}$$

Remark 9. for $h \geq 0$, $\hat{\gamma}(h)$ is approximately equal to the sample covariance of the $n - h$ pairs of observations $(x_1, x_{1+h}), (x_2, x_{2+h}), \dots, (x_{n-h}, x_n)$, with only difference in the denominator (n instead of $n - h$, to ensure that sample covariance matrix $\widehat{\Gamma}_n = [\hat{\gamma}(i - j)]_{i,j=1}^n$ is nonnegative definite) and the subtraction of the overall mean $\bar{x}$ from each factor of the summand

Definition 1.5.3 (Sample autocorrelation function). Again defined for h such that $-n < h < n$ as

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}$$

Remark 10. Like sample covariance matrix, sample correlation matrix $\widehat{R}_n = [\widehat{\rho}(i - j)]_{i,j=1}^n$ is nonnegative definite and each of its diagonal elements is equal to 1

Important remark 4. The sample autocovariance and autocorrelation functions can be computed for any data set and are not restricted to observations from a stationary time series.

For data

- containing a trend, $|\widehat{\rho}(h)|$ will exhibit slow decay as h increases
- with a substantial deterministic periodic component, $|\widehat{\rho}(h)|$ will exhibit similar behavior with the same periodicity.

Thus $\widehat{\rho}$ can be useful as indicator of nonstationarity

Chapter 2

Linear processes

2.1 Linear process

Remark 11. We now see a family of processes which encompass AR and MA under some conditions.

Relevance of this process is related to Wold decomposition theorem and Wold representation

Definition 2.1.1 (Linear process (MA(∞))). A process defined like infinite linear combination of variables that form a white noise process:

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} \quad (2.1)$$

where

- $\varepsilon_t \sim \text{WN}(0, \sigma^2)$
- $\psi_0 = 1$ and $\sum_{j=0}^{\infty} \psi_j^2 < \infty$

Linear processes are also referred to as MA(∞) processes

Proposition 2.1.1 (Properties). *A linear process has*

$$\begin{aligned} \mathbb{E}[X_t] &= 0 \\ \gamma_0 = \text{Var}[X_t] &= \sigma^2 \sum_{j=0}^{\infty} \psi_j^2 < \infty \text{ (by def)} \\ \gamma_k = \text{Cov}(X_t, X_{t+k}) &= \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+|k|} \end{aligned}$$

Thus linear process are stationary process.

Proof. Respectively for mean

$$\mathbb{E}[X_t] = \mathbb{E}\left[\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}\right] = \sum_{j=0}^{\infty} \psi_j \mathbb{E}[\varepsilon_{t-j}] = 0$$

where at the last, we needed the sum of squared coefficients to be convergent.
For variance:

$$\begin{aligned}
\gamma_0 &= \text{Var}[X_t] = \mathbb{E}[X_t X_t] = \mathbb{E}\left[\left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}\right)^2\right] \\
&= \mathbb{E}[(\varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \dots)(\varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \dots)] \\
&= \mathbb{E}[\varepsilon_t^2 + \varepsilon_t(\psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \dots) + \psi_1 \varepsilon_{t-1} \varepsilon_t + \psi_1^2 \varepsilon_{t-1}^2 + \psi_1 \varepsilon_{t-1}(\psi_1 \varepsilon_{t-2} + \dots) + \dots] \\
&= \sigma^2 + 0 + 0 + \psi_1^2 \sigma^2 + 0 + 0 + \psi_2^2 \sigma^2 + \dots \\
&= \sigma^2 \sum_{j=0}^{\infty} \psi_j^2
\end{aligned}$$

which is $< \infty$ by assumption of definition.

For covariance

$$\begin{aligned}
\gamma_1 &= \mathbb{E}[X_t X_{t+1}] \\
&= \mathbb{E}[(\varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \dots)(\varepsilon_{t+1} + \psi_1 \varepsilon_t + \psi_2 \varepsilon_{t-1} + \dots)] \\
&= \mathbb{E}[\varepsilon_t \varepsilon_{t+1} + \psi_1 \varepsilon_t^2 + \psi_2 \varepsilon_t \varepsilon_{t-1} + \dots + \psi_1 \varepsilon_{t-1} \varepsilon_{t+1} + \psi_1^2 \varepsilon_{t-1}^2 + \psi_1 \psi_2 \varepsilon_{t-1}^2 + \dots + \psi_2 \varepsilon_{t-2} \varepsilon_{t+1} + \psi_2 \psi_1 \varepsilon_{t-2}^2 + \dots] \\
&= \mathbb{E}[\psi_1 \varepsilon_t^2 + \psi_1 \psi_2 \varepsilon_{t-1}^2] + \psi_2 \psi_3 \varepsilon_{t-1} + \dots \\
&= \psi_1 \sigma^2 + \psi_1 \psi_2 \sigma^2 + \psi_2 \psi_3 \sigma^2 + \dots \\
&= \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+1}
\end{aligned}$$

and the last due to $\psi_0 = 1$.

For the second level

$$\gamma_2 = \mathbb{E}[X_t X_{t+2}] = \dots = \sigma^2(\psi_0 \psi_2 + \psi_1 \psi_3 + \dots) = \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+2}$$

□

Theorem 2.1.2 (Wold decomposition theorem). *Any stationary stochastic process $\{Z_t : t \in \mathbb{Z}\}$ can be written/decomposed as the sum of two processes:*

$$Z_t = X_t + D_t \quad t \in \mathbb{Z}$$

where:

- $\{X_t : t \in \mathbb{Z}\}$ is a linear process
- $\{D_t : t \in \mathbb{Z}\}$ is a deterministic process, defined as $D_t = a + b \cdot t$, with $a, b \in \mathbb{R}$ known (basically it's a process with zero variance at each time)

Example 2.1.1 (Wold representation of MA(1) with zero mean). The process is

$$X_t = \varepsilon_t + \theta \varepsilon_{t-1}$$

with $\varepsilon_t \sim \text{WN}(0, \sigma^2)$. The linear Wold representation requires that a process is writable as

$$X_t = \psi_0 \varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \dots$$

Thus for an MA(1) we can just equate above and the wold representation of a MA(1) is

$$\begin{aligned}\psi_0 &= 1 \\ \psi_1 &= \theta \\ \psi_k &= 0, \quad \forall k > 1\end{aligned}$$

This above is the *Wold representation of a MA(1)*

By using general results regarding linear processes one can retrieve what already found previously, for example $\text{Cov}(X_t, X_{t+1})$

$$\begin{aligned}\gamma_1 &= \sigma^2 \left(\sum_{j=0}^{\infty} \psi_j \cdot \psi_{j+1} \right) \\ &= \sigma^2 (\psi_0 \psi_1 + \psi_1 \psi_2 + \psi_2 \psi_3 + \dots) \\ &= \sigma^2 (\theta + \theta \cdot 0 + 0 + 0 + \dots) \\ &= \sigma^2 \theta\end{aligned}$$

while for variance

$$\begin{aligned}\gamma_0 &= \sigma^2 \left(\sum_{j=0}^{\infty} \psi_j \cdot \psi_{j+1} \right) = \sigma^2 (1 + \psi_1^2 + \psi_2^2 + \dots) \\ &= \sigma^2 (1 + \theta^2 + 0 + \dots) = \sigma^2 (1 + \theta^2)\end{aligned}$$

Example 2.1.2 (Wold representation of AR(1) with zero mean). For MAs is easy, it will be more useful for AR processes. For AR(1) we hypothesized it's a stationary process; being able to write it in its Wold will give us the condition under which it's stationary.

The model is

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad \varepsilon \sim \text{WN}(0, \sigma^2)$$

To put it in Wold representation we have to rewrite it as linear combination of white noise (and not as function of past terms of the series itself):

$$X_t = \psi_0 \varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \dots$$

To do so we can at least in first instance use recursion, by having

$$X_{t-1} = \phi X_{t-2} + \varepsilon_{t-1}$$

and by replacement we get

$$X_t = \phi(\phi X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t = \phi^2 X_{t-2} + \phi \varepsilon_{t-1} + \varepsilon_t$$

Going on with recursion

$$X_{t-2} = \phi X_{t-3} + \varepsilon_{t-2}$$

and thus again by substitution in X_t formula

$$X_t = \phi^2(\phi X_{t-3} + \varepsilon_{t-2}) + \phi \varepsilon_{t-1} + \varepsilon_t = \phi^3 X_{t-3} + \phi^2 \varepsilon_{t-2} + \phi \varepsilon_{t-1} + \varepsilon_t$$

And going like that, going ahead in general we obtain:

$$X_t = \phi^n X_{t-n} + \sum_{j=0}^{n-1} \phi^j \varepsilon_{t-j}$$

This above can be the representation of a linear process (and thus being a stationary process) if and only if $|\phi| < 1$ (what we've asked for in the AR definition). Infact if that is the case, for $n \rightarrow \infty$ we have that $\phi^n \rightarrow 0$ and AR(1) becomes

$$X_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$$

Obtaining its Wold representation is now easy by having

$$\psi_j = \phi^j$$

So under these condition even AR(1) is a linear process and we can find what already found via linear process properties. Eg for autocorrelatio γ_k

$$\begin{aligned} \gamma_k &= \sigma^2 \left(\sum_{j=0}^{\infty} \psi_j \psi_{j+|k|} \right) = \sigma^2 \sum_{j=0}^{\infty} \phi^j \phi^{j+|k|} = \sigma^2 \sum_{j=0}^{\infty} \phi^j \phi^j \phi^{|k|} \\ &= \sigma^2 \phi^{|k|} \cdot \sum_{j=0}^{\infty} \phi^{2j} = \sigma^2 \phi^{|k|} \cdot \sum_{j=0}^{\infty} (\phi^2)^j \stackrel{(1)}{=} \sigma^2 \frac{\phi^{|k|}}{1 - \phi^2} \\ &= \frac{\sigma^2}{1 - \phi^2} \cdot \phi^{|k|} \end{aligned}$$

where in (1) $\sum_{j=0}^{\infty} (\phi^2)^j \rightarrow \frac{1}{1-\phi^2}$ (geometric series) if $|\phi| < 1$

2.2 Polynomials with lag operator

Remark 12. There's a way to write processes in a bit more compact form, it involves the use of backshift operator (indicated by B), sometimes called also lag operator (L)

Definition 2.2.1 (Lag operator). Applied to a random variable of random process it returns the previous one

$$L \cdot X_t = X_{t-1}$$

Proposition 2.2.1. *In notation, it works similarly to algebra*

$$\begin{aligned} L^k X_t &= X_{t-k} \\ L^{-k} X_t &= X_{t+k} \end{aligned}$$

Definition 2.2.2 (Difference operator). Defined using the lag as $(1 - L)$ that is

$$(1 - L)X_t = X_t - X_{t-1}$$

Remark 13. It's used to remove trend and autocorrelation, but we're not limited to use it for a single period; eg if seasonality is available, value from the same month could be correlated and we could be interested in looking at seasonal difference operator

Definition 2.2.3 (Seasonal difference operator). Defined as $(1 - L^s)$ with s length of the season (typically $s = 12$ for monthly data or $s = 4$ for quarterly data).

$$(1 - L^{12})X_t = X_t - X_{t-12}$$

Proposition 2.2.2 (Second order difference operator). *For further differentiation difference operator works like algebra; to do a double differentiation¹ we take the first difference and then difference again*

$$\begin{aligned} (1 - L)^2 X_t &= (1 - L) \underbrace{(1 - L)X_t}_{\text{first}} \\ &= (1 - 2L + L^2)X_t \\ &= X_t - 2X_{t-1} + X_{t-2} \end{aligned}$$

Pay attention that $(1 - L)^2 \neq (1 - L^2)$

2.3 Polynomials in the lag operator

Remark 14. This simplify a lot computations

Important remark 5. A linear process can be rewritten using the lag operator as

$$\begin{aligned} X_t &= \varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \dots \\ &= L^0 \varepsilon_t + \psi_1 L \varepsilon_t + \psi_2 L^2 \varepsilon_t + \dots \\ &= (1 + \psi_1 L + \psi_2 L^2 + \dots) \cdot \varepsilon_t \\ &= \Psi(L) \cdot \varepsilon_t \end{aligned}$$

where

$$\Psi(L) = 1 + \psi_1 L + \psi_2 L^2 + \dots$$

with $\Psi(L)$ a polynomial (order ∞) in the lag operator, with coefficients $1, \psi_1, \psi_2, \dots$

¹Similarly to second order derivative we obtain a constant if say we start from a parabola; second order differences allow us to make stationary a series which is non stationary because of the presence of a polynomial trend typical of order 2. It's not that frequent to apply a second order difference

Example 2.3.1 (MA(1) rewritten). we have

$$X_t = \varepsilon_t + \theta\varepsilon_{t-1} = \varepsilon_t + \theta L\varepsilon_t = (1 + \theta L)\varepsilon_t$$

Example 2.3.2 (AR(1) rewritten). We have

$$X_t = \phi X_{t-1} + \varepsilon_t X_t - \phi L X_t = \varepsilon_t (1 - \phi L) X_t = \varepsilon_t$$

2.4 ARMA process

Example 2.4.1 (MA(q)). We have

$$\begin{aligned} X_t &= \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q} = (1 + \theta_1 L + \theta_2 L^2 + \dots + \theta_q L^q) \varepsilon_t \\ &= \Theta_q(L) \varepsilon_t \end{aligned}$$

with

$$\Theta_q(L) = 1 + \theta_1 L + \theta_2 L^2 + \dots + \theta_q L^q$$

Example 2.4.2 (AR(p)).

$$\begin{aligned} X_t &= \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \varepsilon_t \\ (1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p) X_t &= \varepsilon_t \\ \Phi_p(L) X_t &= \varepsilon_t \end{aligned}$$

with

$$\Phi_p(L) = 1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p$$

Remark 15. We may have both components

Example 2.4.3 (ARMA(1,1)). Defined as

$$(1 - \phi L) X_t = (1 + \theta L) \varepsilon_t$$

That developed leads to

$$X_t = \phi X_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$$

where $\phi X_{t-1} + \varepsilon_t$ is the AR(1) part and $\varepsilon_t + \theta \varepsilon_{t-1}$ the MA(1)

Example 2.4.4 (ARMA(p,q)). Defined as

$$\Phi_p(L) X_t = \Theta_q(L) \varepsilon_t$$

that developed leads to

$$X_t = \underbrace{\phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p}}_{AR(p)} + \varepsilon_t + \underbrace{\theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q}}_{MA(q)}$$

2.5 Invertibility

Definition 2.5.1. A stochastic process $\{X_t : t \in \mathbb{Z}\}$ is invertible if it can be expressed as function of its past values plus an *innovation term* $\varepsilon_t \sim \text{WN}(0, \sigma^2)$

$$X_t = g(X_{t-1}, X_{t-2}, \dots; \varepsilon_t) \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

2.5.1 AR(1) and MA(1)

Example 2.5.1 (Invertibility of AR(1)). It's clearly an invertible

$$X_t = \omega + \phi X_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

Example 2.5.2 (Invertibility of MA(1)). Less clearly, the MA:

$$X_t = \omega + \theta \varepsilon_{t-1} + \varepsilon_t, \quad X_t \sim \text{WN}(0, \sigma^2)$$

it is not directly written as an invertible process (since does not depend on X_{t-1} etc). We could do the following by substitution

$$\begin{aligned} \varepsilon_t &= X_t - \omega - \theta \varepsilon_{t-1} \\ \varepsilon_{t-1} &= X_{t-1} - \omega - \theta \varepsilon_{t-2} \end{aligned}$$

having

$$X_t = \omega + \theta(X_{t-1} - \omega - \theta \varepsilon_{t-2}) + (X_t - \omega - \theta \varepsilon_{t-1})$$

and going on with substitution of $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots$

Remark 16. In general:

- AR process are *invertible by construction*/definition but not stationary unless we impose conditions on coefficients ϕ
- MA(1) are stationary by construction but are not directly invertible.

We however need a better general tool to derive invertibility and stationarity conditions for stochastic processes.

Invertibility of a stochastic process and/or stationarity of a stochastic process are related to the mathematical invertibility of the polynomial in the lag operator that is associated with the process.

Proposition 2.5.1 (Invertibility di MA(1)). *Let's tackle invertibility conditions for MA(1) having mean $\omega = 0$. Considering $\omega = 0$*

$$\begin{aligned} X_t &= \varepsilon_t - \theta \varepsilon_{t-1}, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2) \\ &= (1 - \theta L) \varepsilon_t \end{aligned}$$

Here we've expressed the process a function of a polynomial in the lag operator. Now

1. in general, adopting the Wold representation we can write the process with a generic MA(∞) representation

$$X_t = \Theta(L) \cdot \varepsilon_t$$

in practice we have a polynomial in the lag operator $\Theta(L)$ that acts on ε_t so one end up with X_t being a linear combination of current and past values of ε_t

NB: qui adottato la sua definizione malata di MA con il segno meno del coefficiente perché non ho tempo

2. but for invertibility one would need MA to be writable as

$$\Pi(L)X_t = \varepsilon_t$$

because in that case it would be easy to move elements from left (function of past values of X_t) to the right.

This above is called $AR(\infty)$ representation of MA with $\Pi(L)$ a polynomial acting on X_t like

$$\Pi(L) = 1 + \pi_1 L + \pi_2 L^2 + \dots$$

If exists the inverse $\Theta^{-1}(L)$, let's say $(1 - \theta L)^{-1}$ then we could use it as $\Pi(L)$ by pre-multiplying both members of process definition, eg

$$(1 - \theta L)^{-1} X_t = \varepsilon_t$$

Who can $(1 - \theta L)^{-1}$ be? This is like saying

$$\frac{1}{(1 - \theta L)} X_t = \varepsilon_t$$

Basically, similarly to a geometric series, if $|\theta| < 1$ we have that $\frac{1}{1 - \theta L}$ is the limit to which the geometric series of reason θL converges, for $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \sum_{j=0}^n \theta^j L^j = \sum_{j=0}^{\infty} \theta^j L^j = \frac{1}{1 - \theta L}, \quad \text{with } |\theta| < 1$$

For this reason if $|\theta| < 1$ MA(1) is invertible because we can write

$$\begin{aligned} \frac{1}{1 - \theta L} X_t &= \varepsilon_t \\ \left(\sum_{j=0}^{\infty} \theta^j L^j \right) X_t &= \sum_{j=0}^{\infty} \theta^j L^j X_t \varepsilon_t \end{aligned}$$

thus we have

$$\begin{aligned} X_t + \theta X_{t-1} + \theta^2 X_{t-2} + \dots &= \varepsilon_t \\ X_t = \varepsilon_t - \theta X_{t-1} - \theta^2 X_{t-2} - \dots \end{aligned}$$

so above

- we have the $AR(\infty)$ representation of a MA, it's an infinite sum but we have just one parameter which is θ
- the coefficient of the inverse $\Theta^{-1}(L)$ that is $\Pi(L)$ are simply $\pi_j = \theta^j$
- we have represented even MA as function of past own values and innovation term, thus it's invertible
- the condition of invertibility of MA(1) $|\theta| < 1$ are the same of stationarity of AR(1) process ($|\phi| < 1$)

2.5.2 Implications of invertibility of MA(1)

Remark 17. On the practical/interpretational level invertibility is important for identification and for asymptotic forgetting of initial conditions (at a certain stage the past become irrelevant).

2.5.2.1 Identification

Focusing again on MA(1) (with zero mean, $\omega = 0 \implies \mathbb{E}[X_t] = 0$)

$$\begin{aligned} X_t &= \varepsilon_t - \theta \varepsilon_{t-1}, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2) \\ &= (1 - \theta L)\varepsilon_t \end{aligned}$$

we've said it is

- always stationary
- invertible if $|\theta| < 1$ (condition of convergence of geometric series)
- the *theoretical ACF* of a MA(1) is

$$\rho_k = \begin{cases} 1 & \text{if } k = 0 \\ -\frac{\theta}{1+\theta^2}, & \text{if } |k| = 1 \\ 0 & \text{if } |k| > 1 \end{cases}$$

NB: qui è la sua notazione con MA a definta con $-\theta$

Two points regarding ACF and identification:

- the $|\theta| < 1$ condition is needed otherwise setting $\theta = x$ (eg $x = 0.4$) or setting $\theta = 1/x = 1/0.4$ would lead to the same theoretical autocorrelation (check it), and the solution would be non invertible.
So first implication of invertibility is identification
- ACF characterize MA process: once one see the acf of a stationary process going to 0 after a certain time t it's an $MA(t)$.
This can be a tool to identify the data generating process: basically one estimates the *empirical ACF* looking at data and if the process is stationary and the autocorrelation goes to zero after tot it's an MA(tot).
This will tell the number of parameter of interest to estimate (eg MA(1) has only one θ and we want to estimate it)

2.5.2.2 Asymptotic forgetting/discounting the past

Back to $AR(\infty)$ representation of a $MA(1)$ process

$$X_t = \varepsilon_t - \theta X_{t-1} - \theta^2 X_{t-2} - \dots$$

if $|\theta| < 1$ then past values of X_t have smaller and smaller impact on X_t .

2.5.2.3 Implication for forecast

With the classical MA(1) representation

$$X_t = \varepsilon_t - \theta \varepsilon_{t-1}, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

What one typically observe is the X_t . If one want to make any kind of forecast one need to estimate ε_t as a function of what is observed. Via the $AR(\infty)$ representation of the MA(1) we can write the process as a function of its past observations (by def of invertibility).

2.5.3 Concluding remarks

In summary we've discussed about *properties*

- stationarity
- invertibility

According to the process we define there are *representations*:

- we would like a *stationary* process to be represented as $AR(\infty)$
- we want a Wold representation for processes which are invertible

The point is that

- $AR(\infty)$ are crucial for forecasts and identification
- Wold representation is essential for stationarity but even more for deriving the theoretical properties of a process (a linear combination of uncorrelated variable makes all the calculation easy wrt to $AR(\infty)$ representation where all the variable involved are correlated)

2.5.4 AR(p) and MA(q)

General condition (regarding parameters $\theta_1, \theta_2, \dots$) under which $MA(q)$ is invertible, that is writable like

$$X_t = (1 - \theta_1 L - \theta_2 L^2 - \dots - \theta_q L^q) \varepsilon_t$$

equivalently we see conditions (involving $\phi_1, \dots$) under which $AR(p)$ is writable as

$$(1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p) X_t = \varepsilon_t$$

and *stationary*

NB: credo scrittura come Wold che implica che un processo sia stazionario

In both cases the invertibility of $MA(q)$ and stationarity of $AR(p)$ process depend on the invertibility of polynomials of order q and p respectively in lag operators with parameters $\theta_1, \dots, \theta_q$ and $\phi_1, \dots, \phi_p$.

The structure of the above polynomial is the same (duality of AR and MA processes)

2.5.4.1 Invertibility of a finite polynomial

A polynomial of degree n in x (with coefficients $a_0, \dots, a_n$)

$$P_n(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$$

with leading term equal to one $a_0 = 1$ (convention important for us).

The characteristic equation is $P_n(x) = 0$ and has solutions/roots $z_1, z_2, \dots, z_n$ so that we can rewrite the polynomial as the product of following terms²

$$P_n(x) = (z_1 - x)(z_2 - x) \dots (z_n - x)$$

²eg $P_2(x) = (1 - 2x + x^2) = (x - 1)(x - 1)$ with $z_1 = 1, z_2 = 1$

Our aim is to find (if exists) the inverse of the polynomial. If we assume that roots are different from 0 $z_i \neq 0$, $i = 1, \dots, n$ we can equivalently write it as

$$P_n(x) \propto \left(1 - \frac{x}{z_1}\right) \left(1 - \frac{x}{z_2}\right) \cdots \left(1 - \frac{x}{z_n}\right)$$

the inverse of the polynomial will be

$$(P_n(x))^{-1} \propto \left(1 - \frac{x}{z_1}\right)^{-1} \left(1 - \frac{x}{z_2}\right)^{-1} \cdots \left(1 - \frac{x}{z_n}\right)^{-1} = \prod_{i=1}^n \left(1 - \frac{x}{z_i}\right)^{-1}$$

All the factor are of the form $\left(1 - \frac{x}{z_i}\right)$. If we replace the variable x with the lag operator each of these term can be considered as the limit to which the geometric series of reason $\frac{1}{z_i}$ converge provided that the modulus of the reason is < 1 $\left|\frac{1}{z_i}\right| < 1$ (equivalently $|z_i| > 1$). We know by the MA(1) example that:

$$\left(1 - \frac{L}{z_i}\right)^{-1} = \left(1 - \frac{1}{z_i}L\right)^{-1} = \sum_{j=0}^{\infty} \left(\frac{1}{z_i}\right)^j L^j$$

if $\left|\frac{1}{z_i}\right| < 1$ eg $|z_i| > 1$.

Important remark 6. So we say that a polynomial of order n is invertible if all the roots z_i of the characteristic equation associated with the polynomial are in modulus greater than 1 (or are external to the circle of unit radius in complex terms).

This was in terms of roots of equation but we want a condition on the parameters

Example 2.5.3 (MA(1)).

$$X_t = (1 - \theta L)\varepsilon_t$$

The polynomial is

$$P_1(L) = \Theta_1(L) = 1 - \theta L$$

its char equation associated is

$$1 - \theta L = 0$$

$$\theta L = 1$$

$$L = \frac{1}{\theta}$$

this last is a root/solution (and is the only one being P_1 of order 1).

The process is invertible if $P_1(L)$ is invertible and in turn $P_1(L)$ is invertible if the root $\left|\frac{1}{\theta}\right| > 1 \iff |\theta| < 1$

Example 2.5.4 (AR(1)). Derive the condition of stationarity of an AR(1) process (based on the roots of the AR(1) polynomial), it's the same

To summarize stationarity and invertibility based on roots of characteristic equation associated with the polynomial in the lag operator, in general:

- AR(p) process are **stationary** if all the roots of $\Phi_p(L) = 0$ are > 1 in absolute value
- MA(q) process are **invertible** if all the roots of $\Theta_q(L) = 0$ are > 1 in absolute value
- mixed process: this hold as well, due to the multiplicative structure of ARMA(p, q) processes, stationarity and invertibility applies *separately* to the AR and MA polynomial.

Example 2.5.5 (ARMA(1,1)).

$$(1 - \phi L)X_t = (1 - \theta L)\varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

this is stationary if $|\phi| < 1$ and invertible if $|\theta| < 1$

Roots are estimated by software and shouldn't be greater than 1

2.5.5 Random walk a cazzo

Brief summary on AR(1) processes:

- always invertible
- stationary if $|\phi| < 1$
- if $|\phi| > 1$: brockwell davis ITSF pag 54 and 56 and remark 2 pag 57 plus problem 4.10 (credo)
- if $\phi = 1$ of AR(1) we have as special important case the random walk process, which is a *non stationary process* (given that $|\phi| < 1$ is excluded) with lot of application in analysis of stochastic trends (many series do have)

Important remark 7 (Differencing a non-stationary process). The process is an AR(1) with $\phi = 1$ (so it's non stationary and its variance is infinite) that is

$$X_t = X_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

suppose a random walk with t noise ($\nu = 6$ centered on 0)

$$X_t = X_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim t_6(0, 1)$$

It's non stationary, but if we take $Y_t = X_t - X_{t-1}$

$$Y_t = X_t - X_{t-1} = \varepsilon_t$$

then Y_t is stationary since ε_t is a stationary process. So if we take the differences and obtain the process $\varepsilon_t \sim t_6(0, 1)$ the realization will have the white noise shape. So when we take the differences of a nonstationary process (similar to take the derivative of a continuous function) what we find is a stationary process (in case of RW)

Let's define

$$X_t = x_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim NID(0, \sigma^2)$$

It's a non stationary process being an AR(1) with $\phi = 1$ (root on the unit circle). The differenced process is stationary:

$$(1 - L)X_t = \varepsilon_t$$

If we fix an initial condition, $X_0 = 0$ then

$$X_1 = X_0 + \varepsilon_1 = \varepsilon_1$$

$$X_2 = X_1 + \varepsilon_2 = \varepsilon_1 + \varepsilon_2$$

$$X_3 = X_2 + \varepsilon_3 = \varepsilon_1 + \varepsilon_2 + \varepsilon_3$$

and thus

$$X_t = \sum_{k=1}^t \varepsilon_k$$

Now with the above formulation (as sum of IID gaussian) it's easy to see $\mathbb{E}[X_t] = 0$ and $\text{Var}[X_t] = t\sigma^2$ so the variance *does depend* on t (so it's a non stationary process with path produced with tends to spread out)

As exercise derive $\text{Cov}(X_t, X_{t+k})$ and $\text{Cov}(X_t, X_{t-k})$ (since the process is not stationary we should not expect that the two covariances are equal)

2.6 AR(2) and MA(2)

We complete the picture of basic AR and MA processes (stationarity and invertibility)

Definition 2.6.1 (zero mean AR(2)). Defined adding $\psi_2 X_{t-2}$ to an AR(1)

$$X_t = \phi_1 X_{t-1} + \psi_2 X_{t-2} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

rewritable as

$$(1 - \phi_1 L - \phi_2 L^2)X_t = \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

Remark 18. It's

- always *invertible* by construction (function of its past value and innovation term)
- *stationary* if the roots z_1, z_2 of the characteristic equation associated to the polynomial $1 - \phi_1 L - \phi_2 L^2 = 0$ are in modulus greater than one

To get this we go in a non straightforward way. We derive the covariance function

We now derive the conditions of stationarity. To derive them

- we derive γ_0 and γ_k . Yule walker recursive equations with initial conditions

$$\gamma_0 = \frac{\sigma^2(1 - \phi_2)}{(1 + \phi_2)[(1 - \phi_2)^2 - \phi_1^2]}$$

$$\gamma_1 = \frac{\phi_1 \gamma_0}{1 - \phi_2}$$

- consider that $|z_1| |z_2| > 1 \implies |z_1, z_2| > 1$ and the fact that (Yule Walker equations) $\rho_1 = \phi_1 \rho_0 + \phi_2 \rho_1$ with $\rho_1 = \frac{\phi_1}{1 - \phi_2}$

For the Wold representation solve

$$(1 - \phi_1 L - \phi_2 L^2) X_t = \varepsilon_t$$

we search for condition such that X_t can be written as

$$X_t = (1 - \phi_1 L - \phi_2 L^2)^{-1} \varepsilon_t$$

which means that the product

$$\underbrace{(1 - \phi_1 L - \phi_2 L^2)}_{\Phi_2(L)} \underbrace{(1 + \psi_1 L + \psi_2 L^2 + \dots)}_{\Psi(L)} = 1$$

what does this product give

$$\begin{aligned} 1 + \psi_1 L + \psi_2 L^2 + \dots - \phi_1 L - \phi_1 \psi_1 L^2 - \phi_1 \psi_2 L^3 - \dots - \phi_2 L^2 - \phi_2 \psi_1 L^3 - \phi_2 \psi_2 L^4 + \dots = 1 \\ (\psi_1 - \phi_1) L + (\psi_2 - \phi_1 \psi_1 - \phi_2) L^2 + (\psi_3 - \phi_1 \psi_2 - \phi_2 \psi_1) L^3 + \dots = 0 \end{aligned}$$

For all this to be 0 we have to have

$$\begin{aligned} \psi_1 &= \phi_1 \\ \psi_2 &= \phi_1 \psi_1 - \phi_2 \psi_0 = \phi_1^2 - \phi_2 \\ \psi_3 &= \phi_1 \psi_2 - \phi_2 \psi_1 \\ &\dots \end{aligned}$$

Important remark 8. In general the **wold representation of an AR(2) process**

$$\psi_j = \phi_1 \psi_{j-1} - \phi_2 \psi_{j-2}$$

with $\psi_0 = 1, \psi_1 = \phi_1$. Not easy to derive conditions based on these representation

Covariance AR(2) Let's go to the covariance function of AR(2).

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$$

one way is to use wold representation but we see another way giusto per non annoiarci. The technique is the same

$$\begin{aligned} \gamma_0 &= \mathbb{E} \left[X_t \underbrace{(\phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t)}_{X_t} \right] \\ &= \phi_1 \gamma_1 + \phi_2 \gamma_2 + \sigma^2 \end{aligned}$$

warning: we're assuming that X_t is stationary.
Doing the same

$$\begin{aligned}\gamma_1 &= \mathbb{E}[X_t(\phi_1 X_t + \phi_2 X_{t-1} + \varepsilon_{t+1})] \\ &= \phi_1 \gamma_0 + \phi_2 \gamma_1 \\ \implies \gamma_1 &= \frac{\phi_1 \gamma_0}{1 - \phi_2}\end{aligned}$$

and

$$\begin{aligned}\gamma_2 &= \mathbb{E}[X_t(\phi_1 X_{t+1} + \phi_2 X_t + \varepsilon_{t+2})] \\ &= \phi_1 \gamma_1 + \phi_2 \gamma_0\end{aligned}$$

from which we develop more γ_0 by substitution

$$\gamma_0 = \frac{\phi_1^2}{1 - \phi_2} \gamma_0 + \frac{\phi_1^2 \phi_2}{1 - \phi_2} \gamma_0 + \phi_2^2 \gamma_0 + \sigma^2$$

and we do some algebra to collect for γ_0 (actually assuming $\phi_2 \neq 0$)

$$\begin{aligned}\left(1 - \frac{\phi_1^2}{1 - \phi_2} - \frac{\phi_1^2 \phi_2}{1 - \phi_2} - \phi_2^2\right) \gamma_0 &= \sigma^2 \\ (1 - \phi_2 - \phi_1^2 - \phi_1^2 \phi_2 - (1 - \phi_2) \phi_2^2) \gamma_0 &= \sigma^2 (1 - \phi_2) \\ (1 - \phi_2 - \phi_1^2 (1 + \phi_2) - (1 - \phi_2) \phi_2^2) \gamma_0 &= \sigma^2 (1 - \phi_2)\end{aligned}$$

from which doing all the calculation we end up with an expression of

$$\gamma_0 = \frac{\sigma^2 (1 - \phi_2)}{(1 + \phi_2)((1 - \phi_2)^2 - \phi_1^2)}$$

Then we can go back and obtain

$$\gamma_1 = \frac{\phi_1}{1 - \phi_2} \gamma_0$$

and then

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} \quad \text{with initial conditions } \gamma_0, \gamma_1$$

this above is the **autocovariance function of AR(2)**. We don't want the denominator of γ_0 to be 0.

2.6.1 Stationary conditions of AR(2)

So up to now, for AR(2) process we've done

- Wold representation
- ACV assuming stationarity; Now we want to get the stationary conditions.

An AR(2) process is stationary if the roots of the characteristic equation associated with $\Phi_2(L)$ that is

$$(1 - \phi_1 L - \phi_2 L^2) = 0$$

that is the following roots

$$z_{1,2} = \frac{-\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2\phi_2} = \begin{cases} z_1 = \frac{-\phi_1 + \sqrt{\dots}}{\dots} \\ z_2 = \frac{-\phi_1 - \sqrt{\dots}}{\dots} \end{cases}$$

are all, in modulus, greater than 1, that is

$$|z_1| > 1, |z_2| > 1 \text{ (thus } |z_1 z_2| > 1)$$

To have $|z_1| > 1, |z_2| > 1$ as function of ϕ -s we can setup a system

$$z_1 < -1 \vee z_1 > 1, z_2 < -1 \vee z_2 > 1$$

to find solutions by substituting $\frac{-\phi_1 \pm \sqrt{\dots}}{\dots}$ is quite involved. But since if $|z_1| > 1, |z_2| > 1$ this implies $|z_1 z_2| > 1$, this helps because the products of the two is the type $(a+b)(a-b) = a^2 - b^2$. The product of the two will be $\frac{\phi_1^2 - \phi_1^2 - 4\phi_2}{4\phi_2^2}$ so the conditions become

$$\left| \frac{\phi_1^2 - \phi_1^2 - 4\phi_2}{4\phi_2^2} \right| > 1$$

$$|\phi_2| < 1$$

and this above, $|\phi_2| < 1$, is the first condition. But it's not enough because we need condition on ϕ_1 as well.

Remembering that we've said

$$\begin{aligned} \gamma_k &= \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} \\ \rho_k &= \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} \end{aligned}$$

If $k = 1$ then

$$\rho_1 = \phi_1 \rho_0 + \phi_2 \rho_{-1}$$

assuming stationarity so $\cong \phi_1 \rho_0 + \phi_1 \rho_1$ so

$$\rho_1 = \phi_1 \underbrace{\rho_0}_1 + \phi_2 \rho_1 \implies \rho_1 = \frac{\phi_1}{1 - \phi_2}$$

and since for sure

$$-1 < \rho_1 < 1$$

and $|\phi_2| < 1$ we get two additional conditions for stationarity of an AR(2) process (from the double inequality above by substituting ρ_1 developed before)

$$\frac{\phi_1}{1 - \phi_2} < 1 \implies \phi_2 + \phi_1 < 1$$

On the other hand

$$\frac{\phi_1}{1 - \phi_2} > -1 \implies \phi_2 - \phi_1 < 1$$

Thus summarizing AR(2) is stationary if

$$\begin{cases} |\phi_2| < 1 \\ \phi_2 + \phi_1 < 1 \\ \phi_2 - \phi_1 < 1 \end{cases}$$

2.6.2 Stationary conditions of MA(2)

The process is

$$X_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

or

$$X_t = (1 - \theta_1 L - \theta_2 L^2) \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

The stationarity condition shown for will be same as condition of invertibility for MA(2) given that polynomial is the similar, that is

$$\begin{cases} |\theta_2| < 1 \\ \theta_2 + \theta_1 < 1 \\ \theta_2 - \theta_1 < 1 \end{cases}$$

Example 2.6.1. Do As an exercise

- Show it is stationary (by direct calculation of $\mathbb{E}[X_t]$)
- Derive and plot the ACF, γ_0, γ_k (expect γ_k becomes 0 after second lag)
- Derive the Wold representation ($\psi_1 = -\theta_1, \psi_2 = -\theta_2, \psi_0 = 1, \psi_j = 0, \forall j > 2$)
- State (and show) the invertibility conditions

Chapter 3

Prediction

Here we use linearity of stationary processes in order to start making prediction of X_t given past values: this is based on *conditioning* and *conditional expectation* will be our predictor (best predictor in the MSE sense).

3.1 Premises

Example 3.1.1 (ex 1.1 ITSF). Let X and Y be two random variables with $\mathbb{E}[Y] = \mu$ and $\mathbb{E}[Y^2] < \infty$

1. show that the constant c that minimizes $\mathbb{E}[Y - c]$ is $c = \mu$
2. deduce that the random variable $f(X)$ that minimizes $\mathbb{E}[(Y - f(X))^2|X]$ is $f(X) = \mathbb{E}[Y|X]$.
3. deduce that the random variable $f(X)$ that minimizes $\mathbb{E}[(Y - f(X))^2]$ (no conditioning) is also $f(X) = \mathbb{E}[Y|X]$.

In order

1. to prove $\min_{c \in \mathbb{R}} \mathbb{E}[(Y - c)^2] = \mu$ the objective function to be minimized is

$$\begin{aligned}\phi(c) &= \mathbb{E}[Y^2 - 2cY + c^2] = \mathbb{E}[Y^2] - 2c\mathbb{E}[Y] + \mathbb{E}[c^2] \\ &= \mathbb{E}[Y^2] - 2c\mu + c^2\end{aligned}$$

It's derivative with respect to c is

$$\frac{\partial}{\partial c}\phi(c) = -2\mu + 2c$$

and equating to 0

$$\frac{\partial}{\partial c}\phi(c) = 0 \implies c = \mu$$

2. to deduce that the rv $f(X)$ (with f measurable function) that minimize $\mathbb{E}[(Y - f(X))^2|X]$ (where the conditioning means the information/sigma

algebra $\sigma(X)$) is $\mathbb{E}[Y|X]$, the objective function to be minimized

$$\begin{aligned}\phi(f(X)) &= \mathbb{E}[Y^2 - 2f(X)Y + f^2(X)|X] \\ &= \mathbb{E}[Y^2|X] - 2\mathbb{E}[f(X) \cdot Y|X] + \mathbb{E}[f^2(X)|X] \\ &= \mathbb{E}[Y^2|X] - 2f(X) \cdot \mathbb{E}[Y|X] + f^2(X)\end{aligned}$$

The derivative with respect to $f(X)$

$$\frac{\partial}{\partial f(X)} \phi(f(X)) = -2\mathbb{E}[Y|X] + 2f(X)$$

and equating to zero we have $f(X) = \mathbb{E}[Y|X]$.

Hence the conditional expectation of Y given X is the function of X that minimises the square distance between Y and $f(X)$ (on average, conditional to $X \dots$, minimizes also unconditionally)

3. todo based on the *law of iterated expectation* will se

Example 3.1.2 (Ex 1.2 ITSF). Generalization of the problem to all the past

Proposition 3.1.1. *Two useful results on conditioning the first of which is the law of iterated expectations (l.i.e.)*

$$\begin{aligned}\mathbb{E}[Y] &= \mathbb{E}[\mathbb{E}[Y|X]] \\ \text{Var}[Y] &= \mathbb{E}[\text{Var}[Y|X]] + \text{Var}[\mathbb{E}[Y|X]]\end{aligned}$$

Remark 19. For the law of iterated expectations, the unconditional expectation of the conditional expectation of Y given X is equal to the unconditional expectation of Y Here

- conditional means: based on / given some information (it will be natural for us to assume that the *information* is the *past* in the context of time series).¹
- unconditional: no information available (one condition on the trivial sigma algebra)

Law of iterated expectation. If we have two rv X, Y and some mild condition about

$$\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y]$$

where $\mathbb{E}[Y|X]$ is a measurable function of X , the conditioning variable.

$$\begin{aligned}\mathbb{E}[\mathbb{E}[Y|X]] &\stackrel{(1)}{=} \int_{R_X} \mathbb{E}[Y|X] \cdot f_X(x) \, dx \\ &\stackrel{(2)}{=} \int_{R_X} \int_{R_Y} y \cdot f_{Y|X}(y|x) \, dy \cdot f_X(x) \, dx \\ &\stackrel{(3)}{=} \int_{R_Y} y \int_{R_X} f_{XY}(x, y) \, dx \, dy \\ &\stackrel{(4)}{=} \int_{R_Y} y \cdot f_Y(y) \, dy \\ &= \mathbb{E}[Y]\end{aligned}$$

¹The concept of information is crucial in general: for example with the fisher information the biggest the information is the smaller the variance of an unbiased estimation

where

- in (1) the outside expectation in $\mathbb{E}[\mathbb{E}[Y|X]]$ is with respect to X , so we'll have an integral with respect to the domain of X (we suppose $R_X = \mathbb{R}, R_Y = \mathbb{R}$)
- (2) we develop the inside integral attached to $\mathbb{E}[Y|X]$ that is
- (3) we assume we can move a little bit of quantities and especially that the product of $f_{Y|X}(y|x)$ for $f_X(x)$ is equal to the joint density
- (4) integrating out the joint we have the marginal

□

Proof. The other proof of variance as exercise

□

3.2 Prediction

Definition 3.2.1 (Predictor). It's defined as conditional expectation of tomorrow's rv given all the past

$$X_{n+1|n} = \mathbb{E}[X_{n+1}|X_1, \dots, X_n]$$

Important remark 9. The predictor according to what we've seen (ex 1.1 ITSF) is the best predictor in the sense that minimises square distance between future value X_{n+1} and its prediction $X_{n+1|n}$ given the past information (conditional MSE) which is the loss function we want to minimize in making predictions

$$\mathbb{E}[(X_{n+1} - X_{n+1|n})^2 | X_1, \dots, X_n]$$

and by the law of iterated expectation (l.i.e) also minimises the unconditional MSE

$$\mathbb{E}[(X_{n+1} - X_{n+1|n})^2]$$

ie $X_{n+1|n}$ is the minimum mean square error predictor of X_{n+1} given the past information set (sigma algebra generated by $X_1, \dots, X_n$, called filtration)

$$\mathcal{F}_n = \{X_1, X_2, \dots, X_n\}$$

Since by the l.i.e

$$\mathbb{E}[X_{n+1} - X_{n+1|n}] = 0$$

that is the predictor is *unbiased* then the minimum MSE is also the minimum variance, that is $X_{n+1|n}$ also minimises the *prediction error variance*:

$$\mathbb{E}[(X_{n+1|n} - X_{n+1})^2]$$

where the quantity $X_{n+1|n} - X_{n+1}$ is the *prediction error*

Remark 20. The predictor error variance is an important quantity in time series

Example 3.2.1 (One step ahead prediction error variance of $X_{n+1|n}$ when X is a linear process). If $\{X_t\}_{t \in T}$ is a linear process that is

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad \text{where } \psi_0 = 1, \sum_{j=0}^{\infty} \psi_j^2 < \infty, \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

The predictor is developed as follow

$$\begin{aligned} X_{t+1|t} &= \mathbb{E}[X_{t+1}|X_t, X_{t-1}, \dots, X_1] \\ &= \mathbb{E}\left[\sum_{j=0}^{\infty} \psi_j \varepsilon_{t+1-j} | X_t, \dots, X_1\right] \\ &= \mathbb{E}\left[\varepsilon_{t+1} + \psi_1 \varepsilon_t + \psi_2 \varepsilon_{t-1} + \dots \mid \underbrace{X_t, \dots, X_1}_{\mathcal{F}_t}\right] \\ &= \mathbb{E}[\varepsilon_{t+1} | \mathcal{F}_t] + \psi_1 \mathbb{E}[\varepsilon_t | \mathcal{F}_t] + \psi_2 \mathbb{E}[\varepsilon_{t-1} | \mathcal{F}_t] + \dots \\ &= \mathbb{E}[\varepsilon_{t+1}] + \psi_1 \varepsilon_t + \psi_2 \varepsilon_{t-1} + \dots \end{aligned}$$

where in the last passage

- $\mathbb{E}[\varepsilon_{t+1} | \mathcal{F}_t] = \mathbb{E}[\varepsilon_{t+1}]$ because ε_{t+1} is uncorrelated with past observation
- if X_t is measurable with respect to $\mathcal{F}_t$ the same occurs to the ε_t and so these are taken out of conditional expectation

The last equation is our predictor: basically we add the expected value

Now the **prediction error** is the difference between future and our the prediction

$$X_{t+1} - X_{t+1|t}$$

It's expected value given the past

$$\begin{aligned} \mathbb{E}[X_{t+1} - X_{t+1|t} | \mathcal{F}_t] &= \mathbb{E}[X_{t+1} | \mathcal{F}_t] - \mathbb{E}[X_{t+1|t} | \mathcal{F}_t] \\ &\stackrel{(1)}{=} X_{t+1|t} - X_{t+1|t} \\ &= 0 \end{aligned}$$

where in (1) $\mathbb{E}[X_{t+1} | \mathcal{F}_t] = X_{t+1|t}$ by definition of the predictor while $\mathbb{E}[X_{t+1|t} | \mathcal{F}_t] = X_{t+1|t}$ by construction (in the sense that being a function of past variable up to t so is known at time t and thus we can get it out of expectation).

So the prediction error has null expected value

If we take the unconditional expectation

$$\mathbb{E}[\mathbb{E}[X_{t+1} - X_{t+1|t} | \mathcal{F}_t]] = \mathbb{E}[0] = 0$$

The prediction error variance (for a prediction error which is unbiased/has zero mean)

$$\begin{aligned} \mathbb{E}[(X_{t+1} - X_{t+1|t})^2 | \mathcal{F}_t] &= \mathbb{E}\left[\left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{t+1-j} - \sum_{j=1}^{\infty} \psi_j \varepsilon_{t+1-j}\right)^2 \mid \mathcal{F}_t\right] \\ &= \mathbb{E}[\varepsilon_{t+1}^2 | \mathcal{F}_t] = \mathbb{E}[\varepsilon_{t+1}^2] \\ &= \sigma^2 \end{aligned}$$

This σ^2 is the prediction error variance.

Summarizing we have

- $X_1, \dots, X_n$ past variable which we call $\mathcal{F}_n$
- predictor $X_{n+1|n} = \mathbb{E}[X_{n+1}|X_1, \dots, X_n]$ which is best in the sense that minimise the meas square error
- prediction error is $X_{n+1|n} - X_{n+1}$: it has conditional and unconditional zero mean, $\mathbb{E}[X_{n+1|n} - X_{n+1}|\mathcal{F}_n] = 0$ and $\mathbb{E}[X_{n+1|n} - X_{n+1}] = 0$
- **prediction error variance** $\mathbb{E}[(X_{n+1|n} - X_{n+1})^2]$.
For a linear process it's always equal to σ^2 and it's **smaller than the unconditional variance** $\gamma_0 = \sigma^2 \sum_{j=0}^{\infty} \psi_j^2$:
 - eg for AR(1) $\gamma_0 = \frac{\sigma^2}{1-\psi^2} > \sigma^2$ (with $|\phi| < 1$)
 - for MA(1) $\gamma_0 = \sigma^2(1 + \theta^2) > \sigma^2$

So having information on the past implies smaller variability

3.3 Best linear predictors

Previously we seen that

$$X_{n+1|n} = \mathbb{E}[X_{n+1}|X_n, \dots, X_1]$$

is the best predictor. Now we want to restrict/focus on the best *linear* predictor

$$X_{n+1|n} = \mathbb{E}[X_{n+1}|X_n, \dots, X_1] = \phi_{n1}X_n + \phi_{n2}X_{n-1} + \dots + \phi_{nn}X_1$$

which is a linear combination of past observation with coefficient $\phi_{n1}, \dots, \phi_{nn}$ such that the distance between future value and linear predictor

$$\mathbb{E}[(X_{n+1} - (\phi_{n1}X_n + \phi_{n2}X_{n-1} + \dots + \phi_{nn}X_1))^2|\mathcal{F}_n]$$

is minimum. The minimization can be carried out (as in previous class) algebraically by derive wrt to all the parameters and equating to 0

$$\frac{\partial}{\partial \phi_{n1}} \frac{\partial}{\partial \phi_{n2}} \dots \frac{\partial}{\partial \phi_{nn}} f() = 0$$

with f some kind of objective function.

Alternatively, equivalently we use a geometric interpretation/method

$$\mathbb{E}[(X_{n+1} - X_{n+1|n})X_j] = 0, \quad \forall j = 1, \dots, n$$

ie $X_{n+1|n}$ is the orthogonal projection of X_{n+1} onto the (infinite Hilbert) vector space spanned by $X_1, \dots, X_n$. That is we want to find coefficients

$$X_{n+1|n} = \phi_{n1}X_n + \phi_{n2}X_{n-1} + \dots + \phi_{nn}x_1$$

such that is minimum the following quantity

$$\mathbb{E}[(X_{n+1} - X_{n+1|n})^2]$$

We can think of a vector space created by the past values $\mathcal{C} = \text{span}\{X_1, \dots, X_n\}$ (∞ -dimensional hilbert space) basically it's a vector space of function but we can think it); we have that the value of tomorrow does not belong to this vector space ($X_{n+1} \notin \mathcal{C}$) otherwise the value of tomorrow would be just a linear combination of values up to today. We look for a vector in $X_{n+1|n} \in \mathcal{C}$ (so which can thus be obtained by linear combination of past values) which minimizes the (squared) distance between the prediction (our vector) and the future real observation X_{n+1} . To have it with minimum distance we have to project the future observation orthogonally on our $\mathcal{C}$ and find the corresponding vector.

All the vector belonging to $\mathcal{C}$ are *orthogonal* to the *distance* between future value and its projection; thus even each single past value should be orthogonal (set the coefficient of the other elements of the basis to 0) with inner product with distance being 0:

$$\langle X_j, (X_{n+1} - X_{n+1|n}) \rangle = 0, \quad \forall j = 1, \dots, n$$

In this space the inner product is represented by the expectation of the product of the two vectors (basically it's saying rv with inner product null aren't correlated i guess). So this above is equivalent to having

$$\mathbb{E}[(X_{n+1} - X_{n+1|n}) \cdot X_j] = 0, \quad \forall j = 1, \dots, n$$

Which are the implication here? Let's consider the case $j = n$ (assume process stationary with $\mathbb{E}[X_t] = 0, \forall t$, with $\gamma_k = \gamma_{-k}$)

$$\begin{aligned} \mathbb{E}[(X_{n+1} - X_{n+1|n})X_n] &= 0 \\ \mathbb{E}[(X_{n+1} - (\phi_{n1}X_n + \phi_{n2}X_{n-1} + \dots + \phi_{nn}X_1)) \cdot X_n] &= 0 \\ \underbrace{\mathbb{E}[X_{n+1} \cdot X_n]}_{\gamma_1} &= \phi_{n1} \mathbb{E}[X_n X_n] + \phi_{n2} \mathbb{E}[X_n X_{n-1}] + \dots + \phi_{nn} \mathbb{E}[X_1 \cdot X_n] \\ \gamma_1 &= \phi_{n1}\gamma_0 + \phi_{n2}\gamma_1 + \dots + \phi_{nn}\gamma_{n-1} \end{aligned}$$

If $j = n - 1$ developing the same way and starting from the last point we have

$$\begin{aligned} \mathbb{E}[X_{n+1}X_n] &= \phi_{n1} \mathbb{E}[X_n X_{n-1}] + \dots + \phi_{nn} \mathbb{E}[X_1 X_{n-1}] \\ \gamma_2 &= \phi_{n1}\gamma_1 + \phi_{n2}\gamma_0 + \dots + \phi_{nn}\gamma_{n-2} \end{aligned}$$

We can go ahead up to $j = 1$ and putting everything together, the above (orthogonality or least squares) condition is equivalent to:

$$\begin{aligned} \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \dots \\ \gamma_n \end{bmatrix} &= \begin{bmatrix} \gamma_0 & \gamma_1 & \gamma_2 & \dots & \gamma_{n-1} \\ \gamma_1 & \gamma_0 & \gamma_1 & \dots & \gamma_{n-2} \\ \gamma_2 & \gamma_1 & \gamma_0 & \dots & \gamma_{n-3} \\ \vdots & & & & \\ \gamma_{n-1} & \gamma_{n-2} & \gamma_{n-3} & \dots & \gamma_0 \end{bmatrix} \begin{bmatrix} \phi_{n1} \\ \phi_{n2} \\ \dots \\ \phi_{nn} \end{bmatrix} \\ \gamma_n &= \Gamma_n \phi_n \end{aligned}$$

where

- γ_n are the autocovariances of X_t for $k = 1, \dots, n$
- $\mathbf{\Gamma}_n$ is the so called Toeplitz matrix associate with this system, ie the variances-autocovariances matrix of $\mathbf{X} = [X_n, X_{n-1}, \dots, X_1]^\top$, $\mathbf{\Gamma}_n$ is a $n \times n$ matrix defined with $\mathbf{\Gamma}_n = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]$ (assumin the vector is zero meaned), the main diagonal is composed of variance (constant) and symmetric covariance out of diagonal

Again we want to find ϕ_n which are the coefficient of the best linear predictor we're searching to obtain by minimization; the coefficients can be obtained by solving the linear system

$$\phi_n = \mathbf{\Gamma}_n^{-1} \gamma_n \quad (3.1)$$

Solving the system can be done by inverting $\mathbf{\Gamma}_n$ (this can be done whenever the process is stationary); we can divide both member of equation above by γ_0 obtaining something related to autocorrelation

$$\frac{1}{\gamma_0} \gamma_n = \frac{1}{\gamma_0} \mathbf{\Gamma}_n \phi_n$$

$$\begin{bmatrix} \rho_1 \\ \rho_2 \\ \dots \\ \rho_n \end{bmatrix} = \begin{bmatrix} 1 & \rho_1 & \dots & \dots & \rho_{n-1} \\ \vdots & & & & \\ \vdots & & & & \\ \vdots & & & & \\ \rho_{n-1} & \rho_{n-2} & \rho_{n-3} & \dots & 1 \end{bmatrix} \begin{bmatrix} \phi_{n1} \\ \phi_{n2} \\ \dots \\ \phi_{nn} \end{bmatrix}$$

then invert this last autocorrelation matrix to obtain the coefficient in the equivalent system.

Important remark 10 (Practical). The best linear predictor $\hat{x}_{n+1|n}$ of X_{n+1} given the past information set can be then obtained as follows:

- given an observed stationary time series $x_1, \dots, x_n$, estimate the autocovariance function as

$$\hat{\gamma}_k = \frac{1}{n} \sum_{t=1}^{n-k} (x_t - \bar{x}_n)(x_{t+k} - \bar{x}_n)$$

where

- note we're using n as denominator which is biased (the sample acv should have $n - k$ instead)
- the quantity

$$\bar{x}_n = \frac{1}{n} \sum_{t=1}^n x_t$$

is the sample mean of the serie

- construct the sample vector $\hat{\gamma}_n$ and the sample covariance matrix $\hat{\mathbf{\Gamma}}_n$ to obtain the estimated coefficients

$$\hat{\phi}_n = \hat{\mathbf{\Gamma}}_n^{-1} \hat{\gamma}_n$$

- obtain

$$\hat{x}_{n+1|n} = \hat{\phi}_{n1}x_n + \hat{\phi}_{n2}x_{n-1} + \dots \hat{\phi}_{nn}x_1$$

Moreover we can obtain an estimation of σ^2 (variance of the prediction error)

$$\begin{aligned}\hat{\sigma}^2 &= \mathbb{E}[(X_{n+1} - X_{n+1|n})^2] = \mathbb{E}[X_{n+1}^2 - 2\phi_n^\top \mathbf{X}_n \mathbf{X}_{n+1} + (\phi_n^\top \mathbf{X}_n)^2] \\ &= \gamma_0 - 2\phi_n^\top \gamma_n + \mathbb{E}[\phi_n^\top \mathbf{X}_n \mathbf{X}_n^\top \phi_n] \\ &= \gamma_0 - 2\phi_n^\top \gamma_n + \phi_n^\top \Gamma_n \phi_n \\ &= \hat{\gamma}_0 - \hat{\phi}_n^\top \hat{\gamma}_n\end{aligned}$$

which follows by 3.1.

Example 3.3.1 (The best linear predictor for an AR(1) process (ex 2.5.1 ITSF)). if

$$X_t = \omega + \phi X_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

with $\mathbb{E}[X_t] = \frac{\omega}{1-\phi}$ if $\omega = 0$ it's zero mean ar(1) stationary if $|\phi| < 1$ and $\rho_k = \phi^{|k|}$.

Construct the best linear predictor based on the formula seen before (here we use the autocorrelation version/ premultiplied by $\frac{1}{\gamma_0}$)

$$\begin{aligned}\gamma_n &= \Gamma_n \phi_n \\ \gamma_n \frac{1}{\gamma_0} &= \frac{1}{\gamma_0} \Gamma_n \phi_n\end{aligned}$$

Doing calculation and remembering for AR(1) $\rho_k = \phi^{|k|}$ the system becomes

$$\begin{bmatrix} \phi \\ \phi^2 \\ \dots \\ \phi^n \end{bmatrix} = \begin{bmatrix} 1 & \phi & \phi^2 & \dots & \phi^{n-1} \\ \phi & \phi^2 & \phi & \dots & \phi^{n-2} \\ \dots & \dots & \dots & \dots & \dots \\ \phi^{n-1} & \phi^{n-2} & \dots & \dots & 1 \end{bmatrix} \begin{bmatrix} \phi_{n1} \\ \phi_{n2} \\ \dots \\ \phi_{nn} \end{bmatrix}$$

We observe that a possible solution is the vector (equality above is satisfied with following substitution)

$$\begin{bmatrix} \phi_{n1} \\ \phi_{n2} \\ \dots \\ \phi_{nn} \end{bmatrix} = \begin{bmatrix} \phi \\ 0 \\ \dots \\ 0 \end{bmatrix}$$

So the best linear predictor for AR(1)

$$\begin{aligned}X_{n+1|n} &= \phi_{n1}X_n + \phi_{n2}X_{n-1} + \dots \phi_{nn}X_1 \\ &= \phi X_n + 0X_{n-1} + \dots 0X_1 \\ &= \phi X_n\end{aligned}$$

This is somehow expected since in AR(1) with $\omega = 0$

$$X_t = \phi X_{t-1} + \varepsilon_t, \quad |\phi| < 1, \varepsilon_t \in \text{WN}(0, \sigma^2)$$

if we take the conditional expectation (the best overall predictor)

$$\begin{aligned}
X_{n+1|n} &= \mathbb{E}[X_{n+1}|X_n, X_{n-1}, \dots, X_1] \\
&= \mathbb{E}[\phi X_n + \varepsilon_{n+1}|X_n, X_{n-1}, \dots, X_1] \\
&= \phi \mathbb{E}[X_n|X_n, X_{n-1}, \dots, X_1] + \mathbb{E}[\varepsilon_{n+1}|X_n, X_{n-1}, \dots, X_1] \\
&= \phi X_n + \mathbb{E}[\varepsilon_{n+1}] \\
&= \phi X_n
\end{aligned}$$

So the above, which is the best linear predictor (MMSE) is also a linear predictor of $X_n, \dots, X_1$ with weights $\phi, 0, \dots, 0$, so it must be the best linear predictor (as developed above).

So in case of AR(1) the best predictor $X_{n+1|n} = \mathbb{E}[X_{n+1}|\mathcal{F}_n]$ is also the best *linear* predictor $X_{n+1|n} = \phi_{n1}X_n + \phi_{n2}X_{n-1} + \dots + \phi_{nn}x_1 = \phi X_n$.

This is not always true, eg we could have a best predictor that is non linear (which is not)

Important remark 11 (MA(q)). Warning: for a MA(q) process

$$X_t = \varepsilon_t - \theta \varepsilon_{t-1}$$

(with ε_t being $\mathcal{F}_t$ measurable) to construct a predictor using $x_1, \dots, x_n$ (which is what we observe)

$$\begin{aligned}
x_t &= \varepsilon_t - \theta \varepsilon_{t-1} \\
\varepsilon_t &= x_t + \theta \varepsilon_{t-1} \\
&= x_t + \theta(x_{t-1} + \theta \varepsilon_{t-2}) \\
&= x_t + \theta x_{t-1} + \theta^2(x_{t-2} + \theta \varepsilon_{t-3}) \\
&= \sum_{j=0}^{\infty} \theta^j x_{t-j}
\end{aligned}$$

in order to construct a predictor as a function of the observable variables

Chapter 4

Partial autocorrelation function

Remark 21. It's an useful tool to (among other) identify MA or AR process of order q/p from a stationary time series.

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Definition 4.0.1 (The formal definition). the partial autocorrelation α_k is not defined for $k = 0$ (differently from the global autocorrelation seen so far which is)

$$a_1 = \text{Corr}(X_2, X_1) = \rho_1$$
$$a_k = \text{Corr}(X_{k+1} - \mathbb{E}[X_{k+1}|X_2, \dots, X_k], X_1 - \mathbb{E}[X_1, X_2, \dots, X_k])$$

The peculiar is the last one where the partial autocorrelation for a lag of k is the autocorrelation between X_{k+1} and X_1 after we have subtracted from both each the conditional expectations/best prediction given $X_2, \dots, X_k$, that are the variable between X_1 and X_{k+1}

Definition 4.0.2 (Equivalent definition). A much more intuitive definition we typically use is the following

$$a_k = \phi_{kk}$$

where

$$\mathbb{E}[X_{k+1}|X_1, \dots, X_k] = \phi_{k1}X_k + \phi_{k2}X_{k-1} + \dots + \phi_{kk}X_1 +$$

Here basically is the regression coefficient of X_1 on the prediction/regression of X_{k+1} after having considered all the other times $X_2, \dots, X_k$.

If we take the derivative ϕ_{kk} is the increment to $\mathbb{E}[X_{k+1}|X_1, \dots, X_k]$ following unity increase in X_1

Tool to calculate ϕ_{kk} : were we use the same techniques used to determine coefficient of the best linear predictor. We know that for the best linear predictor, the difference $X_{k+1} - \mathbb{E}[X_{k+1}|X_1, \dots, X_k]$ is *orthogonal* to all the vector spanning from $\{X_1, \dots, X_k\}$

$$X_{k+1} - \mathbb{E}[X_{k+1}|X_1, \dots, X_k] \perp\!\!\!\perp X_1, \dots, X_k$$

and thus for orthogonality what we have to have is that the inner product

$$\mathbb{E}[X_{k+1} - \phi_{k1}X_k - \phi_{k2}X_{k-1} - \dots - \phi_{kk}X_1]X_j = 0 \quad \forall j = 1, \dots, k$$

Taking expectations and dividing both the members by γ_0

$$\rho_{k+1-j} = \phi_{k1}\rho_{k-j} + \phi_{k2}\rho_{k-1-j} + \dots + \phi_{kk}\rho_{1-j}, \quad j = 1, \dots, k$$

In matrix form

$$\underbrace{\begin{bmatrix} \rho_1 \\ \rho_2 \\ \dots \\ \rho_k \end{bmatrix}}_{\rho_k} = \underbrace{\begin{bmatrix} \rho_0 & \rho_1 & \dots & \rho_{k-1} \\ \rho_1 & \rho_0 & \dots & \rho_{k-2} \\ \dots & \dots & \dots & \dots \\ \rho_{k-1} & \rho_{k-2} & \dots & \rho_0 \end{bmatrix}}_{R_k} \underbrace{\begin{bmatrix} \phi_{k1} \\ \phi_{k2} \\ \dots \\ \phi_{kk} \end{bmatrix}}_{\phi_k}$$

and ϕ_{kk} can be obtained recursively or by the Cramer rule.

Here we have the same linear system as the one used to determine the coefficients of the BLP but differently here k does vary and $\forall k = 1, 2, \dots$ there is a different system of dimension $k \times k$.

Note that here we only need ϕ_{kk} ; there is no need to solve the whole system but we can use cramer rule where, to have the last element of the vector ϕ_k we can do the ratio of determinant of following matrices

$$\phi_{kk} = \frac{|R_k^{(k)}|}{|R_k|}$$

with

- R_k as in the previous, the acf matrix associated with the process X_t . The determinant at the denominator is always different from 0 for stationary processes
- $R_k^{(k)}$ is R_k but with the last column replaced by the vector of known terms

$$R_k^{(k)} = \begin{bmatrix} \rho_0 & \rho_1 & \dots & \rho_1 \\ \rho_1 & \rho_0 & \dots & \rho_2 \\ \dots & \dots & \dots & \dots \\ \rho_{k-1} & \rho_{k-2} & \dots & \rho_k \end{bmatrix}$$

Example 4.0.1 (AR(1) stationary process with zero mean). With $X_t = \phi X_{t-1} + \varepsilon_t$, with $|\phi| < 1$ and $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ we have that

$$\rho_k = \phi^{|k|}, \quad k = 0, \pm 1, \pm 2, \dots$$

and

$$\phi_{kk} = \begin{cases} \phi_{11} = \rho_1 = \phi & \text{if } k = 1 \\ \phi_{22} = \frac{\begin{vmatrix} \rho_0 & \rho_1 \\ \rho_1 & \rho_2 \end{vmatrix}}{\begin{vmatrix} \rho_0 & \rho_1 \\ \rho_1 & \rho_0 \end{vmatrix}} = \frac{\begin{vmatrix} 1 & \phi \\ \phi & \phi^2 \end{vmatrix}}{\begin{vmatrix} 1 & \phi \\ \phi & 1 \end{vmatrix}} = \frac{\phi^2 - \phi^2}{1 - \phi^2} = 0 & \text{if } k = 2 \\ \phi_{33} = \frac{\begin{vmatrix} \rho_0 & \rho_1 & \rho_1 \\ \rho_1 & \rho_0 & \rho_2 \\ \rho_2 & \rho_1 & \rho_3 \end{vmatrix}}{|R_3|} = \frac{\begin{vmatrix} 1 & \phi & \phi \\ \phi & 1 & \phi^2 \\ \phi^2 & \phi & \phi^3 \end{vmatrix}}{|R_3|} = \frac{0}{R_3} = 0 & \text{if } k = 3 \end{cases}$$

In general for an AR(1) process

$$\phi_{kk} = \begin{cases} \phi, & k = 1 \\ 0 & k > 1 \end{cases}$$

In general for an AR(p) processes PACF start in ϕ for $k = 1$ then has an decay and becomes zero after lag p . So we can use this to identify the order of the process.

With a mixed ARMA we can't use this because of the following

Example 4.0.2 (MA). Derive ϕ_{kk} , $k = 1, 2$ for a MA(1). (*it doesnt get to 0*)

Chapter 5

Sample stats

We're left with doing inference on the *moments* of a *stationary* stochastic process: we're interested

- estimation of parameters μ , γ_k and ρ_k
- deriving asymptotic distribution of sample mean $\bar{x}_n$ and correlation $\hat{\rho}_k$
- testing for serial correlation

of a stationary stochastic process. References are in ITSF chapter 2. Other things that can be interesting are (to read if needed on slides)

5.1 Mean

5.1.1 Estimation

Assuming we've observed the time series $x_1, x_2, \dots, x_n$ coming from a stationary stochastic process with mean μ , variance γ_0 and covariance γ_k ; we can estimate the sample mean is the moment estimator we will consider

$$\bar{x}_n = \frac{1}{n} \sum_{t=1}^n x_t$$

is an unbiased estimator for μ (if we take expectation of $\bar{x}_n$ we'll get μ if the process is stationary), the variance of $\bar{x}_n$ is

$$\text{Var} [\bar{x}_n] = \mathbb{E} [(\bar{x}_n - \mu)^2]$$

suppose now that $\mu = 0$ ¹. We can write the variance as (we're assuming that $\mathbb{E}[X_t] = 0$)

$$\begin{aligned} \text{Var}[\bar{x}_n] &= \mathbb{E} \left[\left(\frac{1}{n} \sum_{t=1}^n x_t \right) \left(\frac{1}{n} \sum_{s=1}^n x_s \right) \right] = \frac{1}{n^2} \mathbb{E} \left[\left(\sum_{t=1}^n x_t \right) \left(\sum_{s=1}^n x_s \right) \right] \\ &\stackrel{(1)}{=} \frac{1}{n^2} (n\gamma_0 + 2(n-1)\gamma_1 + 2(n-2)\gamma_2 + \dots + 2\gamma_{n-1}) = \frac{1}{n} \left(\gamma_0 + 2 \sum_{k=1}^n \frac{n-k}{n} \gamma_k \right) \\ &= \frac{1}{n} \sum_{|k| < n} \frac{n-|k|}{n} \gamma_k \end{aligned}$$

with (1) coming from basic algebra.

This above is the variance of the sample mean estimator

5.1.2 Ergodicity of $\bar{x}_n$

In general ergodicity is a property of an estimator to converge to the actual moment.

Here we want to prove that $\bar{x}_n \rightarrow \mu$:

- almost surely: ergodic theorem for strongly stationary processes
- or in means square (ergodic theorem for weakly or second order stationary processes).

These are law of large numbers for dependent data, which require a restriction on the memory of the process. One of these restrictions is ergodicity, sort of average asymptotic independence

Definition 5.1.1. A weakly stationary stochastic process $\{X_t\}_{t \in T}$ is ergodic with respect to moment $\mathbb{E}[g(X_t)]$ if the sample moment $\frac{1}{n} \sum_{t=1}^n g(x_t)$ converges (in probability or almost surely) to $\mathbb{E}[g(X_t)]$ that is

$$\frac{1}{n} \sum_{t=1}^n g(x_t) \rightarrow \mathbb{E}[g(X_t)]$$

Remark 22. As we said $\{X_t\}_{t \in T}$ ergodic with respect to the mean if

$$\frac{1}{n} \sum_{t=1}^n x_t \rightarrow \mu$$

Important remark 12. The property of ergodicity ensures the *consistency* of estimators of the moments.

Theorem 5.1.1. If $\{X_t\}_{t \in T}$ is stationary with mean μ and covariance γ_k then as $n \rightarrow \infty$

$$\begin{aligned} \text{Var}[\bar{x}_n] &= \mathbb{E}[(\bar{x}_n - \mu)^2] \rightarrow 0 \quad \text{if } \gamma_n \rightarrow 0 \\ n \text{Var}[\bar{x}_n] &\rightarrow \sum_{k=-\infty}^{\infty} \gamma_k = 2\pi f(0) \quad \text{if } \sum_{k=-\infty}^{\infty} |\gamma_k| < \infty \end{aligned}$$

¹Remind that even though the mean of a process is zero we shall be going to estimate it.

Proof. Dimostrazione di $\text{Var}[\bar{x}_n] \rightarrow 0$ as $n \rightarrow \infty$ Pagina 10 di 14 haandout 7 sample mean $\square$

Remark 23. Thus the variance of the sample mean converges to 0 if the covariance at lag n converges to zero.

In large sample by multiplying per n converges to the sum of autocovariances from $-\infty$ to $+\infty$ (if the sequence of autocovariances is absolutely summable), where f is the spectral density evaluated in zero

Important remark 13. if $\{x_t\}_{t \in T}$ is a linear process then the variance of sample mean multiplied by n does converge to $\sigma^2 \left(\sum_{j=0}^{\infty} \psi_j \right)^2$

$$n \text{Var}[\bar{x}_n] \rightarrow \sum_{k=-\infty}^{\infty} \gamma_k = 2\pi f(0) = \sigma^2 \left(\sum_{j=0}^{\infty} \psi_j \right)^2$$

Theorem 5.1.2. If $\{X_t\}_{t \in T}$ is the linear process $X_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$ with $\sum_{j=0}^{\infty} |\psi_j| < \infty$, $\varepsilon_t \sim \text{IID}(0, \sigma^2)$ then

$$\sqrt{n}(\bar{x}_n - \mu) \xrightarrow{d} N(0, v)$$

where

$$v = \sum_{k=-\infty}^{\infty} \gamma_k = 2\pi f(0) = \sigma^2 \left(\sum_{j=0}^{\infty} \psi_j^2 \right)$$

Important remark 14. If the linear process $\{x_t\}_{t \in T}$ is also gaussian then

$$\sqrt{n}(\bar{x}_n - \mu) \sim N \left(0, \sum_{|k| < n} \frac{n - |k|}{n} \gamma_k \right)$$

Remark 24. Based on this we are able to construct asymptotic interval for the sample mean estimator of a stationary time series

5.2 Autocovariance and autocorrelation

For the estimate

$$\hat{\gamma}_k = \frac{1}{n} \sum_{t=1}^{n-k} (x_t - \bar{x}_n)(x_{t+k} - \bar{x}_n)$$

and

$$\hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0}$$

Notice that $\hat{\rho}_k = \hat{\rho}_{-k}$

Important remark 15. $\hat{\gamma}_k$ is biased for γ_k but asymptotically unbiased and more efficient than the unbiased estimator obtained by dividing by $n - k$ rather by n

Theorem 5.2.1. *If $\{x_t\}_{t \in T}$ is the linear process $x_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$, with $\sum_{j=0}^{\infty} |\psi_j| < \infty$, $\varepsilon_t \sim IID(0, \sigma^2)$, and $\mathbb{E}[x_t^4] < \infty$ then*

$$\sqrt{n}(\hat{\rho}(h) - \rho(h)) \xrightarrow{d} N(0, W)$$

where

$$\begin{aligned}\hat{\rho}(h) &= [\hat{\rho}_1, \hat{\rho}_2, \dots, \hat{\rho}_h]^\top \\ \rho(h) &= [\rho_1, \rho_2, \dots, \rho_h]^\top\end{aligned}$$

and W has generic element (Bartlett formula), with $i, j = 1, \dots, h$

$$w_{ij} = \sum_{k=-\infty}^{\infty} (\rho_{k+i}\rho_{k+j} + \rho_{k-i}\rho_{k+j} + 2\rho_i\rho_j\rho_k^2 - 2\rho_i\rho_k\rho_{k+j} - 2\rho_j\rho_k\rho_{k+i})$$

Important remark 16. straightforward algebra shows that

$$w_{ij} = \sum_{k=-\infty}^{\infty} (\rho_{k+i} + \rho_{k-i} - 2\rho_i)(\rho_{k+j} + \rho_{k-j} - 2\rho_j)\rho_k$$

5.2.1 Bartlett formula for a white noise process

If $\{x_t\}$ is a white noise process then

$$w_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

and an approximate confidence interval for ρ_k is

$$I_{0.95}(\rho) = \left(-1.96 \frac{1}{\sqrt{n}}, 1.96 \frac{1}{\sqrt{n}} \right)$$

This can be used as a check for the significance of the single estimated autocorrelation coefficients $\hat{\rho}_k$

5.2.2 Testing the estimated noise sequence

These are all statistical test which test if a group of autocorrelations of a time series are different from zero.² The hypotheses are:

- H_0 : The data is **not correlated** (i.e. the correlations in the population from which the sample is taken are 0, so that any observed correlations in the data result from randomness of the sampling process).
- H_1 : The data exhibit serial correlation

²Instead of testing randomness at each distinct lag, it tests the "overall" randomness based on a number of lags, and is therefore a **portmanteau test**

1. the Box-Pierce: under H_0

$$Q_{BP} = n \sum_{j=1}^h \hat{\rho}(j)^2 \sim \chi^2(h)$$

The Box–Pierce test statistic is a simplified version of the Ljung–Box statistic for which subsequent simulation studies have shown poor performance³ so the following has to be preferred

2. **Ljung-box test**

$$Q_{LB} = n(n+2) \sum_{j=1}^h \frac{\hat{\rho}(j)^2}{n-j} \sim \chi^2(h)$$

where n is the sample size $\rho(j)$ is the sample autocorrelation at lag j and h is the *number of lags being tested*. The Box-Ljung is an omnibus test of independence at all lags up to the one you specify. Hyndman and Athanasopoulos in their forecasting book⁴ recommend using as h :

- $h = 10$ for non seasonal data
- $h = 2m$ for seasonal data, where m is the period of seasonality

Under H_0 the statistics follow asymptotically follows $\chi^2_{(h)}$. For significance level α , the critical region for rejection of the hypothesis of randomness is $Q > \chi^2_{1-\alpha, h}$ where $\chi^2_{1-\alpha, h}$ is the $(1 - \alpha)$ -quantile of the chi-squared distribution with h degrees of freedom.

The Ljung-Box test is widely applied in econometrics and other applications of time series analysis. A similar assessment can be also carried out with the Breusch-Godfrey test and the Durbin–Watson test.

3. MCleod Li

$$Q_{ML} = n(n+2) \sum_{j=1}^h \frac{\hat{\rho}_{y^2}(j)^2}{n-j} \sim \chi^2(h)$$

5.3 Higher order moment estimation

Definition 5.3.1 (Skewness). is

$$S = \frac{\mathbb{E}[(y_t - \mu)^3]}{\sigma^3}$$

with moment estimator

$$\hat{S} = \frac{\frac{1}{n} \sum_{t=1}^n (y_t - \bar{y})^3}{\hat{\sigma}^3}$$

where under H_0 $\hat{\sigma}^2 = \sqrt{\gamma_0}$

$$\sqrt{n}(\hat{S}) \xrightarrow{d} N(0, 6)$$

³Ljung-Box statistic is closer to a $\chi^2(h)$ distribution than is the distribution for the Box-Pierce statistic for all sample sizes. See https://en.wikipedia.org/wiki/Ljung%E2%80%9993Box_test

⁴See <https://robjhyndman.com/hyndsight/ljung-box-test/>

Definition 5.3.2 (Kurtosis). The kurtosis and the excess kurtosis ($K - 3$)

$$K = \frac{\mathbb{E}[(y_t - \mu)^4]}{\sigma^4}$$

moment estimator

$$\hat{K} = \frac{\frac{1}{n} \sum_{t=1}^n (y_t - \bar{y})^4}{\hat{\sigma}^4}$$

finally

$$\sqrt{n}(\hat{K} - 3) \xrightarrow{d} N(0, 24)$$

5.4 Testing normality

Jarque bera test for normality

$$BJ = n \left(\frac{\hat{S}^2}{6} + \frac{(\hat{K} - 3)^2}{24} \right) \sim_{H_0} \chi^2(2)$$

Chapter 6

Transformations

6.1 Stationarity in mean

How to make a series stationary in mean

- Estimate a polynomial trend with a linear filter (weighted average) and then subtract it from the series
- Use an exponential smoothing filter
- Apply an high pass filter (designed in the frequency domain)
- Apply a difference operator (for deterministic and stochastic trends): Differencing $(1 - L)^d X_t$ stationary process

6.1.1 Linear filters for trend estimation

Let

$$X_t = m_t + Z_t$$

where m_t trend component and Z_t stationary zero mean stochastic process. We are interested in estimating m_t .

- for *linear trends and at the interior of the series*, simple 2-sided moving average

$$\hat{m}_t = \frac{1}{2h+1} \sum_{j=-h}^h X_{t-j}$$

- weighted averages

$$\hat{m}_t = \sum_{j=-h}^h w_j X_{t-j}$$

λ	Transformation
-2	$1/x^2$
-1	$1/x$
-0.5	$1/\sqrt{x}$
0	$\log x$
0.5	$\sqrt{x}$
1	x (no transformation)
2	x^2

Table 6.1

6.2 Stationarity in variance

How to make a series stationary in variance? **Box-Cox** transform: it's a series of parametric family of transformation to stabilize the variance of x_t

$$z_t(\lambda) = \begin{cases} \frac{x_t^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \log(x_t) & \lambda = 0 \end{cases}$$

where λ is a parameter to be determined. Given the estimated parameter we apply a transformation to the series to make it stable 6.1.

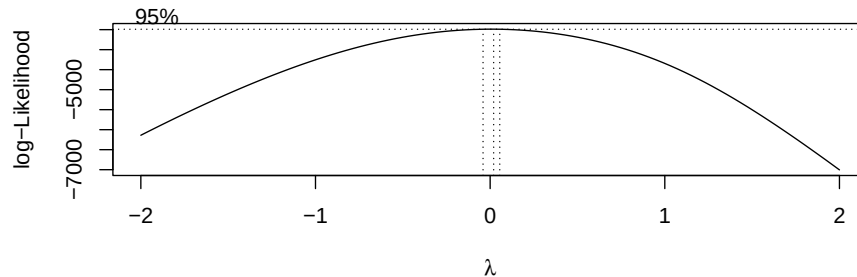
In practise, it is advised to choose the value from the table rather than the precise value if the estimated transformation parameter is close to one of the values of the previous table because the value from the table is simpler to understand. For example

- $\lambda = 1 \implies \sigma$ is constant $\implies$ no need to transform
- $\lambda = 0 \implies \sigma$ is linearly proportional to μ , $\implies$ log transform is needed
- $\lambda = \frac{1}{2} \implies \sigma = k\sqrt{\mu}$ (slow variation) $\implies$

```
library(MASS)

##
## Caricamento pacchetto: 'MASS'
## Il seguente oggetto è mascherato da 'package:lbts':
##
##      deaths

set.seed(1)
x <- exp(rnorm(1000))
bc <- boxcox(x~1) # suggested lambda=0, log(x) to make it normal
```



```
bc$x[which.max(bc$y)]

## [1] 0.0202

# non è l'intervallo di confidenza ma meglio che niente
# da https://stackoverflow.com/questions/34550744
bc$x[bc$y > max(bc$y) - 1/2 * qchisq(.95,1)]

## [1] -0.0202  0.0202

# Reference: Linear Models With R by Julian Faraway Section 8.1.
```

6.3 Box-Jenkins procedure

Iterative procedure for identification, estimation and forecasting of seasonal ARIMA models

1. **Preliminary analysis:** graphical analysis. Plot the series and check for stationarity and seasonality:
 - **stationarity** can be assessed looking at the plot. This should show constant location and scale, suggesting that mean and variance of the time series are constant. Furthermore, *non*-stationarity can also be detected from an autocorrelation plot with a very slow decay
 - **seasonality:** can usually be assessed from an **autocorrelation plot**, a **seasonal subseries** (da gennaio a dicembre si plotta la media dei diversi mesi negli anni) or a **spectral density plot**. If it exists, one need to identify the order for the seasonal autoregressive and seasonal moving average terms.
For many series, the period is known and a single seasonality term is sufficient: for example, for monthly data one would typically include either a seasonal AR 12 term or a seasonal MA 12 term.
2. **Transformation** for making the series **stationary** in mean (detrending or differencing) and in variance (Box-Cox transform)

- **differencing:** Box and Jenkins recommend the differencing approach to achieve stationarity. However, fitting a curve and subtracting the fitted values from the original data can also be used in the context of Box–Jenkins models.
 - **seasonal differencing:** For Box–Jenkins models, one does not explicitly remove seasonality before fitting the model. Instead, one includes the order of the seasonal terms in the model specification to the ARIMA estimation software. However, it may be helpful to apply a seasonal difference to the data and regenerate the autocorrelation and partial autocorrelation plots. This may help in the model identification of the non-seasonal component of the model. In some cases, the seasonal differencing may remove most or all of the seasonality effect
 - **box-cox transform:** see above
3. **Identification:** identify p and q . Once stationarity and seasonality have been addressed, the next step is to identify the order (i.e. the p and q) of the autoregressive and moving average terms. This can be done on the basis of criteria

- **estimated ACF and PACF:** The sample autocorrelation plot and the sample partial autocorrelation plot are compared to the theoretical behavior of these plots.

In practice, the sample autocorrelation and partial autocorrelation functions are random variables and do not give the same picture as the theoretical functions. This makes the model identification more difficult. In particular, mixed models can be particularly difficult to identify. Although experience is helpful, developing good models using these sample plots can involve much trial and error.

However

- an **AR(1)** process, the **sample autocorrelation** function should have an exponentially decreasing appearance
- **higher-order AR:** processes are often a mixture of exponentially decreasing and damped sinusoidal components. The sample autocorrelation needs to be supplemented with a **partial autocorrelation plot**. The partial autocorrelation of an **AR(p)** process becomes zero at lag $p + 1$ and greater, so we examine the sample partial autocorrelation function to see if there is *evidence of a departure from zero*: this is usually determined by placing a 95% confidence interval on the sample partial autocorrelation plot (which is approximately $\pm 2/\sqrt{n}$ with n denoting the sample size)
- **MA(q):** the autocorrelation function becomes zero at lag $q + 1$ and greater, so we examine the sample autocorrelation function to see where it essentially becomes zero, as above.
The sample partial autocorrelation function is generally not helpful for identifying the order of the moving average process.

Table 6.2 summarizes how one can use the sample autocorrelation

Shape	Indicated model
Exponential, decaying to zero	Autoregressive model. Use the partial autocorrelation plot to identify the order of the autoregressive model.
Alternating positive and negative, decaying to zero	Autoregressive model. Use the partial autocorrelation plot to help identify the order.
One or more spikes, rest are essentially zero (or close to zero)	Moving average model, order identified by where plot becomes zero.
Decay, starting after a few lags	Mixed autoregressive and moving average (ARMA) model.
All zero or close to zero	Data are essentially random.
High values at fixed intervals	Include seasonal autoregressive term.
No decay to zero (or it decays extremely slowly)	Series is not stationary.

Table 6.2: Autocorrelation function shape

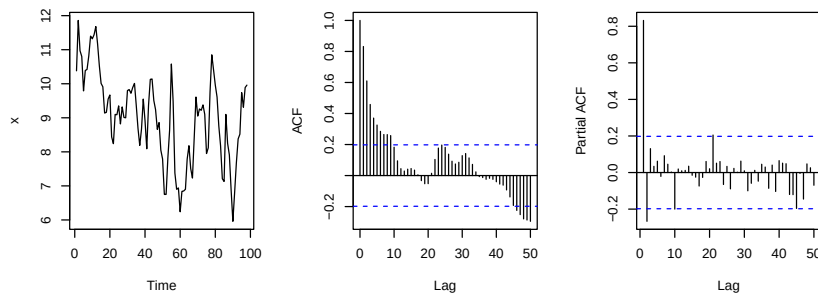
function for model identification.¹

- using **information criteria** (Brockwell Davis usano come metodo primario l'AIC):

$$AIC(p, q) = \log \hat{\sigma}^2 + 2 \frac{p+q}{n}$$

$$BIC(p, q) = \log \hat{\sigma}^2 \log n \frac{p+q}{n}$$

```
library(lbts)
ts_plot(lake)
```



¹Finally, Hyndman & Athanasopoulos suggest the following

- The data may follow an ARIMA(p,d,0) model if the ACF and PACF plots of the differenced data show the following patterns:
 - * the ACF is exponentially decaying or sinusoidal;
 - * there is a significant spike at lag p in PACF, but none beyond lag p.
- The data may follow an ARIMA(0,d,q) model if the ACF and PACF plots of the differenced data show the following patterns:
 - * the PACF is exponentially decaying or sinusoidal;
 - * there is a significant spike at lag q in ACF, but none beyond lag q.

```
# https://ionides.github.io/531w20/05/notes05-annotated.pdf
aic_table <- function(data,P=5,Q=5){
  table <- matrix(NA,(P+1),(Q+1))
  for(p in 0:P) {
    for(q in 0:Q) {
      # assuming stationarity ... 0 for I component
      table[p+1,q+1] <- arima(data,order=c(p,0,q))$aic
    }
  }
  dimnames(table) <- list(
    paste("AR",0:P, sep=""),
    paste("MA",0:Q, sep=""))
  table
}

(aics <- aic_table(lake))

##          MA0   MA1   MA2   MA3   MA4   MA5
## AR0 333.3 253.3 228.9 220.1 220.5 220.7
## AR1 217.2 212.5 214.5 215.9 217.3 219.3
## AR2 213.3 214.5 216.4 217.5 218.3 220.2
## AR3 214.0 215.8 217.7 219.2 220.2 222.2
## AR4 215.6 217.2 218.4 219.8 221.3 223.0
## AR5 217.6 218.3 220.3 221.3 223.3 224.9

# lowest AIC is obtained with Arma(1,1)

# however we should also look at diagnostic plots for these models (eg look at
# residuals of estimation).

# Are there outliers:
```

4. Estimation (ML): estimating the parameters for Box-Jenkins models involves numerically approximating the solutions of nonlinear equations. Maximum likelihood estimation is generally the preferred technique

```
(lake_arma11 <- arima(lake, order=c(1,0,1)))

##
## Call:
## arima(x = lake, order = c(1, 0, 1))
##
## Coefficients:
##          ar1      ma1  intercept
##          0.745  0.321         9.055
## s.e.    0.078  0.114         0.350
##
## sigma^2 estimated as 0.475:  log likelihood = -103.2,  aic = 212.5
```


5. **Diagnostic checking:** on the estimated coefficients and on the basis of the residual analysis.

- **estimated coefficients:** credo si riferisca a se dei coefficienti di un modello specificato non sono diversi da 0
- **residuals:** the error term is assumed to follow the assumptions for a stationary univariate process. The residuals should be *white noise* (or independent when their distributions are normal) drawings from a fixed distribution with a constant mean and variance.

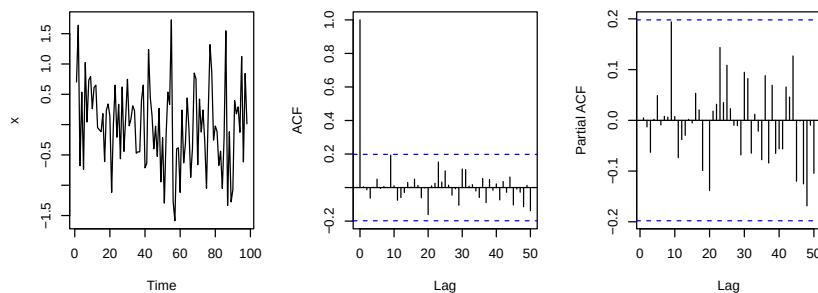
Way to assess if the residuals from the Box–Jenkins model follow the assumptions:

- do graphics (including an autocorrelation plot) of the residuals
- value of the Box–Ljung statistic.

If these assumptions are not satisfied, one needs to fit a more appropriate model. That is, go back to the model identification step and try to develop a better model. Hopefully the analysis of the residuals can provide some clues as to a more appropriate model

```
## look at residuals: they seems not autocorrelated looking at ACF and white
## noise (within the confidence interval)
```

```
ts_plot(lake_arma11$residuals)
```



```
# fuck hyndyman facciamo su tutto e ignoriamo seasonality
## boxlung test # set looking at periodicity i guess
```

```
library(stats)
sapply(1:15, function(l)
  Box.test(lake_arma11$residuals,
    type = "L", # specificarlo altrimenti fa il box-pierce
    lag=l)$p.value)
```

```
## [1] 0.9626 0.9904 0.9342 0.9800 0.9833 0.9946 0.9983 0.9995 0.8488 0.9015 0.9054 0.921
## [13] 0.9464 0.9640 0.9780
```

6. Forecasting

If one step fails, then the procedure starts from the beginning.

Part II

Advanced time series

Chapter 7

Advanced intro

7.1 Introduction

Definition 7.1.1 (Stochastic process). Considering $\{Y_t\}_{t \in T}$ as stochastic process with each single random variable being a measurable function of type $Y_t : (\Omega, \mathcal{A}, \mathbb{P}) \rightarrow (\mathbb{R}, \beta(\mathbb{R}), F_y)$ (so being measurable $Y_t^{-1}(B) \in \mathcal{A}, \forall B \in \beta(\mathbb{R})$).

Important remark 17. If $\mathcal{F}$ is the natural filtration of the process (a collection of $\mathcal{F}_t = \sigma\{Y_0, Y_1, \dots, Y_t\}$ with $\mathcal{F}_t \subset \mathcal{F}_{t+1}$ and $\mathcal{F} = \{\mathcal{F}_t\}_{t \in T}$) we can define

- **unconditional expectation** of Y_t

$$\mathbb{E}[Y_t] = \int_{\mathbb{R}} y \, dF_y(y) = \mu_t$$

unconditional mean depend on time (when the process is stationary this is not the case, it's a constant).

- in **conditional** expectation we condition on some information, and in time series this is the past, constituted by the filtration $\mathbb{E}[Y_t | \mathcal{F}_{t-1}]$

Remark 25 (Main idea of the course). If in a basic course we deal with *stationary* time series for which mean and variance does not change with time; in this course we drop this assumption and we're interested in models with time-varying mean or time-varying variances, conditional to past information.

We're interested in dynamics of the time varying parameters (be them mean or variance), how they do vary in time (and thus say how the conditional density varies in time); basically we want to recover the time varying parameter based on observations $\{y_1, \dots, y_T\}$ of the time series.

Definition 7.1.2 (Conditional density of a stochastic process). Considering a stochastic process $\{Y_t\}_{t \in \mathbb{Z}}$ formed by a sequence of rvs, and its natural filtration up to $\mathcal{F}_{t-1} = \sigma\{Y_{t-1}, Y_{t-2}, \dots\}$, the conditional density is the parametric distribution given the information at time $t - 1$

$$Y_t | \mathcal{F}_{t-1} \sim p(Y_t, \lambda_t; \theta)$$

where

- p is some density (eg Gaussian, t_ν , Cauchy which is a t with $\nu = 1$); so given the past information, y_t is generated by a distribution Y_t which is actually unknown, but we will assume (as in classic parametric inference) that this is known (we specify it to go with estimation);
- p is function of y_t as any distribution, but depends as well on a **time varying parameter** λ_t , which characterize p .
The time varying parameter can be the conditional expectation ($\mathbb{E}[Y_t|\mathcal{F}_{t-1}] = \lambda_t$) or conditional variance ($\text{Var}[Y_t|\mathcal{F}_{t-1}] = \lambda_t$).
Our problem is to specify a model for λ_t as a function of $\mathcal{F}_{t-1}$
- $\theta \in \Theta \subset \mathbb{R}^m$ are **static parameters** which are needed to characterize the density (eg ν for a t distribution, σ for Gaussian, or constant parameters that are used to specify the evolution of λ_t)
- the only thing we observe is $y_t, y_{t-1}, \dots$ and we want to recover/estimate all the other components. Thus models are *specified*, dynamic/time varying parameters recovered and *static parameters* estimated mainly by likelihood methods.

NB: reference Durbin e
Koopman cap 1 e 2

Definition 7.1.3 (State space and reduced form). Any model specified through a conditional distribution can be recast using two equations (**state space form**) where:

1. the first is observation or **measurement equation**, involving the only variable we're able to observe, y_t ;
2. a **transition equation** that regulates how the unobserved component of our interest λ_t updates itself;

If then we put the measurement equation and transition equation all into 1 by plugging transition equation in measurement equation we obtain the **reduced form**

Important remark 18 (Observation vs parameter driven model). In any model we see we need to specify *the way the time varying parameter λ_t get updated*:

- if the dynamics depends on an error term (aside/independently from the past) too (with its exogenous parameters) we're in a **parameter driven model**
- if the dynamics of the time varying parameter depends on past observation y_{t-1}/λ_{t-1} , but *no additional error terms*, these are called **observation driven model**

This has implication for estimation, indeed

- for **parameter-driven models**
 - for linear and gaussian error models we can use likelihood inference with the Kalman filter: this models are easier to specify and amenable to estimate (Kalman filter easy algorithm and optimal in case of gaussianity)

- for non-linear or non-Gaussian models we can use numerical methods (score driven model I guess) : kalman filter still can be applied but it's not optimal in case of non-gaussian error (eg gaussian AR(1) buried in student t noise with $\nu = 3$): one has too much sensitivity to spikes of t-error and thus estimated λ_t will be too sensitive to extreme observations
- for **observation driven models** where time varying parameter depends on past only, conditional *likelihood is available at each time in closed-form* and we can apply MLE to estimate parameters (compared to parameter driven: PRO easier to estimate, CONS “harder” as models, regularity conditions for MLE hard to establish for non-linear filters)

Example 7.1.1 (Model for conditional time-varying mean). An example is

$$Y_t | \mathcal{F}_{t-1} \sim N(Y_t, \mu_t; \theta) \quad (7.1)$$

with Gaussian conditional density having mean μ_t and static scale σ that is

$$f_{Y_t | \mathcal{F}_{t-1}}(y_t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\left(-\frac{1}{2} \left(\frac{y_t - \mu_t}{\sigma}\right)^2\right)}$$

we have that

- $\lambda_t = \mu_t$ is the time varying parameter (mean), whose evolution depends on some set parameter $\mathbf{v}$. For example

$$\lambda_{t+1} = \omega + \alpha\lambda_t + \beta(y_t - \lambda_t) \quad (7.2)$$

where $\mathbf{v} = (\omega, \alpha, \beta)$

- $\theta = (\sigma, \mathbf{v}) \in \Theta \subset \mathbb{R}^m$ is the static parameter (in this case $m = 3 + 1$)
- an equivalent representation for models with conditional distribution of our process following a distribution $Y_t | \mathcal{F}_{t-1} \sim p$ (for some distribution p) with time varying mean is the following:

$$Y_t = \lambda_t + \sigma\varepsilon_t, \quad \varepsilon_t \sim \text{IID}(0, 1)$$

this is the so called *signal plus noise decomposition* with λ_t latent unobserved signal and ε_t noise component.

Example 7.1.2 (Model for conditional time-varying variance). If the model is

$$Y_t | \mathcal{F}_{t-1} \sim t_\nu(Y_t, \sigma_t; \theta)$$

differently from 7.1 we have

- t_ν is the conditional distribution with constant location μ so

$$f_{Y_t | \mathcal{F}_{t-1}} = c_\nu \left(1 + \frac{(y_t - \mu)^2}{\nu \sigma_t^2} \right)^{-\frac{\nu+1}{2}}$$

with c_ν a constant associated to the student t density for ν degrees of freedom

- $\lambda_t = \sigma_t$ is now the dynamic/time varying parameter whose evolution depends on some set of parameters $\mathbf{v}$, eg we could have

$$\sigma_t^2 = \omega + \alpha y_{t-1}^2$$

Note that $\lambda_t = \sigma_t$ is not actually the variance of the distribution (which for t is $\sigma_t \frac{\nu}{\nu-2}$, with $\nu > 2$ for the variance to exist) but proportional to it

- μ is static here
- $\theta = (\mu, \nu, \mathbf{v}) \in \Theta \subset \mathbb{R}^m$ with the first two parameters characterizing the student t distribution and $\mathbf{v} = (\omega, \alpha)$ the parameters regarding the dynamic update (credo sia un arch 1 model)
- an equivalent representation for models with time varying variance:

$$Y_t = \mu + \lambda_t \varepsilon_t \quad \varepsilon_t \sim \text{IID}(0, 1)$$

this is the so-called *location-scale* representation (with μ location and λ_t scale)

7.2 Model seen

If we consider a general measurement equation of the state space form as the following (a signal and a noise both depending on time):

$$Y_t = \mu_t + \varepsilon_t$$

we can encapsulate all the models we'll see.

TODO: eventuali suoi errori da copia incolla

Model on varying mean Can be achieved

1. AR(1): $\mu_t = \phi y_{t-1}$ with $\varepsilon_t \sim \text{WN}(0, \sigma^2)$
2. RW: $\mu_t = y_{t-1}$ with $\varepsilon_t \sim \text{IID}(0, \sigma^2)$
3. MA(1): $\mu_t = \theta \varepsilon_{t-1}$ with $\varepsilon_t \sim \text{WN}(0, \sigma^2)$
4. LLM: $\mu_{t+1} = \mu_t + \eta_t$ with $\varepsilon_t \sim \text{N}(0, \sigma_\varepsilon^2)$ and $\eta_t \sim \text{N}(0, \sigma_\eta^2)$
5. AR(1) + noise (ARMA(1,1)): $\mu_{t+1} = \phi \mu_t + \eta_t$ with $\varepsilon_t \sim \text{N}(0, \sigma_\varepsilon^2)$ and $\eta_t \sim \text{N}(0, \sigma_\eta^2)$
6. IFKF: $\mu_{t+1|t} = \omega + \phi \mu_{t|t-1} + k v_t$, with $v_t = y_t - \mu_{t|t-1}$ $\eta_t \sim \text{N}(0, \sigma_\eta^2)$
7. score driven model: $\mu_{t+1|t} = \omega + \phi \mu_{t|t-1} + u_t$ with u_t score and ε_t heavy tailed

Model on varying scale/variance Achieved through

1. ARCH: $\mu_t = 0$, $\varepsilon_t = \sigma_t z_t$, with $z_t \sim \text{IID}(0, 1)$, $\sigma_t^2 = \omega + \alpha y_{t-1}^2$
2. GARCH $\mu_t = 0$, $\varepsilon_t = \sigma_t z_t$, with $z_t \sim \text{IID}(0, 1)$, $\sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2$

7.2.1 Zero mean AR(1)

Example 7.2.1 (Zero mean AR(1)). Letting

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t \quad \varepsilon_t \sim \text{WN}(0, \sigma^2) \\ \mu_t &= \phi y_{t-1} \end{aligned}$$

Plugging in we obtain

$$Y_t = \phi Y_{t-1} + \varepsilon_t$$

It's an AR(1) with zero mean, which is a specific form ($\omega = 0$) stationary AR(1) process

$$Y_t = \underbrace{\omega}_{=0} + \phi Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

ω is not the expected value of the process which is $\frac{\omega}{1-\phi}$ (with $|\phi| < 1$ implying stationarity and that expectation is also finite).

By definition the WN is defined as

$$\begin{aligned} \mathbb{E}[\varepsilon_t] &= 0 \\ \mathbb{E}[\varepsilon_t^2] &= \sigma^2 < \infty \\ \mathbb{E}[\varepsilon_t \varepsilon_s] &= 0, \forall t \neq s \end{aligned}$$

Here for

- unconditional¹ and conditional² mean we have

$$\mathbb{E}[Y_t] = 0$$

$$\mathbb{E}[Y_t | \mathcal{F}_{t-1}] = \phi \mathbb{E}[Y_{t-1} | \mathcal{F}_{t-1}] + \mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}] = \phi Y_{t-1} + \mathbb{E}[\varepsilon_t] = \phi Y_{t-1}$$

Note that conditional expectation is dynamic/time varying (depends on t as index via Y_{t-1}). So we can say an *AR(1) model is a model for the conditional mean* (or same said for the time-varying mean or dynamic mean) because it gives us a time varying conditional mean of Y_t .

Furthermore conditional expectation is our *best prediction*

- unconditional³ and conditional variance are

$$\text{Var}[Y_t] = \frac{\sigma^2}{1 - \phi^2}$$

$$\text{Var}[Y_t | \mathcal{F}_{t-1}] = \mathbb{E}[(Y_t - \mathbb{E}[Y_t | \mathcal{F}_{t-1}])^2 | \mathcal{F}_{t-1}] = \mathbb{E}[\varepsilon_t^2 | \mathcal{F}_{t-1}] = \sigma^2$$

In the latter case (going from unconditional to conditional) what does information brings in? there is an *improvement* in terms of variability (since $\phi^2 < 1$). Differently from the conditional expected value, the conditional variance is not time varying/a function of the past information.

In general, in linear models $\text{Var}[Y_t | \mathcal{F}_{t-1}] = \sigma^2$.

Conditional variance is the *variability* of our *best prediction*.

Linear models such as AR(1) are *not* good models for time-varying variance, they do not capture any time variation in the conditional variance.

¹To derive the unconditional mean of an AR(1) process either assume stationarity or use the wold representation (do it as an exercise)

²easier to calculate

³Calculated again assuming stationarity or by using wold representation

- regarding autocovariance:

$$\begin{aligned}\gamma_1 &= \mathbb{E}[Y_t \cdot Y_{t+1}] = \mathbb{E}[Y_t \cdot (\phi Y_t + \varepsilon_{t+1})] \\ &= \phi \gamma_0 + \underbrace{\mathbb{E}[Y_t \cdot \varepsilon_{t+1}]}_{=0} = \phi \gamma_0 \\ \gamma_2 &= \mathbb{E}[Y_t \cdot Y_{t+2}] = \mathbb{E}[Y_t \cdot (\phi Y_{t+1} + \varepsilon_{t+2})] = \phi \gamma_1\end{aligned}$$

So in general, for the autocovariance

$$\gamma_k = \phi \gamma_{k-1} \iff \gamma_k = \phi^k \gamma_0$$

with $\gamma_k = \gamma_{-k}$ and $\gamma_0 = \frac{\sigma^2}{1-\phi^2}$

- for autocorrelation

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \phi^{|k|}$$

when plotting the autocorrelation (for $\phi > 0$) it never goes to zero but is linearly decreasing

- The Wold representation of AR(1) (assuming $|\phi| < 1$, condition for existence of the inverse of the polynomial in the lag operator)

$$(1 - \phi L)Y_t = \varepsilon_t Y_t = \frac{1}{1 - \phi L} \varepsilon_t = \sum_{j=0}^{\infty} \phi^j L^j \varepsilon_t$$

7.2.2 Random walk

NB: fuck yea

Example 7.2.2 (random walk). letting

$$\begin{aligned}y_t &= \mu_t + \varepsilon_t \quad \varepsilon_t \sim \text{IID}(0, \sigma^2) \\ \mu_t &= y_{t-1}\end{aligned}$$

Plugging in we obtain

$$y_t = y_{t-1} + \varepsilon_t$$

So this case is that $\{Y_t\}_{t \in T}$ is a **random walk process**.

A RW can be seen also like by putting

$$\begin{aligned}Y_0 &= 0 \\ Y_1 &= Y_0 + \varepsilon_1 = \varepsilon_1 \\ Y_2 &= Y_1 + \varepsilon_2 = \varepsilon_1 + \varepsilon_2 \\ Y_t &= \sum_{k=1}^t \varepsilon_k\end{aligned}$$

Here

- the unconditional moments

$$\begin{aligned}\mathbb{E}[Y_t] &= 0 \\ \text{Var}[Y_t] &= t \cdot \sigma^2\end{aligned}$$

and the latter implies that the process is not weakly stationary (variance depending on t). Finally

$$\begin{aligned}\mathbb{E}[Y_t \cdot Y_{t+k}] &= t \cdot \sigma^2, \quad k > 0 \\ \mathbb{E}[Y_t \cdot Y_{t-k}] &= (t-k) \cdot \sigma^2, \quad k > 0\end{aligned}$$

Covariance function is non symmetric (the process is not stationary so we're not surprised ...). The autocovariance function is linearly decreasing for k increasing (but never goes to 0)

- the unconditional moments; to calculate conditional moments

$$\mathbb{E}[Y_t | \mathcal{F}_{t-1}] = \mathbb{E}[Y_{t-1} | \mathcal{F}_{t-1}] + \mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}] = Y_{t-1}$$

and we conclude that RW is a martingale process.

- If we take the differences

$$(1-L)Y_t = Y_t - Y_{t-1} = \varepsilon_t$$

which is a *martingale difference sequence* (IID in this case as here ε_t is IID)

Definition 7.2.1 (MDS). A stochastic process X is a martingale difference sequence if its expectation with respect to the past is zero. That is X_t is an MDS if it satisfies the following two conditions for any t

$$\begin{aligned}\mathbb{E}[|X_t|] &< \infty \\ \mathbb{E}[X_t | \mathcal{F}_{t-1}] &= 0\end{aligned}$$

By construction, this implies that if Y_t is a martingale, then $X_t = Y_t - Y_{t-1}$ will be an MDS. Hence the name

Remark 26. The MDS is an extremely useful construct in modern probability theory because it implies much milder restrictions on the memory of the sequence than independence, yet most limit theorems that hold for an independent sequence will also hold for an MDS.

7.2.3 MA(1)

Example 7.2.3. let

$$\begin{aligned}y_t &= \mu_t + \varepsilon_t \\ \mu_t &= -\theta \varepsilon_{t-1}\end{aligned}$$

with $\varepsilon_t \sim \text{WN}(0, \sigma^2)$. Plugging in we obtain

$$y_t = \varepsilon_t - \theta \varepsilon_{t-1}$$

NB: naa

TODO: qui fa casino col MA(1) e sto - di merda

which is a MA(1).

$$\mathbb{E}[Y_t] = 0$$

$$\rho_k = \begin{cases} 1 & k = 0 \\ \frac{-\theta}{1+\theta^2} & |k| = 1 \\ 0 & |k| > 1 \end{cases}$$

Here the autocorrelation if $\theta < 0$ is decreasing and becomes zero after the q-th lag

$$Y_t = (1 - \theta L)\varepsilon_t$$

$$\varepsilon_t = \sum_{j=0}^{\infty} \theta^j y_{t-j} = y_t + \theta y_{t-1} + \dots$$

$$y_t = - \sum_{j=1}^{\infty} \theta^j y_{t-j} + \varepsilon_t$$

and thus

$$\mathbb{E}[Y_t | \mathcal{F}_{t-1}] = - \sum_{j=1}^{\infty} \theta^j y_{t-j}$$

which can be done if $|\theta| < 1$ (condition of invertibility)

Remark 27. above some processes already seen in the past, below we start looking at new processes

7.2.4 LLM (local level model): random walk + noise

Definition 7.2.2 (LLM, Random walk + noise). If we specify the model like

$$y_t = \mu_t + \varepsilon_t \quad \varepsilon_t \sim \text{NID}(0, \sigma_\varepsilon^2)$$

$$\mu_{t+1} = \mu_t + \eta_t \quad \eta_t \sim \text{NID}(0, \sigma_\eta^2)$$

with

- ε_t and η_t independent for all $t, s \in \mathbb{Z}$ ($\mathbb{E}[\varepsilon_t \eta_s] = 0$, so uncorrelated but also gaussian so independent);
- $\mu_1 \sim N(a_1, P)$ (an initialization step, can be a constant parameter actually or have some distribution with known properties)

we have that

- y_t has a *random walk + noise* shape (this is what this process is) where μ_t is a random walk
- the time varying parameter μ_{t+1} depends on its own *source of error* η_t , with its distribution and parameters, but *not on past value of the process* (eg y_t) which makes this a **parameter driven model**

Remark 28. We now start studying properties of the model; in time series mainly means autocovariance, autocorrelation and in essence the dependence relation among variables $\{Y_t\}$, and between $\{Y_t\}$ and $\{\mu_t\}$ (here we have actually two stochastic processes).

In practice we analyze:

- the behavior at limit cases
- workout the moments and especially autocovariance function

Limit cases As special cases, if

- $\sigma_\varepsilon^2 = 0$ we obtain random walk

$$\begin{aligned} y_t &= \mu_t \\ \mu_{t+1} &= \mu_t + \eta_t \end{aligned}$$

thus $y_t \sim RW$

- $\sigma_\eta^2 = 0$ we have a independent white noise with/plus a constant mean μ_t

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t \\ \mu_{t+1} &= \mu_t = c \end{aligned}$$

thus $y_t \sim \text{NID}(c, \sigma_\eta^2)$

Moments To obtain it we rewrite the process as a single equation (the *reduced form*) in order to apply what needed to get autocovariance. By replacing the transition equation in the measurements equation we have

$$\begin{aligned} y_{t+1} &= \mu_{t+1} + \varepsilon_{t+1} \\ &= \mu_t + \eta_t + \varepsilon_{t+1} \end{aligned}$$

from which by subtracting $y_t = \mu_t + \varepsilon_t$ from both members

$$\begin{aligned} y_{t+1} - y_t &= \mu_t + \eta_t + \varepsilon_{t+1} - \mu_t - \varepsilon_t \\ &= \eta_t + \varepsilon_{t+1} - \varepsilon_t \end{aligned}$$

So in general if $\Delta = (1 - L)$ (with L lag operator and $LX_t = X_{t-1}$ and $L^k X_t = X_{t-k}$) we can write the reduced form of a LLM (going back of 1 wrt to previous btw ...) as

$$\begin{aligned} \Delta y_t &= \eta_{t-1} + \varepsilon_t - \varepsilon_{t-1} \\ &= \varepsilon_t - \varepsilon_{t-1} + \eta_{t-1} \end{aligned}$$

So above we see that the (lagged) reduced form of a LLM is a MA(1). We can now easily calculate (unconditional) expected value, variance, covariance of lag 1

$$\begin{aligned} \mathbb{E}[\Delta y_t] &= \mathbb{E}[\varepsilon_t - \varepsilon_{t-1} + \eta_{t-1}] = \mathbb{E}[\varepsilon_t] - \mathbb{E}[\varepsilon_{t-1}] + \mathbb{E}[\eta_{t-1}] = 0 \\ \mathbb{E}[(\Delta y_t)(\Delta y_t)] &= \mathbb{E}[(\varepsilon_t - \varepsilon_{t-1} + \eta_{t-1})(\varepsilon_t - \varepsilon_{t-1} + \eta_{t-1})] = \sigma_\varepsilon^2 + \sigma_\varepsilon^2 + \sigma_\eta^2 = 2\sigma_\varepsilon^2 + \sigma_\eta^2 \\ \mathbb{E}[(\Delta y_t)(\Delta y_{t-1})] &= \mathbb{E}[(\varepsilon_t - \varepsilon_{t-1} + \eta_{t-1})(\varepsilon_{t-1} - \varepsilon_{t-2} + \eta_{t-2})] = -\sigma_\varepsilon^2 \\ \mathbb{E}[(\Delta y_t)(\Delta y_{t-2})] &= \mathbb{E}[(\varepsilon_t - \varepsilon_{t-1} + \eta_{t-1})(\varepsilon_{t-2} - \varepsilon_{t-3} + \eta_{t-3})] = 0 \end{aligned}$$

Thus to summarize expected value and autocovariance (with variance included)

$$\mathbb{E}[\Delta y_t] = 0$$

$$\gamma_k = \begin{cases} 2\sigma_\varepsilon^2 + \sigma_\eta^2 & k = 0 \\ -\sigma_\varepsilon^2 & k = 1, -1 \\ 0 & |k| > 1 \end{cases}$$

The implication of the features above is that the (lagged) reduced form of a LLM is a MA(1) ($X_t = \varepsilon_t - \theta\varepsilon_{t-1}$). Thus is stationary and invertible if its underlying $|\theta| < 1$

Finally, for autocorrelation we have in general that $\rho_k = \frac{\gamma_k}{\gamma_0}$. Here:

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{-\sigma_\varepsilon^2}{2\sigma_\varepsilon^2 + \sigma_\eta^2} = -\frac{1}{\frac{\sigma_\eta^2}{\sigma_\varepsilon^2} + 2} = -\frac{1}{q + 2}$$

Thus for autocorrelation, finally

$$\rho_k = \begin{cases} 1 & k = 0 \\ -\frac{1}{q+2} & k = 1, -1 \\ 0 & |k| > 1 \end{cases}$$

The $q = \frac{\sigma_\eta^2}{\sigma_\varepsilon^2}$

- is called signal-to-noise ratio (ratio between the variance of the signal μ_t , and the variance of the noise ε_t)
- tells us how much of the overall variability that we observed in a series is explained by variance of signal or of noise.
- has range $0 < q < +\infty$ and thus $\rho_1 = -\frac{1}{q+2}$ is not in $[-1, 1]$ but in $-1/2 < \rho_1 < 0$ (for $q \rightarrow \infty$ and $q \rightarrow 0$ respectively)

Thus

- the reduced form of δy_t is a MA(1) model with $-1/2 < \rho_1 < 0$. So considering the MA(1)'s ρ_1 it must be that

$$-\frac{1}{2} < -\frac{\theta}{(1 + \theta^2)} < 0 \implies \theta \in (0, 1)$$

so it's a MA(1) with restriction on θ

- the previous point implies that (if we not consider the differencing) the reduced form of y_t is an ARIMA(0, 1, 1) model with $-1/2 < \rho_1 < 0$ and $\theta \in (0, 1)$

7.2.5 AR(1) + noise (ARMA(1,1))

NB: fuck yeah

Remark 29. Similar to the previous one but changes only 1 parameter (from random walk + signal, for which $\phi = 1$, to a an AR(1) stationary)

Definition 7.2.3. Model is specified like

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t, & \varepsilon_t &\sim \text{NID}(0, \sigma_\varepsilon^2) \\ \mu_{t+1} &= \phi \mu_t + \eta_t, & \eta_t &\sim \text{NID}(0, \sigma_\eta^2) \end{aligned}$$

with

- ε_t and η_t independent for all $t, s \in \mathbb{Z}$ ($\mathbb{E}[\varepsilon_t \eta_s] = 0$, so uncorrelated but also gaussian so independent);
- $\mu_1 \sim N(a_1, P)$ (an initialization step, can be a constant parameter actually or have some distribution with known properties)
- the only difference wrt to previous is ϕ . This is an AR(1) + noise (already in state space form), where the signal μ_t is an AR(1) process.
- The reduced form is an ARMA(1,1) (quite involved to get) with restriction on the parameters
- This again is a parameter driven model: here the *static parameters* are $\phi, \sigma_\eta^2, \sigma_\varepsilon^2$.
If $|\phi| < 1$ the process is *stationary*; but needs to be $\phi \neq 0$ to be identifiable.

Remark 30 (Special cases). If:

- $\sigma_\varepsilon^2 = 0$, we obtain a pure AR(1) process
- $\sigma_\eta^2 = 0$, then $y_t \sim N(c, \sigma_\varepsilon^2)$

```
set.seed(1)
library(lbts)

# simulating it
n <- 300

# variances of the error terms
sigma_epsilon <- 1 # variance of the noise
sigma_eta <- 1 # scale (sd) of error of the signal

# error terms
eta <- sigma_eta * rnorm(n) # sigma cdot N(0,1)
epsilon <- sigma_epsilon * rnorm(n)

# AR parameter (the bigger the more persistent)
# if negative then then a lot erratic
phi <- 0.8
mu <- 0 # unconditional mean of an AR(1) process with no intercept (its mu[1])
y <- 0

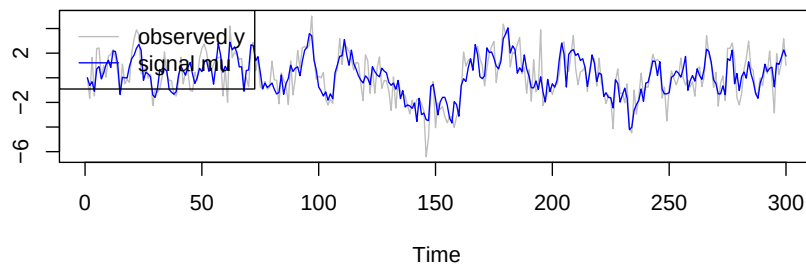
for(t in 1:(n-1)){
  # we start updating the prediction of next signal with initialization parameters
  mu[t+1] <- phi * mu[t] + eta[t]
```

```

# obtain the observed signal + noise
y[t+1] <- mu[t+1] + epsilon[t+1]
}

## plot observed y and underlying signal together
lbs::mts_plot(list(y, mu), col = c("grey", "blue"))
legend("topleft", col = c("grey", "blue"), lty=1,
       legend = c("observed y", "signal mu"))

```



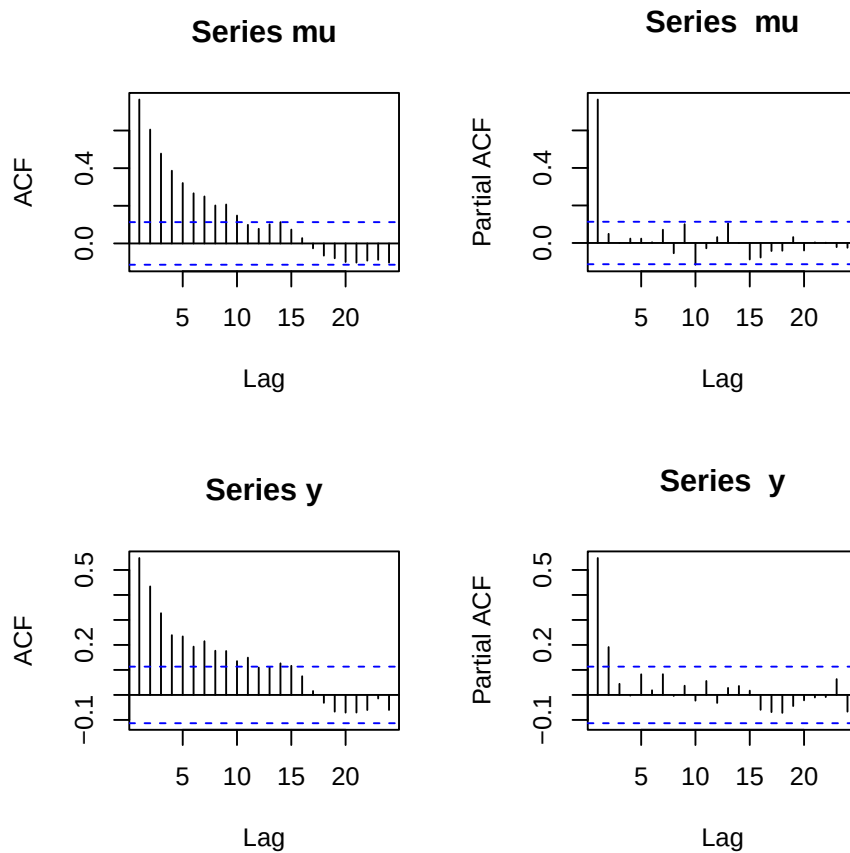
check the correlation structure of the simulated process to be sure that we are properly simulating an AR(1) signal plus noise process.

```

par(mfrow=c(2,2))
# Looking at the acf and pacf of mu should be those of an AR(1) process.
acf(mu)
pacf(mu)

# acf and pacf of y are those of an ARMA(1,1) process, as expected
acf(y)
pacf(y)

```

```
## lei dice indeed, the acf and pacf of mu are those of an AR(1) process, while the acf
# and pacf of y are those of an ARMA(1,1) process, as expected
```

```
## mah non so rick
```

To simulate data are generated by a heavy tailed distribution (instead of Gaussina, this does break the assumption of KF which is optimal (in MMSE sense) for linear Gaussian models)

```
nu <- 3
e <- sqrt((nu-2)/nu) * sigma_epsilon * rt(n, nu)
```

7.2.6 Local linear trend model (LLTM)

Remark 31. The following is not in state space form yet since three equation **NB: naa** are displayed

Definition 7.2.4 (Local linear trend model). Defined as a LLM plus a slope

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t, & \varepsilon_t &\sim \text{NID}(0, \sigma_\varepsilon^2) \\ \mu_{t+1} &= \beta_t + \mu_t + \eta_t, & \eta_t &\sim \text{NID}(0, \sigma_\eta^2) \\ \beta_{t+1} &= \beta_t + \xi_t, & \psi_t &\sim \text{NID}(0, \sigma_\xi^2) \end{aligned}$$

with all error terms independent at all lags and leads.

The *slope* β_t is defined as a random walk

Remark 32 (Special cases). If

- $\sigma_\xi^2 = 0$ we have a random walk with constant drift $\beta_1 x$
- if also $\sigma_\eta^2 = 0$ we have a trend straight line with slope β_1 and intercept μ_1
- $\sigma_\xi^2 > 0, \sigma_\eta^2 = 0$ we have a trend smooth curve on IRW(integrated random walk)

7.2.7 Innovation form of the Kalman filter

Remark 33. Now we see **observation driven model**.

With notation $\mu_{t+1|t}$ we emphasize that given the information at time t , $\mathcal{F}_t$, μ_{t+1} is known

Example 7.2.4 (IFKF). The acronym is “innovation form of Kalman filter” and is

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t \\ \mu_{t+1|t} &= \omega + \phi\mu_{t|t-1} + kv_t \end{aligned}$$

where $v_t = y_t - \mu_{t|t-1}$ is what is called *one step ahead innovation error*
In this specification/representation μ_t just depends on past observations

7.2.8 ARCH(1)

Remark 34. Now model for time varying variance

Example 7.2.5 (ARCH(1)).

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t \\ \mu_t &= 0 \\ \varepsilon_t &= \sigma_t z_t \\ \sigma_t^2 &= \omega + \alpha y_{t-1}^2 \end{aligned}$$

so summarizing

$$\begin{aligned} y_t &= \sigma_t z_t \\ \sigma_t^2 &= \omega + \alpha y_{t-1}^2 \end{aligned}$$

with $z_t \sim \text{IID}(0, 1)$.

If we calculate the conditional expectation (σ_t is F_{t-1} measurable by its definition)

$$\mathbb{E}[Y_t | \mathcal{F}_{t-1}] = \mathbb{E}[\sigma_t z_t | \mathcal{F}_{t-1}] = \sigma_t \mathbb{E}[z_t | \mathcal{F}_{t-1}] = 0$$

The conditional variance is time varying

$$\mathbb{E}[Y_t^2 | \mathcal{F}_{t-1}] = \mathbb{E}[\sigma_t^2 z_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2 \mathbb{E}[z_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2 \cdot 1 (= \omega + \alpha y_{t-1}^2)$$

What does this mean? How do we create a model for time varying variance? we have scaled an iid random variable z_t by multiplying it by σ_t which is $\mathcal{F}_{t-1}$ -measurable (it's the square root of $\omega + \alpha y_{t-1}^2$ so it's $\mathcal{F}_{t-1}$ -measurable). That implies that the original model has (one step ahead) conditional distribution

$$p(y_t | \mathcal{F}_{t-1}) \sim \text{IID}(0, \sigma_t^2)$$

with σ_t^2 time varying and depends on y_{t-1}^2 .

This is an ARCH(1) model for σ_t^2 : AR stands for autoregressive and CH for conditional heteroskedasticity (conditional variance changes in time).

If we calculate the unconditional mean, by the law of iterated expectation

$$\mathbb{E}[Y_t] = \mathbb{E}[\mathbb{E}[Y_t | \mathcal{F}_{t-1}]] = 0$$

while the unconditional variance (we will see) is

$$\text{Var}[Y_t] = \frac{\omega}{1 - \alpha}$$

7.2.9 GARCH

Example 7.2.6 (GARCH). G stands for *generalized*. Summarizing it

$$\begin{aligned} y_t &= \sigma_t \varepsilon_t \quad \varepsilon_t \sim \text{NID}(0, 1) \\ \sigma_t^2 &= \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2 \end{aligned}$$

with $\omega > 0$, $\alpha > 0$, $\beta > 0$ (this imply positivity)

Again σ_t^2 is $\mathcal{F}_{t-1}$ -measurable and

$$\mathbb{E}[Y_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2$$

Here σ_t^2 has a recursion and depends on its past values and past values of the series

Chapter 8

State space form models

8.1 Examples

8.2 SSM in more detail

Remark 35. Let's describe the general definition of a SSF model. SSM, under gaussianity, and the Kalman filter gives us a general methodology to treat unobserved component models

Definition 8.2.1 (SSM model). Model defined using two equations (below y_t can be univariate or multivariate and is the only thing observed)

$$\begin{aligned} y_t &= Z_t \alpha_t + G_t \varepsilon_t, & \varepsilon_t &\sim N(0, \mathbf{I}) \\ \alpha_{t+1} &= T_t \alpha_t + H_t \eta_t, & \eta_t &\sim N(0, Q_t) \end{aligned}$$

with ε_t, η_s are independent for all $t, s \in \mathbb{Z}$ and $\alpha_1 \sim N(a_1, P)$.

The above are respectively the *observation/measurement equation* and the *state/transition equation* regarding the time varying parameter.

Z_t, T_t, G_t, H_t, Q_t are system matrices (or parameters):

- here are associated to time but often are static/fixed and we consider them as such here (eg $T_t = T$, $G_t = G$, $Z_t = Z$, $T_t = T$);
- Z_t is a matrix of coefficient allowing to write the model in state space form;
- T_t is the transition matrix which dictates how time varying parameter evolves;
- G_t, H_t are scale matrices

Remark 36. Examples are the UC models (LLM, LLT, AR(1) + noise) but even ARMA models

Example 8.2.1 (LLM). The LLM is already defined in state space form so it's

easy to find out that can be defined just by setting

$$\begin{aligned} Z_t &= 1 \\ \alpha_t &= \mu_t \\ G_t &= \sigma_\varepsilon \\ T_t &= 1 \\ H_t &= \sigma_\eta \\ Q_t &= 1 \end{aligned}$$

Example 8.2.2 (AR(1)+noise). Similar to above except T_t

$$\begin{aligned} Z_t &= 1 \\ \alpha_t &= \mu_t \\ G_t &= \sigma_\varepsilon \\ T_t &= \phi \\ H_t &= \sigma_\eta \\ Q_t &= 1 \end{aligned}$$

Example 8.2.3 (LLTM). It is not in SSF by default but defined in three equation like

$$\begin{aligned} y_t &= \mu_t + \varepsilon_t^*, & \varepsilon_t^* &\sim \text{NID}(0, \sigma_\varepsilon^2) \\ \mu_{t+1} &= \beta_t + \mu_t + \eta_t^*, & \eta_t^* &\sim \text{NID}(0, \sigma_\eta^2) \\ \beta_{t+1} &= \beta_t + \xi_t, & \xi_t &\sim \text{NID}(0, \sigma_\xi^2) \end{aligned}$$

with all error terms independent at all lags. The signal μ_{t+1} is basically a RW + noise + a *slope* β_t , which is defined itself as a random walk as well.

To have all this shit in SSF we need to wrap the last two equations (state vector α_t) as follows

$$\begin{aligned} y_t &= \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{Z_t} \underbrace{\begin{bmatrix} \mu_t \\ \beta_t \end{bmatrix}}_{\alpha_t} + \underbrace{\sigma_\varepsilon}_{G_t} \underbrace{\varepsilon_t^*}_{\varepsilon_t} \\ \underbrace{\begin{bmatrix} \mu_{t+1} \\ \beta_{t+1} \end{bmatrix}}_{\alpha_{t+1}} &= \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}}_{T_t} \underbrace{\begin{bmatrix} \mu_t \\ \beta_t \end{bmatrix}}_{\alpha_t} + \underbrace{\begin{bmatrix} \sigma_\eta & 0 \\ 0 & \sigma_\xi \end{bmatrix}}_{H_t} \underbrace{\begin{bmatrix} \eta_t^* \\ \xi_t \end{bmatrix}}_{\eta_t} \end{aligned}$$

with $Q_t = 1$. Notices that all system matrices are all static (not depending on t)

Remark 37. These examples to see that many stochastic processes can be casted on the same common ground; we can cast a lot of model in this form by setting the proper matrix as above.

This will enable us, as long as we have gaussianity in the error term, to apply the kalman filter and to estimate all the parameters of the model and recover all the evolution of the states just based on a simple recursion

Example 8.2.4 (ARMA(2,1)). Beside unobserved component models, also arma processes can be cast in SSF. Here an example (one can check it holds as exercise). Model defined as

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \zeta_t + \theta \zeta_{t-1}, \quad \zeta_t \sim N(0, \sigma_\zeta^2)$$

Which is casted in SSF with the following

$$\alpha_t = \begin{bmatrix} y_t \\ \phi_2 y_{t-1} + \theta \zeta_t \end{bmatrix}, \quad T_t = \begin{bmatrix} \phi_1 & 1 \\ \phi_2 & 0 \end{bmatrix}, \quad Z_t = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$H_t = \begin{bmatrix} 1 \\ \theta \end{bmatrix}, \quad Q_t = \sigma_\zeta^2, \quad \eta_t = \zeta_{t+1}, \quad G_t = 0$$

If we have a model cast in state space form:

- which is linear (as we specified above) and Gaussian error: we derive properties and results based on properties of multivariate normal distribution
- state vector α_t evolves as a VAR(1) process (basically α_t evolves as an AR(1) but its components are a vector)
- system matrices typically contain unknown parameters; estimation involves
 1. recovering the dynamic state (reconstruct α_t)
 2. estimating static parameters (done in same step by using MLE)

8.3 The Kalman motherfucking filter

It's a methodology to find unknown in a SSF model. We start from the model (whatever it is) casted in SSF definition

$$y_t = Z_t \alpha_t + G_t \varepsilon_t, \quad \varepsilon_t \sim N(0, \mathbf{I})$$

$$\alpha_{t+1} = T_t \alpha_t + H_t \eta_t, \quad \eta_t \sim N(0, Q_t)$$

with ε_t, η_s are independent for all $t, s \in \mathbb{Z}$ and $\alpha_1 \sim N(a_1, P)$.

The transition equation in the SSF cast of the model is a recursive equation so if we know α_1 we know the following sequence.

If $\mathcal{F}_t$ represent the information set up to time t , we can define two quantities: the conditional (one-step ahead) expectation and variance of the (unobserved) state given the past:

$$\alpha_{t+1|t} = \mathbb{E}[\alpha_{t+1} | \mathcal{F}_t]$$

$$P_{t+1|t} = \text{Var}[\alpha_{t+1} | \mathcal{F}_t]$$

Assume some initial conditions are given, that is $\alpha_{1|0} = a_1$ and $P_{1|0} = P_1$.

The **prediction error** (crucial quantity) is defined as the difference between y_t and its best (MSE sense) predictor (which is the conditional expectation given

the past, $\mathbb{E}[y_t|\mathcal{F}_{t-1}]$)

$$\begin{aligned}
 v_t &= y_t - \mathbb{E}[y_t|\mathcal{F}_{t-1}] \\
 &\stackrel{(1)}{=} y_t - \mathbb{E}[Z_t\alpha_t + G_t\varepsilon_t|\mathcal{F}_{t-1}] \stackrel{(2)}{=} y_t - Z_t \mathbb{E}[\alpha_t|\mathcal{F}_{t-1}] \\
 &\stackrel{(3)}{=} y_t - Z_t\alpha_{t|t-1} \\
 &\stackrel{(4)}{=} Z_t\alpha_t + G_t\varepsilon_t - Z_t\alpha_{t|t-1} \\
 &= Z_t(\alpha_t - \alpha_{t|t-1}) + G_t\varepsilon_t
 \end{aligned}$$

where in the rewriting of prediction error using the SSM model equation to have useful (for subsequent calculations) expressions, in:

- (1) we replaced y_t from the observation/measurement equation
- (2) taken out Z_t (set of design coefficient independent from past observations) and removed $\mathbb{E}[G_t\varepsilon_t|\mathcal{F}_{t-1}] = G_t \mathbb{E}[\varepsilon_t|\mathcal{F}_{t-1}] = 0$
- (3) just replaced by definition $\alpha_{t|t-1}$ of above
- (4) we replaced again $y_t = Z_t\alpha_t + G_t\varepsilon_t$

From above one can find the conditional expectation and variance of the prediction error

$$\begin{aligned}
 \mathbb{E}[v_t|\mathcal{F}_{t-1}] &= 0 \\
 \text{Var}[v_t|\mathcal{F}_{t-1}] &= Z_t P_{t|t-1} Z_t^\top + G_t G_t^\top
 \end{aligned}$$

Idea of KF We have that

- suppose we are at time $t - 1$: our **best prediction** of y_t given the past information set $\mathcal{F}_{t-1}$ is

$$\mathbb{E}[y_t|\mathcal{F}_{t-1}] = \mathbb{E}[Z_t\alpha_t + G_t\varepsilon_t|\mathcal{F}_{t-1}] = Z_t\alpha_{t|t-1}$$

- when the actual observation y_t arrives, we can calculate the **prediction/innovation error**

$$v_t = y_t - Z_t\alpha_{t|t-1}$$

and its variance, let's callit F_t

$$F_t = Z_t P_{t|t-1} Z_t^\top + G_t G_t^\top$$

(btw v_t and F_t are the crucial component for ML estimation too)

- now we're able to update: the **new best estimate** of α_t is then updated and now based on the old estimate $\alpha_{t|t-1}$ and the new information brought by v_t (new info is y_t incorporated into v_t)

$$\alpha_{t+1|t} = T_t\alpha_{t|t-1} + K_tv_t$$

so we have a way of going from the old estimate $\alpha_{t|t-1}$ to the updated version $\alpha_{t+1|t}$ by considering v_t .

Similarly for the variance (*we will see how later*)

$$P_{t+1|t} = T_t P_{t|t-1} T_t^\top + H_t H_t^\top - K_t F_t K_t^\top$$

In both equation above the **Kalman gain** K_t , is the optimal weighting matrix for the new evidence

$$K_t = T_t P_{t|t-1} Z_t^\top F_t^{-1}$$

Definition 8.3.1. Kalman filter is particularly useful for linear and gaussian model

1. we will have initial conditions $\alpha_{1|0} = a_1$, $P_{1|0} = P_1$
2. the sequence then should be

$$\begin{aligned} v_1 &= y_1 - Z \alpha_{1|0} = y_1 - Z_1 a_1 \\ F_1 &= Z P_1 Z^\top + G G^\top \\ K_1 &= T P_1 Z^\top F_1^{-1} \\ \alpha_{2|1} &= T \alpha_{1|0} + K_1 v_1 \\ P_{2|1} &= T P_1 T^\top + H Q H^\top - K_1 F_1 K_1^\top \end{aligned}$$

3. and in general we go by updating, in sequence, prediction error, its variance, the kalman gain, the updated state estimate and the updated variance

$$\begin{aligned} v_t &= y_t - Z_t \alpha_{t|t-1} \\ F_t &= Z_t P_{t|t-1} Z_t^\top + G_t G_t^\top \\ K_t &= T_t P_{t|t-1} Z_t^\top F_t^{-1} \\ \alpha_{t+1|t} &= T_t \alpha_{t|t-1} + K_t v_t \\ P_{t+1|t} &= T_t P_{t|t-1} T_t^\top + H_t Q_t H_t^\top - K_t F_t K_t^\top \end{aligned}$$

Proof Saltata

Maximum likelihood estimation Supposing system matrices depend on a parameter vector $\theta \in \Theta \subset \mathbb{R}^m$ (which collects all the static parameters, $\sigma_\varepsilon^2, \sigma_\eta^2$), the model is completely Gaussian. The loglikelihood of $y_1, \dots, y_n$ is:

$$\log L(\theta) = \sum_{t=1}^n \log p(y_t | \mathcal{F}_{t-1})$$

In the state space model, $p(y_t | \mathcal{F}_{t-1})$ is a Gaussian density with mean $Z_t \alpha_{t|t-1}$ and variance F_t (that is $p(y_t | \mathcal{F}_{t-1}) = \frac{1}{\sqrt{2\pi F_t}} e^{-\frac{1}{2} \frac{v_t^2}{F_t}}$), from the Kalman filter, so that

$$\log L(\theta) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^n (\log F_t + F_t^{-1} v_t^2)$$

This is called the prediction error decomposition of the likelihood. Estimation proceeds by numerically maximising in $L(\theta)$.

8.4 Innovation form of Kalman filter (IFKF)

Consider the AR(1)+noise model in the form (a **parameter driven model**)

$$\begin{aligned} y_t &= \alpha_t + \varepsilon_t, & \varepsilon_t &\sim N(0, \sigma_\varepsilon^2) \\ \alpha_{t+1} &= \phi\alpha_t + \eta_t, & \eta_t &\sim N(0, \sigma_\eta^2) \end{aligned}$$

here α_{t+1} is a function of α_t and η_t rv with its own distribution. Assuming $\alpha_1 \sim N(\alpha_{1|0}, P_{1|0})$ with

$$\begin{aligned} \alpha_{1|0} &= \mathbb{E}[\alpha_1] = 0 \\ P_{1|0} &= \text{Var}[\alpha_1] = \frac{\sigma_\eta^2}{1 - \phi^2} \end{aligned}$$

that imply

$$\begin{aligned} v_1 &= y_1 - \alpha_{1|0} = y_1 \\ F_1 &= P_{1|0} + \sigma_\varepsilon^2 = \frac{\sigma_\eta^2}{1 - \phi^2} + \sigma_\varepsilon^2 \end{aligned}$$

So that $K_1, \alpha_{2|1}, P_{2|1}$ can be calculated and then the iterations starts:

- Kalman filter gives estimated state $\hat{\alpha}_{t|t-1}$
- ML estimation gives estimates of $\hat{\pi}, \hat{\sigma}_\varepsilon^2, \hat{\sigma}_\eta^2$
- we can write with the **innovation form of KF** which is basically the first and fourth equation of KF procedure¹

$$v_t = y_t - \alpha_{t|t-1} \tag{8.1}$$

$$\alpha_{t+1|t} = \phi\alpha_{t|t-1} + K_t v_t \tag{8.2}$$

$\alpha_{t+1|t}$ is a function of $\alpha_{t|t-1}$ and of the past observation $y_t, y_{t-1}, \dots$ incorporated through the recursion and the prediction error v_t . So we can see the IFKF as an **observation driven form** of the model.

The notation $\alpha_{t+1|t}$ emphasizes the fact that $\alpha_{t+1|t}$ is $\mathcal{F}_t$ -measurable (known at time t).

- The Kalman gain k_t depends on ϕ and $q = \frac{\sigma_\varepsilon^2}{\sigma_\eta^2}$, the signal to noise ratio. In the steady state (after a tot of iterations), k_t is constant. Basically it's determined deterministically

Important remark 19. If in equation 8.1 however y_t comes from heavy tailed distributions, eg t, KF can be applied but through innovation v_t (directly linked to y_t), recovered signal α_t is sensitive to extreme values.

¹The remaining formulas in this case follows

$$\begin{aligned} F_t &= P_{t|t-1} + \sigma_\varepsilon^2 \\ K_t &= \phi P_{t|t-1} F_t^{-1} \\ P_{t+1|t} &= \phi^2 P_{t|t-1} + \sigma_\eta^2 - \phi^2 P_{t|t-1}^2 F_t^{-1} \end{aligned}$$

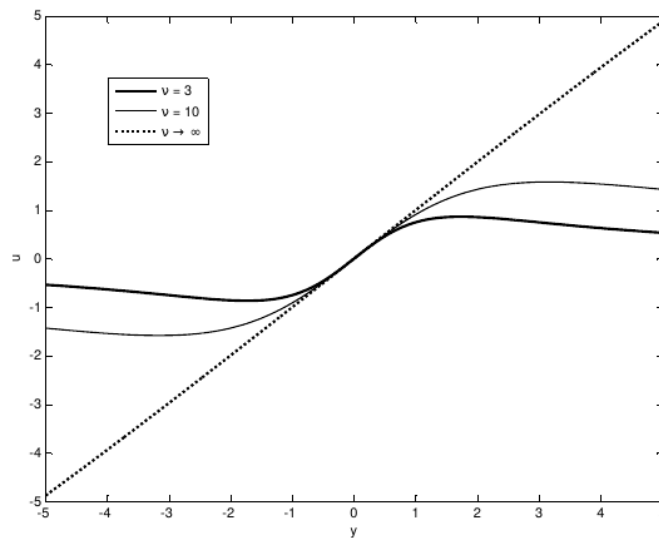


Figure 8.1: response of u_t to a standardised observation y_t for different values of ν

In essence KF is neither optimal nor robust.

Score driven model try to make a robust thing here in order to avoid extreme values to have an effect on estimation: in particular idea here is to replace the innovation error (martingale) with another martingale difference sequence which reacts to y_t in a way that is bounded (in the centre however we dont change the data) 9.1

Remark 38. If error of the y_t is gaussian we dont do score, we go with gaussian and kalman filter.

8.5 Lab tests

we want to run kalman filter on a generated AR(1)+noise

```
#####
# KALMAN filter recursions with our own function
#####

KF_pred <- function(y, phi, sigma_epsilon, sigma_eta){

  # we here separate the arguments of the function that will be estimated -
  # phi, sigma_epsilon, sigma_eta by the arguments of the function that will
  # not - mu_1|0 and P_1|0

  n <- length(y)

  # allocate space
```

```

mu_pred <- 0 # this will be updated as mu_{t|t-1}: initialize it
P <- 1 # this will be updated as P_{t|t-1}: initialize it
v <- rep(NA, n) # the innovation error
K <- 0 # the Kalman gain
F <- 0 # the conditional variance of v_t
dllk <- rep(NA, n) #the log-likelihood value
llk <- 0

# the recursion

for(t in 1:(n-1)){
  v[t] = y[t] - mu_pred[t];
  F[t] = P[t] + sigma_epsilon^2;
  K[t] = (phi * P[t])/F[t]
  P[t+1] = phi^2 * P[t] + sigma_eta^2 - K[t]*F[t]*K[t]
  mu_pred[t+1] = phi * mu_pred[t] + K[t]*v[t]
  dllk[t] = - 0.5 * log(F[t] + (v[t]^2/F[t]))
  llk = llk + dllk[t]
}

## #llk = sum(dllk)

## out <- list(mu_pred = as.ts(mu_pred), llk = llk)
## #out <- list(v, mu_pred)
list("mu_pred" = mu_pred, "llk"=llk)
}

#####
# KALMAN filter likelihood
#####

KF_loglikelihood <- function(par, y){
  # par should be a vector with initial guess of phi, sigma_epsilon and sigma_eta
  phi <- par[1]
  sigma_epsilon <- par[2]
  sigma_eta <- par[3]
  - KF_pred(y, phi=phi, sigma_epsilon=sigma_epsilon, sigma_eta=sigma_eta)$llk
}

#####
# KALMAN filter parameter estimation
#####
# input: the data and the initial values of the parameters that we aim to estimate
# objective function: the KF Gaussian likelihood given from the KF
# (prediction error decomposition), see the function above
# output: the estimated parameters
# the filtered mu will be recovered from the KF recursions

```

```

KF_estimator <- function(y, theta_0=c("phi"=0.8, "sigma_epsilon"=1, "sigma_eta"=1)){
  ## optim(par = theta_0, fn = KF_loglikelihood, y = y)$par
  optim(par = theta_0, fn = KF_loglikelihood, y = y,
        method = "L-BFGS-B",
        lower = c(-1, 0.01, 0.01),
        upper = c(1, 100, 100),
        )$par
}

## -----

library(lbts)
set.seed(2)
ar1n <- sim_ar1noise(n=1000, phi=0.8, sigma_epsilon = 2, sigma_eta=2)
(hat <- KF_estimator(ar1n$y))

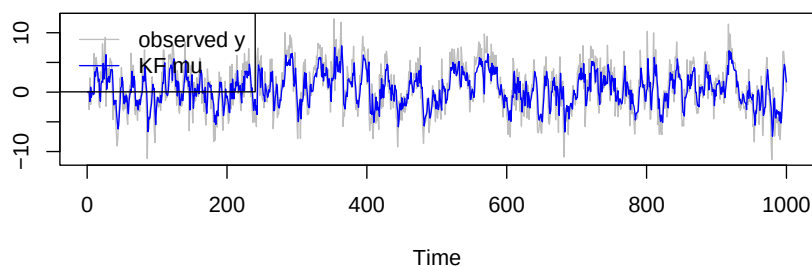
##           phi sigma_epsilon      sigma_eta
##      0.8144      0.7981      0.9737

# estimating kalman signal I suppose

kf_mu <- KF_pred(y=ar1n$y,
                  phi=hat["phi"],
                  sigma_epsilon=hat["sigma_epsilon"],
                  sigma_eta=hat["sigma_eta"])$mu_pred

lbts::mts_plot(list(ar1n$y, kf_mu), col = c("grey", "blue"))
legend("topleft", col = c("grey", "blue"), lty=1, legend = c("observed y", "KF mu"))

```



```

## or
lbts::KF(ar1n$y, plot=FALSE)$hat_theta

##           phi sigma_epsilon      sigma_eta
##      0.8144      0.7981      0.9737

```


Chapter 9

Score-driven models

Remark 39. If error of the y_t is gaussian we dont do score, we go with gaussian and kalman filter.

9.1 Score driven models

Example 9.1.1. Assuming $y_t|\mathcal{F}_{t-1} \sim t_\nu(\lambda_t, \sigma)$ is equivalent to

$$\begin{aligned}y_t &= \lambda_t + v_t \\ \lambda_{t+1} &= \phi\lambda_t + \kappa u_t\end{aligned}$$

with

- κ is a matrix which depends on unknown parameter to be estimated (differently from the KF for which it is almost deterministic determined); eg if univariate it's simple but if multivariate we don't know if its' triangular/diagonal etc
- u_t error proportional to the score of the conditional log-likelihood with respecto to λ_t

$$u_t \propto \frac{y_t - \lambda_t}{1 + \frac{(y_t - \lambda_t)^2}{\nu\sigma^2}}$$

score is nonlinear in y_t has a redeshending shape (robustness) so higher values of y_t do not affect u_t too much and thus our

Above for $\nu \rightarrow \infty$ (tendente alla normale) we have that $u_t \cong v_t = y_t - \lambda_t$

Remark 40. In general if we assume that $Y_t|\mathcal{F}_{t-1} \sim p(y; g(\lambda_t), \theta)$ with

- $g(\lambda_t)$ a link function depending on time varing parameter
- $\theta \in \Theta \subset \mathbb{R}^p$ static parameter

the evolution of the time varying parameter can be described

- assuming an autoregressive process with error term u_t (ω can be zero below)

$$\lambda_t = \omega + \phi\lambda_{t-1} + \kappa u_{t-1}$$

where the error term is proportional to score of the conditional likelihood taken with respect to the time varying parameter

$$u_t \propto \frac{\partial}{\partial \lambda_t} \log p(y_t | F_{t-1})$$

- more in general the update equation can be written as

$$\lambda_{t+1} = \omega + \beta \lambda_t + \alpha S(\lambda_t) s(y_t, \lambda_t) \quad (9.1)$$

where

- $s(y_t, \lambda_t)$ is the score of the assumed predictive logdensity

$$s(y_t, \lambda_t) = \frac{\partial}{\partial \lambda_t} \log p(y_t, \lambda_t, \theta) = \frac{\partial}{\partial \lambda_t} \log p(y_t | \mathcal{F}_{t-1})$$

we can calculate it at each time because λ_t is $\mathcal{F}_{t-1}$ measurable

- $S(\lambda_t)$ is a positive scaling: instead of using the score directly we adapt it
- ω, α, β are static parameters

The aim again is to recover λ_t (function of θ) based on data

Remark 41. In these model the conditional distribution of the data generating process is given, is heavy tailed and the location and/or scale parameter change over time.

Key feature:

- the dynamics of the time varying parametre are updated by recursion which is driven by the score function of the conditional log-likelihood of the parameter of interest
- the score of the conditional distribution has a **nice distribution/property for many heavy tailed distributions**:
 - when the conditional distribution is t and a model for the *location* parameter is specified then the score is *proportional to a beta* distributed rv with parameters depending on ν
 - when the conditional distribution is t and a model for the *scale* parameter is specified then the score is distributed as a beta rv with parameters depending on ν
- as they are parametric methods, likelihoods estimation is straightforward
- as non-liner models, regularity conditions or consistency and asymptotic normality of MLE are often complicate to establish

9.2 Time varying mean

Considering a location model with Student- t innovations

$$y_t = \lambda_t + \varepsilon_t \quad \varepsilon_t \sim t_\nu(0, \sigma^2)$$

and then the density of y_t

$$p(y; \lambda_t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi}\Gamma(\frac{\nu}{2})(\nu\sigma^2)^{1/2}} \left(1 + \frac{(y_t - \lambda_t)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}$$

so that the logdensity is proportional to (considering only the things depending on λ_t cause we're going to differentiate)

$$\log p(y_t; \lambda_t) \propto -\frac{\nu+1}{2} \log \left(1 + \frac{(y_t - \lambda_t)^2}{\nu\sigma^2}\right)$$

and taking derivative

$$s(y_t, \lambda_t) = \frac{\partial}{\partial \lambda_t} \log p(y_t; \lambda_t) = \frac{\nu+1}{\nu\sigma^2} \frac{(y_t - \lambda_t)^2}{1 + \frac{(y_t - \lambda_t)^2}{\nu\sigma^2}}$$

This above is our score; for convenience it's multiplied by the following scaling factor to obtain a scaled score

Scaled score and final update equation Selecting a constant (*arbitrarily*, we shall come back on this)

$$S(\lambda_t) = \nu\sigma^2/(\nu+1)$$

as a scaling function yields

$$S(\lambda_t)s(y_t, \lambda_t) = \frac{y_t - \lambda_t}{1 + (y_t - \lambda_t)^2/\sigma^2\nu}$$

and the final supradupra version of time varying location model is the folloing

$$y_t = \lambda_t + \varepsilon_t, \quad \varepsilon_t \sim t_\nu(0, \sigma^2) \quad (9.2)$$

$$\lambda_{t+1} = \omega + \phi\lambda_t + \kappa \frac{y_t - \lambda_t}{1 + (y_t - \lambda_t)^2/\sigma^2\nu} \quad (9.3)$$

where ϕ, κ correspond to β, α in the general formulation 9.1.

If $\nu \rightarrow \infty$ the student t becomes a normal: the score become linear in the innovation that is:

$$s(\lambda_t, y_t) = y_t - \lambda_t = y_t - \mathbb{E}[y_t | \mathcal{F}_{t-1}]$$

and thus is linear in the observation.

Important remark 20. Now ν is *estimated* in the process; in real case if it is a lot means conditional data comes from a gaussian distribution and we could go back and go with kalman filter.

If the estimate of ν goes to infity means that one dont have too many outliers (more than a gaussian).

9.2.1 Model estimation

Stiamo considerando un modello

$$\begin{aligned} y_t &= \lambda_t + \varepsilon_t, \quad \varepsilon_t \sim t_\nu(0, \sigma) \\ \lambda_{t+1} &= \omega + \phi\lambda_t + \kappa u_t \end{aligned}$$

with

$$u_t = \frac{v_t}{1 + \frac{v_t^2}{\nu\sigma^2}} \propto \frac{\partial}{\partial \lambda_t} p(y_t | \mathcal{F}_{t-1})$$

where $v_t = y_t - \lambda_t$.

Why do we use this kind of model? motivation is empirical and is robustness of filter λ_{t+1} against extremes that contaminate the observation y_t .

Given the specified model above and given the data $y_1, y_2, \dots$ we want to estimate the parameter above. The static parameters are

$$\theta = (\omega, \phi, \kappa, \sigma, \nu)$$

Once estimated θ the time varying parameter is a deterministic recursion and can be recovered easily.

We use ML estimation for static parameters thus: key advanced is that Likelihood is available in closed form so $\log p(y_t | \mathcal{F}_t - 1)$ is available and we can differentiate

we want an estimator that given sample $(y_1, y_2, \dots, y_t)$ observed time series and λ_1 (initial condition) we want theta to maximize the loglikelihood

$$\hat{\theta}_t = \max_{\theta \in \Theta \subseteq \mathbb{R}^5} l_t(\theta)$$

Thing is that actually likelihood is function of θ , which has λ_t inside; for this reason we do estimation numerically, we don't do minimization (putting derivatives=0). What we do in practice is we write the likelihood and *minimize* numerically its reversed sign (because optimizer algorithms minimizes stuff).

ML are consistent and asymptotically normal under some condition (swapping integral with derivative and ...). In these models these mild regularity condition are at border. Two condition we want to have are consistency and asymptotic normality

9.2.2 Properties of the score

Looking at property of u_t

- as a function of the prediction error: **Robustness** (most important)
- as a function of the error term: **Stationarity**
- as a function of the past observations: **Invertibility**

NB: i guess dynamic parameter λ_t is recovered after

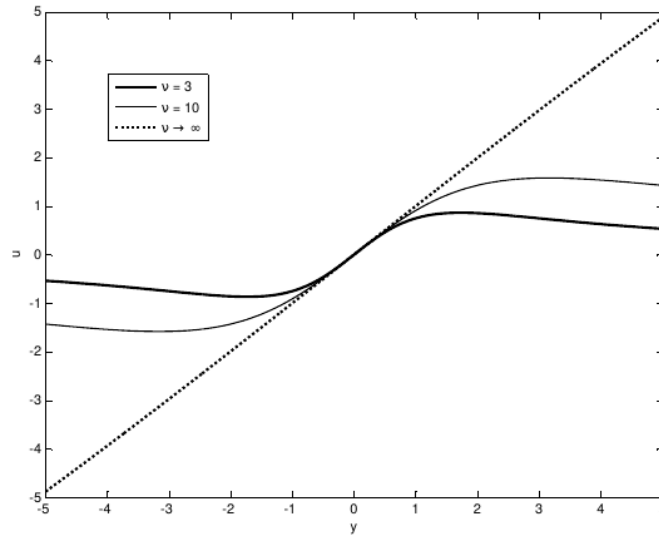


Figure 9.1: response of u_t to a standardised observation y_t for different values of ν

9.2.2.1 Robustness

First observe that the score is this function of the innovation error $v_t = y_t - \lambda_t$ (also called one step ahead prediction error, which is mds)

$$u_t = \frac{v_t}{1 + \frac{v_t^2}{\sigma^2 \nu}} = \frac{v_t}{w_t}$$

that is score u_t can be seen as a “winsorized”/trimmed version of v_t , were w_t downweight v_t (especially with lower values of ν). (fig 9.1). So the impact on the signal is cut down by using the scaled score in case of spikes (this is noise, not signal).

Important remark 21 (κ as further limiter). κ from 9.3 further regulates the impact of u_t on the filtered signal λ_t

Remark 42. At the end of estimation we should have

- y_t observation heavy tailed
- λ_t robust estimate of the signal
- $v_t = y_t - \lambda_t$ heavy tailed innovation/prediction error (heavytailed because its difference between heavy and normal tailed)
- u_t noise associated with signal

if y_t is gaussian than all the remaining are

9.2.2.2 Stationarity

As a function of ε_t (remembering $y_t = \lambda_t + \varepsilon_t$ with $\varepsilon \sim t_\nu(0, \sigma)$) we can rewrite the scaled score

$$u_t = \frac{y_t - \lambda_t}{1 + (y_t - \lambda_t)^2 / \sigma^2 \nu} = \frac{\varepsilon_t}{1 + \frac{\varepsilon_t^2}{\sigma^2 \nu}}$$

the scaled score can be seen as transformation of the random variable ε_t .¹

Proposition 9.2.1. *Even moment of score are*

$$\mathbb{E}[u_t^{2s}] = \sigma^{2s} \nu^s \frac{\text{Beta}(\frac{1}{2} + s, \frac{\nu}{2} + 2)}{\text{Beta}(\frac{1}{2}, \frac{\nu}{2})}$$

while odd moments are null

Proof. I think here she means Y for ε_t .

In general if $Y \sim t$ let $Z = (Y - \lambda)/\sigma \sim t_\nu(0, 1)$

If we then consider the ratio following (ratio between the square of two distributions, which are F , divided by their df, $F(1, \nu)/\nu$ and $1 + F(1, \nu)/\nu$ rv)

$$W = \frac{Z^2/\nu}{1 + Z^2/\nu} \sim \beta \frac{1}{2}, \frac{\nu}{2}$$

and for whatever reason

$$u_t = X = (1 - W)(Y - \lambda)$$

1

Important remark 22 (Ricordella Beta). In general if $W \sim \text{Beta}(\alpha, \beta)$ then

$$f_W(w) = \frac{1}{B(\alpha, \beta)} w^{\alpha-1} (1-w)^{\beta-1}$$

with

- $0 \leq w \leq 1$
- $B(\alpha, \beta)$ is the beta function

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

- $\Gamma(x)$ is the gamma function

$$\Gamma(x) = \int_{-\infty}^{+\infty} t^{x-1} e^{-t} dt$$

Moments of a beta distributed rv: for $h > -\alpha$ and $k > -\beta$

$$\mathbb{E}[W^h (1-W)^k] = \frac{B(\alpha + h, \beta + k)}{B(\alpha, \beta)}$$

from which

$$\mathbb{E}[W] = \frac{\alpha}{\alpha + \beta}$$

which follows by the property of the Gamma function

$$\Gamma(x+1) = x\Gamma(x)$$

that is is a product of a beta ($1 - W$: if W is a beta so is $1 - W$ with parameter inverted) and a t ($Y - \lambda$). It's squared version (and then multiply by ν/ν)

$$X^2 = (1 - W)^2(Y - \lambda)^2 = (1 - W)^2\sigma^2 Z^2 = (1 - W)^2\sigma^2\nu Z^2/\nu$$

where

$$\frac{Z^2}{\nu} = \frac{1}{1 - W} - 1$$

gives that the squared score is the product of two dependent beta rv

$$X^2 = (1 - W)W\sigma^2\nu$$

and then the even power are

$$X^{2s} = (1 - W)^s W^s \sigma^{2s} \nu^s$$

and the result follows by direct integration of the right hand side of the last equality with respect to the density of W

$$f_W(w) = \frac{1}{B(\frac{1}{2}, \frac{\nu}{2})} w^{\frac{1}{2}-1} (1 - w)^{\frac{\nu}{2}-1}$$

□

Important remark 23. Consequences of the proposition 9.2.1 are

- variance of the score

$$V(u_t) = \frac{\sigma^2\nu^2}{(\nu + 1)(\nu + 3)}$$

- Kurtosis of the score is smaller than 3

$$K(u_t) = \frac{3(\nu + 1)(\nu + 2)(\nu + 3)}{\nu(\nu + 5)(\nu + 7)}$$

- score is a MDS if the model is correctly specified $\mathbb{E}[u_t | \mathcal{F}_{t-1}] = 0$
- score and innovation error are correlated

$$\mathbb{E}[u_t v_t] = \nu\sigma^2 \mathbb{E}[W] = \frac{\nu\sigma^2}{\nu + 1}$$

- the variables u_t form an iid sequence

Important remark 24 (Stationarity). For $|\phi| < 1$

$$\lambda_t = \frac{\omega}{1 - \phi} + \kappa \sum_{j=0}^{\infty} \phi^j u_{t-j}$$

is stationary and ergodic (s.e.) (measurable function of the IID sequence u_t)

y_t is a s.e process (continuous function of two ind s.e. process)

Weak stationarity of y_t is implied by strong stationarity (first tow moments of y_t exists)

being λ_t stationary and ergodic and being ε_t stationary and ergodic then y_t is stationary and ergodic

TODO: boh a sto punto
mi sono anche rotto il
cazzo e copio e pace

9.2.2.3 Invertibility

The score is this function of the past variables

$$u_t = \frac{y_t - \lambda_t}{1 + \frac{(y_t - \lambda_t)^2}{\sigma^2 \nu}}$$

$$= \frac{y_t - \omega - \phi \lambda_{t-1} - \kappa \left(\frac{y_{t-1} - \lambda_{t-1}}{1 + \frac{(y_{t-1} - \lambda_{t-1})^2}{\sigma^2 \nu}} \right)}{1 + \left[y_t - \omega - \phi \lambda_{t-1} - \kappa \left(\frac{y_{t-1} - \lambda_{t-1}}{1 + \frac{(y_{t-1} - \lambda_{t-1})^2}{\sigma^2 \nu}} \right) \right]^2 / (\sigma^2 \nu)}$$

and so on backwards

- to the infinite past when we consider the r. v. in the process
- to y_1 (first observations) and to λ_1 (initial condition) in practice

Definition 9.2.1 (invertibility). initial condition on which we start recursion (eg λ_1) does not have impact on asymptotic properties of estimator

Filter invertibility Filtered sequence corresponding to a fixed initial condition λ_1

$$\hat{\lambda}_{t+1} = \omega + \phi \lambda_t + \kappa \frac{y_t - \hat{\lambda}_t}{1 + (y_t - \hat{\lambda}_t)^2 / \sigma^2 \nu}$$

We are interested on the asymptotic behaviour of

$$\sup_{\theta \in \Theta} \left| \hat{\lambda}_{t+1}(\theta) - \lambda_{t+1}(\theta) \right| \quad \text{as } t \rightarrow \infty$$

By the mean value theorem (the mapping from $\hat{\mu}_t$ to $\hat{\mu}_{t+1}$ is the same as the mapping from μ_t to μ_{t+1} and is continuous)

$$\sup_{\theta \in \Theta} \left| \hat{\lambda}_{t+1}(\theta) - \lambda_{t+1}(\theta) \right| \leq \sup_{\theta \in \Theta} \left| \frac{\partial \lambda_{t+1}(\theta)}{\partial \lambda_t(\theta)} \right|_{\lambda_t(\theta) = \lambda} \sup_{\theta \in \Theta} \left| \hat{\lambda}_t(\theta) - \lambda_t(\theta) \right|$$

Setting $x_t = \frac{\partial \lambda_{t+1}(\theta)}{\partial \lambda_t(\theta)}$ and repeatedly applying the mean value theorem

$$\begin{aligned} \sup_{\theta \in \Theta} \left| \hat{\lambda}_{t+1}(\theta) - \lambda_{t+1}(\theta) \right| &\leq \prod_{i=1}^{t-1} \sup_{\theta \in \Theta} \sup_{\lambda} |x_{t-i}(\theta)| \sup_{\theta \in \Theta} |\lambda_1 - \lambda_1(\theta)| \\ &= \exp \left[\sum_{i=1}^{t-1} \log \sup_{\theta \in \Theta} \sup_{\lambda} |x_{t-i}(\theta)| \right] \sup_{\theta \in \Theta} |\lambda_1 - \lambda_1(\theta)| \\ &= \exp \left[\frac{1}{t-1} \sum_{i=1}^{t-1} \log \sup_{\theta \in \Theta} \sup_{\lambda} |x_{t-i}(\theta)| \right]^{t-1} \sup_{\theta \in \Theta} |\lambda_1 - \lambda_1(\theta)| \end{aligned}$$

which converges to zero by the law of large numbers if

$$\mathbb{E} \left[\log \sup_{\theta \in \Theta} \sup_{\lambda} |x_t(\theta)| \right] < 0$$

the above condition is referred to as a contraction condition

Invertibility of the score driven model for the location In the score-driven model for the location one has

$$x_t = \frac{\partial \lambda_{t+1}(\theta)}{\partial \lambda_t(\theta)} = 2(1 - W)W - (1 - W) = (1 - W)(2W - 1)$$

with

$$\mathbb{E}[x_t] = -\frac{\nu}{\nu + 3}$$

so that a sufficient condition for invertibility is $\phi - \kappa < 1$.

Beyond proposition 1 Based on the moments of $u_t X$ distribution is

- symmetric
- bounded
- strictly sub-Gaussian
- if $\nu = 1$ location-scale Beta with values in $(-\sigma/2, \sigma/2)$
- if $\nu \rightarrow \infty$ gaussian

9.3 Lab tests

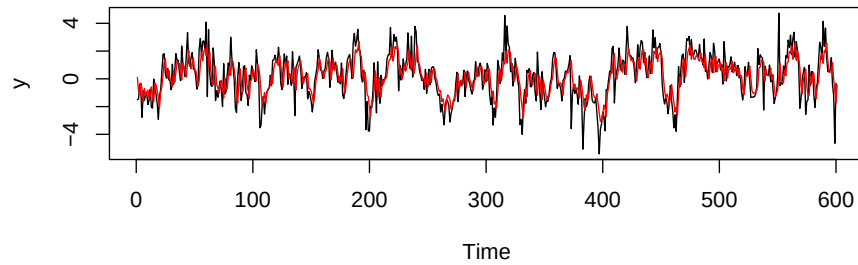
```
library(lbts)
#####
# this file does fit a DCS-t location model to data that ARE SIMULATED
# it uses the functions defined in Simulator, Estimator, Filter
#####

# Simulate a DCS-t location model to data that ARE SIMULATED
T <- 600
omega <- 0
phi <- 0.8
k <- 0.8
varsigma <- 1
nu <- 8

simu <- uDCS_t_model_simulator(T, omega, phi, k, varsigma, nu)
y <- simu$y_t_gen
ts.plot(y)

# estimate the parameter and components
theta_0 <- c(omega, phi, k, varsigma, nu)
est <- uDCS_t_model_estimator(y, theta_0)
filter <- uDCS_t_model_filter(y, est)

ts.plot(y)
lines(filter$Dynamic_Location,col = "red")
```

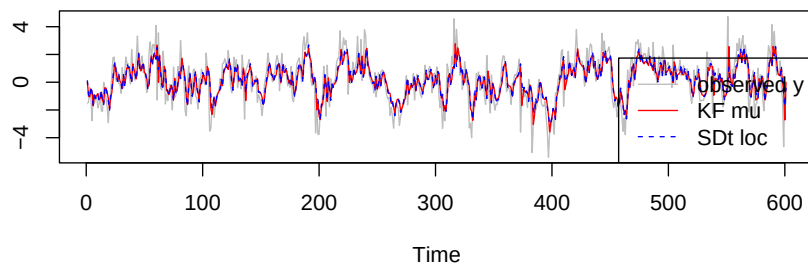


```
## oppure
ats_est <- ats_estimator(y=simu$y_t_gen)
ats_est

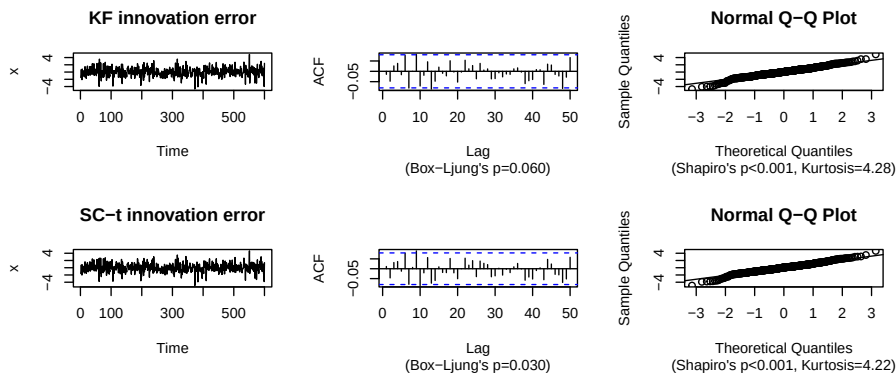
## $kf_hat_theta
##           phi sigma_epsilon      sigma_eta
##         0.8088      0.4653      0.6011
##
## $sct_hat_theta
##      omega      phi      k varsigma      nu
##    0.1037  0.7990  0.8231  1.0472  7.4690

plot(ats_est)

## Warning in (function (..., deparse.level = 1) : number of rows
of result is not a multiple of vector length (arg 1)
```



```
residuals(ats_est)
```

```
#####
# this file does fit a DCS-t location model to data that have to be uploaded
# it uses the functions defined in Estimator, Filter
#####
# initial values of the parameters that will enter into theta (static parameters)
# better call it theta_0
omega <- 0
phi <- 0.8
k <- 0.8
varsigma <- 1
nu <- 8

theta <- c(omega, phi, k, varsigma, nu)

# estimate the parameters (uses functions in Estimator)
# input: data, y, and INITIAL parameter values, theta
est <- uDCS_t_model_estimator(y, theta)
est$theta_list

## Error in est$theta_list: $ operator is invalid for atomic vectors

# recover the estimated time varying signal (uses functions in Filter)
# input: data, y, and ESTIMATED parameter values, theta
filter <- uDCS_t_model_filter(y, est$theta)

## Error in est$theta: $ operator is invalid for atomic vectors

ts.plot(y)
lines(filter$Dynamic_Location,col = "red")

# compare the estimated one-step-ahead prediction error with the score
# both are mds, the score is nonlinear and robust if nu is small,
# if nu->infinity, then the score u_t converges to the one-step-ahead prediction error
# which is then linear and Gaussian

T = NROW(y)
v = 0
for(t in 2:(T+1)){
```

```

v[t-1]=y[t-1]-filter$Dynamic_Location[t-1]
}

filter$Innovation_u_t

## Time Series:
## Start = 1
## End = 600
## Frequency = 1
## [1] -1.2081901 -0.5420145 0.7094080 -0.6449742 -1.3277569 0.7480015 -0.7781466
## [8] -0.0207525 -0.1065774 -0.9141885 -0.1725172 0.4586592 -1.1469606 -0.6793698
## [15] 1.2178729 -0.2953158 -0.6203359 -0.8529599 -1.1876235 0.7616278 0.2177167
## [22] 0.9836602 1.1266057 0.5507248 -0.9749495 -0.3350595 0.6449715 -0.6362365
## [29] -0.1073704 -0.6613205 1.2237480 -0.8455669 0.7351299 1.1063358 0.0691660
## [36] 0.2371566 -0.5740314 -0.5284247 1.3856039 0.0071086 -0.2115737 -0.4100403
## [43] 1.2185813 1.3579580 -0.7944231 -0.0963702 0.7274429 0.6385737 -0.7615058
## [50] 0.1324958 -0.4265075 -0.8182196 0.3404598 1.3713536 1.2888883 -0.2942083
## [57] 1.1122291 0.6890306 -0.1054970 1.3835004 -1.3375556 1.3961318 -1.1535413
## [64] -0.9218654 -0.2819368 1.3963772 0.2526634 -1.3867872 -0.4297611 -0.6340635
## [71] 0.3983533 0.3106567 -0.6787685 -1.1104669 1.3881227 -0.9834378 -0.1921406
## [78] 0.9055391 0.8611541 -0.5632391 1.3981520 0.1362585 -0.3661036 -0.6968059
## [85] -1.3424099 -1.0961821 1.2800753 1.1618317 -0.8696331 -1.1433826 0.1252722
## [92] 1.1970858 0.4716475 -1.2652739 -0.6850046 -0.9254196 1.2371455 -0.0114903
## [99] 0.8937629 -0.1292615 -0.3698599 -0.2978460 1.3651291 0.0259961 -0.6073089
## [106] -1.3415215 -1.3848293 -0.0804563 0.2469105 -1.1827637 0.3499116 0.1681172
## [113] -0.1105822 -0.0061210 0.7000570 -0.6306875 1.2323637 0.5747868 0.5645625
## [120] -0.0138899 0.0766804 1.0745743 -0.3863538 -1.1224369 -0.6606789 1.0750390
## [127] 1.0969972 -0.3172852 0.7893617 -1.2386879 0.3419172 -0.1369004 -0.6089026
## [134] 1.3725409 -0.3963621 -1.3930869 1.2267324 0.6439689 0.4439571 0.5474406
## [141] -0.5348963 -1.0916472 0.1628815 0.2950258 -0.1031285 0.3780646 -1.2791950
## [148] -0.1863612 -0.4242963 -0.6591386 -0.9825175 0.1628012 0.9352797 0.4991257
## [155] 0.5414519 1.1303144 -0.0146740 -0.5613463 0.0751492 -0.4577203 0.5268791
## [162] 0.2447879 0.8216466 0.6833010 -0.0740386 -1.2716492 0.7804941 1.1490540
## [169] -0.6912415 0.4200099 0.3481827 -0.5666003 -0.5709513 0.3412056 -0.5117655
## [176] 1.1235195 -1.2972058 -0.8527466 0.4481006 0.8248727 0.8861464 0.5420214
## [183] 0.4618882 -0.4225289 -0.7370934 1.3888298 1.3361403 0.2584924 0.7885555
## [190] 1.1086664 -0.6006297 0.3499811 -1.1674735 -0.5816208 0.7265104 -0.5607961
## [197] -1.3544755 -0.4834086 -1.3884394 -1.2094854 0.6807577 -0.5008722 0.7569984
## [204] 1.2798139 -0.9210965 -1.3509052 1.3931194 -0.9749637 -0.9284049 1.1884081
## [211] -0.4467258 0.3066778 0.0100349 0.2388428 1.0032444 0.6565148 0.3523698
## [218] 1.3712431 -0.6681542 0.7437195 1.1758957 0.3796870 0.7601884 -0.1957342
## [225] -1.3959194 0.6578643 -0.8030799 0.3847829 0.9561480 -0.3921059 0.4549385
## [232] 0.5890769 1.3489701 -0.5090220 -0.5131579 1.3876353 -1.1729329 0.4903917
## [239] 1.3926695 1.2218484 -1.0055768 0.1049818 -0.4758222 -0.2169105 -0.3072745
## [246] 0.0933075 -0.7392960 1.0421815 -1.1578191 0.3676146 0.7607657 -0.8584776
## [253] 0.8831775 0.3287652 0.1361803 -0.6416812 -0.2915465 -0.6756173 -0.5068786
## [260] -0.1567490 -1.1817461 -0.1892151 -0.9154348 -1.1322589 0.3713826 0.1211463
## [267] -0.4393866 -0.6008687 -1.2164199 -0.3007191 -0.1741822 0.1598362 -0.3887060
## [274] 0.8498998 0.7028302 -1.3262065 0.2686771 0.2226323 -0.3958609 0.2180882

```

```

## [281]  0.6739678 -1.2648283  0.6697831  0.4803519 -0.5258584 -0.3372766 -0.5481708
## [288] -0.4314580  0.6324790  0.0889707  1.0433930 -0.6290895  0.0331879  0.6594995
## [295]  0.4030203 -0.2119869 -1.0917226  0.1628626  0.7566581 -0.6756384  1.0354888
## [302] -0.0161928 -1.2519548 -1.0867855  0.1614246 -1.3871639 -0.2802579 -0.9588797
## [309]  0.8024046  1.3731661  0.4842462 -0.9089892  0.5181268  1.1873780  0.0455582
## [316]  1.3628260 -0.5768797  1.3969773  0.4067119  0.2344363 -1.0758982  0.5105607
## [323] -1.1341509  0.2420200  1.1215378 -0.2589354 -0.6451712 -0.5361578 -1.3980728
## [330] -1.1483164 -1.2992508  0.8945582  0.6122086 -0.9802646  1.2979440  0.3709015
## [337]  0.3510129 -1.0038165 -0.2462562  0.6817230  0.2152722 -1.1039394 -0.8960031
## [344]  1.3905486 -0.0774432 -0.9740506 -1.1820709  1.3236139 -0.5074855 -0.7810500
## [351] -0.9038618 -0.0517541 -0.3792011  0.2095398  0.5184043  0.2518006  1.0750175
## [358]  0.8448428  0.3879095 -0.7529015 -1.0685987 -0.0233574 -0.1020271 -0.4278014
## [365] -0.9038864  0.9435252  0.1735094  0.4953899  0.0252469  0.2226378  0.4836451
## [372]  0.7575782 -1.2019771 -0.9962864  0.2366211 -0.6480700 -0.0179805 -0.4319929
## [379] -0.8765357  1.1998897 -0.5245900 -0.9809223 -1.3248284 -0.8364838  0.7746720
## [386] -0.7865849 -0.1421376  0.9176033  0.4797378 -0.7543781 -0.9495982 -0.5678900
## [393] -0.3863940 -0.0098751 -1.1176136 -1.2684185 -1.3957207 -0.5474932 -0.3398704
## [400]  0.1734034 -0.8107583 -1.2046804  1.3059276  0.2351445  0.5421617  0.5692626
## [407]  0.0221143  1.2653843  0.8549899 -1.3572672 -0.4061459  0.5259172  0.4953469
## [414]  1.1418405 -0.5031119 -0.2934705  0.7972853  0.2277216 -0.6793639  1.2683148
## [421]  1.3880194 -0.0001133 -0.1228373  0.3299683  0.1081748  0.2953685 -0.4901730
## [428] -0.8000822  0.8610504  1.2875564 -0.6961537  1.1719176  1.0276166  0.5194553
## [435] -0.8597760  0.8114405 -0.8311135 -0.4485620  0.6287730  0.9242558 -0.7967636
## [442]  0.8779812  1.3885881  0.2393456  0.9886750 -0.7319828  0.8612522  0.5374636
## [449]  0.6608441 -0.2515325 -1.2813134 -1.1266847 -0.8864277 -0.0413027 -0.5700707
## [456] -1.0045570  0.0375419  0.8573165 -0.9554322 -1.0150807 -1.2840971 -0.1927957
## [463] -1.2533882  1.0719233  1.3417624 -0.7014611  0.9831783  1.0195514  0.5202738
## [470] -1.2470392  1.0950276  1.2875412  0.7850371  0.7907889 -0.6750613  1.1294683
## [477] -0.2756181  0.2589736  0.6372773  0.3992569  0.0478652  0.1594119  1.0231304
## [484] -0.3645626  0.2477278 -0.2113683  1.3363607 -1.0260636  0.1335437 -0.5807907
## [491]  0.7982307  1.0120971  0.2127687 -0.7813539  1.3480292  0.5080422 -0.9666491
## [498]  0.5465354  0.3828294  0.3808315 -0.5092864  0.4657719 -1.0186391  1.3982555
## [505]  0.3577707 -1.0734531  0.7788086  0.1090838  0.3493942  0.6872526 -0.7181192
## [512]  0.2784926  0.5715164 -0.2153574 -0.3801539  0.7086022 -0.6582888  0.1589068
## [519] -0.9913695 -0.8201754  0.3146667  1.1708985 -0.2586903  0.1591459  0.8075529
## [526] -0.8603021 -1.1199079 -0.4660253 -0.2422798 -0.2898819 -0.0696520  0.2338300
## [533]  1.3965890  0.2382538  0.8045589  0.2114433  0.2104550 -1.3666398  0.9202449
## [540] -0.3748762  0.0340528 -0.0202035  0.7284138 -0.8800140  0.4129142 -0.9331238
## [547] -1.0557947 -0.0963805  0.0389076  0.8440898  1.2350790 -1.2027604 -0.1877086
## [554] -0.1918877  1.0281049 -0.3320932  1.1284081  0.3438081  1.1947365  0.2911917
## [561]  1.0712471 -0.6081612  1.1492292  1.0462753 -0.8863298  1.0953263  0.0327776
## [568] -0.6819311 -1.3976158 -0.8346339  0.8756920  0.5559957  0.4342843 -1.3825270
## [575] -0.9233480 -0.3155322  1.3579821 -0.6332898  0.8006095  0.1029229 -1.1983998
## [582] -1.1358714  0.6839311  0.2519630  1.1089032  1.2919013  0.5416066 -0.4313556
## [589]  1.3940270  0.2722846  1.2617622 -0.8990289 -0.7533248  0.1182925  0.1150886
## [596]  0.9059745 -1.1993993 -0.9989690 -1.3328585  1.1276101

# plot of the innnovation error v_t against the robust score u_t
ts.plot(v)
lines(filter$Innovation_u_t,col="green")

```

